To Thine Own σ Be False: Bounding Closure Misspecification against Input Insufficiency on the Activity Axis When Reference σ-Profiles Hurt COSMO-SAC Solubility Prediction

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Abstract

Physics-informed models route a prediction through interpretable physical quantities, on the expectation that grounding those quantities in accurate reference values improves accuracy. Whether it does depends on the fixed model that consumes them, and on which sense of grounding is meant. We make that dependence measurable for solid–liquid solubility. A graph neural network predicts a molecule's charge-density profile, which a fixed thermodynamic model, COSMO-SAC, turns into an activity coefficient and hence a solubility; a quantum-chemical database tabulates the same profiles, giving the learned one an external reference. That reference is an instrument: it splits the fixed model's error into the part the model itself causes and the part its inputs cause. Grounding names two opposite operations on that database, and they carry opposite signs. Supervising the predicted profile against the database during training improves solubility accuracy over three seeds; substituting the reference for the learned profile at inference degrades it, by about twice as much. On infinite-dilution activity data, where the two terms separate, the instrument bounds the deployed COSMO-SAC-2002 model's own error from below and the contribution of its inputs from above, and on the larger of the two sets measured the first bound is the larger; the error localises on hydrogen-bond-donating solvents. Its donor/acceptor-typed 2010 successor lowers the error on that chemistry and shows no improvement on the broader set measured. Trained end to end through the misspecified model, the learned profile departs from its reference, over three seeds, along a direction shared across the molecules it can be compared on; whether that departure cancels the model's own error is not measured. The same reversal of sign occurs for acid dissociation constants predicted through a fixed Hammett relation, where most of the error is carried by the tabulated constants themselves. Grounded latents are not reliably interpretable when the model consuming them is misspecified.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug can be formulated as a tablet or an injection, which solvent a crystallisation is run in, and how a product is purified. The quantity has to be measured one solute, one solvent and one temperature at a time, so the overwhelming majority of the combinations that matter industrially have never been determined, and independent determinations of the same system in different laboratories disagree by enough to put a floor under what any predictor can achieve. Predicting solubility from molecular structure is therefore a long-standing goal, and a hard one. Dissolving a solid trades the cost of breaking up its crystal against the cost of mixing the freed molecules into the solvent, so a predictor has to be right about two unrelated kinds of chemistry at once, and has to stay right for molecules unlike any it was shown.

Predictors of solubility divide by how much physical structure they impose. Direct models regress solubility from molecular descriptors or from a representation learned off the molecular graph: FastSolv[1] does so with a deep neural network and leads on extrapolation to new solutes. They are at present the more accurate there, and they are opaque: the prediction arrives as a single number, with no account of which part of the chemistry it captured. Physics-informed models instead compute solubility from the thermodynamic quantities the physical picture names, an activity coefficient for the solute–solvent interaction and a melting temperature and heat of fusion for the crystal, predicting those quantities and combining them through a fixed equation; SolProp[2] assembles such a thermodynamic cycle out of separately learned components (solvation free energy, vapour pressure, fusion properties). What that provides is decomposability: a prediction arrives with its parts, and each part is a quantity chemists measure and tabulate on its own, so it can be supervised against reference data a direct regressor has no way to use. On accuracy the black box leads and the physics-informed cycle trails it.

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. For one class of intermediate that assumption can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed for itself. It needs testing, because when a predictor of this shape is inaccurate the fixed physical model may be wrong or the learned quantities feeding it may be uninformative, and the two call for opposite remedies: a more faithful physical model in the first case, better representations or more data in the second.

Learned surrogates for the activity coefficient—the quantity a solubility model must supply—have advanced quickly. Winter et al.[3] predict infinite-dilution activity coefficients $\ln \gamma_2^\infty$ directly from SMILES with a language model, and Sanchez-Medina et al.[4] embed a Gibbs–Helmholtz constraint in a graph network to capture their temperature dependence. Both treat the activity coefficient as the regression target rather than retaining a mechanistic model that consumes a predicted molecular property. A second line keeps the mechanistic model and learns its input. COSMO-SAC[5] is a segment-activity reformulation of Klamt's COSMO-RS, and the σ-profile it consumes — screening-charge density over a molecule's cavity — is that framework's construct, tabulated from quantum chemistry.[6] Predicting it from structure removes that calculation from the pipeline; TeNNet-SAC[7] does so while also learning the segment-activity kernel, and gains from truer profiles.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning,[8] whose dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,[9] gray-box hybrids composing learned components with fixed mechanistic maps,[10] and operator learning with a trainable correction to a misspecified equation.[11] The prevailing expectation is that more physics improves generalisation, yet the same structure has documented failure modes in which it makes optimisation or accuracy worse.[12] In neural solvers of partial differential equations, Mohan et al.[13] show that a learnable term added to a fixed discretised operator absorbs truncation artifacts rather than physics, so the program's apparent interpretability is illusory; their error source is numerical and has no external reference.

The same concern appears in concept-bottleneck models as *concept leakage*, in which a learned concept silently encodes information beyond its nominal meaning, so its interpretation misleads.[14–16] Recent work traces one route to leakage to the downstream model: a misspecified head forces the encoder to deform the concept into carrying a dependence the head cannot represent,[17] and overwriting a frozen head's leaked concepts with ground truth can then degrade accuracy.[18] The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan[19] and its calibration critique.[20] Four further literatures bear on it—forecast decomposition, quasi-maximum likelihood, identifiability, and the validity of composed pipelines—and are reviewed in App. L.

What this literature lacks is a way to *measure*, rather than merely diagnose, whether a fixed physical model or its learned inputs limits a composed predictor's accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after; in the concept-bottleneck case there is no external reference against which either question could be settled. This work supplies one—a reference σ-profile for solubility, tabulated substituent constants for a second chemical domain—and holds the mechanistic kernel fixed, so the tabulated value stays available as an external reference for the learned one. That is what turns a qualitative concern into a measurement.

The test and the diagnosis are run here on a solubility pipeline of that shape. Continuum-solvation theory summarises a molecule by the distribution of screening charge over its surface, and one branch of that theory, COSMO-SAC, computes an activity coefficient—and from it a solubility—from a pair of those distributions. They are ordinarily obtained from quantum chemistry and tabulated in a published database; here a graph neural network predicts them instead, which leaves the network's own intermediate with a tabulated counterpart to be held against. The counterpart can enter the pipeline at either of two points: as a target the network is trained toward, or as a replacement for what the network computed, at the moment of prediction (Fig. 1). We implement the thermodynamic model in differentiable form so that both routes are open on a single trained network, and we put the tabulated values to a third use, dividing the fixed model's error between the map itself and the inputs handed to it.

Only one of the two routes helps, and the reason turns out to be a property of the fixed map rather than of the network. Solubility data cannot say which component is at fault, because one measurement constrains the pair jointly; activity coefficients can, because both molecules of a pair carry a tabulated distribution and the two error terms then separate. What separates them is an ordering: the lower bound on the map's own error stands above the upper bound on what its inputs fail to carry. A map in that condition does not become more accurate when it is handed a better input. It becomes more exposed, because an accurate input no longer masks the bias the map itself introduces, and the error surfaces where the map is weakest—here on solvents that donate hydrogen bonds, the chemistry the next version of that map was written to describe.

Two consequences follow for anyone building a predictor of this shape. A learned intermediate trained through a misspecified map cannot be read as the physical quantity it is named after: the predicted distribution drifts away from the tabulated one, and the departures share a common direction across the molecules that can be checked. And substituting a reference value for that intermediate at prediction time is not a safe improvement but a change of regime, which is why the substitution here costs more than the supervision gained. Neither consequence is peculiar to charge distributions: tabulated substituent constants behave the same way inside a fixed Hammett relation for acid dissociation, improving the aromatic series that relation was built for and degrading the ones outside it.

This article is organised as follows. Section 2 sets out the composed predictor, derives the split of its error into a model term and an input term, and states which parts of that split are exact and which can only be bounded. Section 3 describes the data, the network, the thermodynamic layers and the substitution experiment. Section 4 fixes the grades the claims carry and reports the measurements: the two senses of grounding and their signs, the split on the two activity data sets, the effect of raising the fixed model's fidelity, and the drift of the learned profile away from its reference. Section 5 draws out what the results do and do not license and lists the limitations. Section 6 concludes.

2 Setup and theory

For a solute s in solvent v at temperature T, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades two costs against each other—breaking the crystal lattice (Φ) and mixing the freed molecules against non-ideal solute–solvent interactions ($\ln \gamma_2$)—and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\text{fus}}}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right)}_{\Phi(T)\ \text{(crystal)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with the constant-ΔC_p (heat-capacity-of-fusion) correction omitted for clarity. The activity term is produced by a *differentiable closure* g: a graph encoder predicts a σ-profile—the distribution of screening (surface) charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC (the COnductor-like Screening Model–Segment Activity Coefficient,[5] a reformulation of COSMO-RS) maps the pair of profiles to $\ln \gamma_2$. Inside that map, the segment-interaction energy—the electrostatic-misfit and hydrogen-bonding terms that set what two surface segments cost each other—is the closure's *kernel*. It is the part the published revisions of COSMO-SAC rewrite, so a closure variant is named for the year of its kernel: the deployed *2002 kernel*, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write $z^\star$ for the *reference* physical latent, the externally referenced σ-profile from the VT-2005 database,[6] and $g(z^\star)$ for the closure output when fed the reference profile. The model's own head instead predicts a *learned* profile $\hat{z}$ and evaluates $g(\hat{z})$. We call each trained configuration, and each closure variant we evaluate, an *arm*; a *DirectGNN* control arm drops the closure and emits $\ln x_2$ directly, and is the reference against which the cost of the physics bottleneck is measured (schematic in SI, Fig. 4). The two arms were tuned separately, so the gap between them is not attributable to the bottleneck alone (§3); per-arm settings in App. A.

The oracle substitution. Feeding $z^\star$ in place of $\hat{z}$ (a σ-oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^\star)$ should be the best available activity estimate. The paradox is that it is not.

Data. The activity target m is the experimental infinite-dilution activity coefficient ($\ln \gamma_2^\infty$, IDAC) and the reference latent $z^\star$ is the VT-2005 σ-profile; sources, the scaffold split and the matching rules are §3 and App. I.

The results below are stated for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure g consumes a learned physical intermediate $z = h(x)$, and are compared against a matched free predictor that shares the encoder but drops g. Solubility, with g the COSMO-SAC closure and z the σ-profile, is one instance; App. G gives a second (pKa through a fixed Hammett closure), and App. K a controlled synthetic family. Nothing in the decomposition of Lemma 1, or in the results stated in App. B, is specific to COSMO-SAC. Two quantities organise the section: the closure/insufficiency split $B = B_{\text{closure}} + B_{\text{insuff}}$, which is the assumption-light *instrument*, and the observable grounding gap $\mathcal{G}$, which is an *indicator* of closure misspecification whose validity we characterise.

2.1 Why a free head can compensate: structural non-identifiability

The crystal/activity split is not identifiable from solubility alone, which is why an unconstrained head can silently absorb crystal error into the activity term—an indeterminate component that *end-to-end* training of

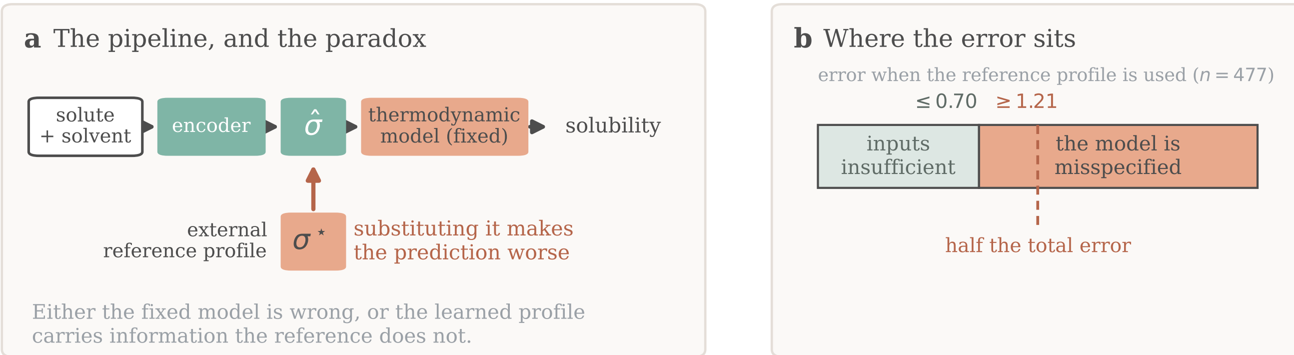

Figure 1: **Overview.** **(a)** A learned encoder predicts a charge-density profile $\hat{\sigma}$; a fixed thermodynamic model turns it into a solubility, and the arrow marks the evaluation-time substitution of the reference $\sigma^\star$, which makes the prediction worse. **(b)** The counterpart error on infinite-dilution activity data, each block a one-sided bound rather than an interval. Training-time supervision is a separate result, not drawn.

the SLE-coupled latent leaves free rather than projecting out, and that is closed only by a rank-completing external constraint (Lemma 2, App. B). It is *enabling background* rather than a principal claim: it explains why grounding an under-determined factor on external single-component data is necessary, and why compensation between the factors is expected, and by itself says nothing about whether grounding helps—which depends on whether the closure consuming the grounded inputs is correctly specified, the question we make measurable next. The design dependence, the graded rather than binary character of the non-identifiability and the operating point that fixes it are in App. B and §F.2.

2.2 The closure–insufficiency decomposition: the instrument

Let m be the activity target (experimental $\ln \gamma_2^\infty$) and $g(z^\star)$ the closure evaluated on the true latent.

Lemma 1 (Exact closure–insufficiency decomposition and its one-sided bounds). *For a* fixed *(non-fitted) closure g, with $z^\star$ denoting the closure's* entire *argument (the profile pair together with the temperature wherever the closure reads one),*

$$\underbrace{\mathbb{E}\big[(m - g(z^\star))^2\big]}_{\textit{ref.-input MSE}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\text{insuff}}\ (\textit{irreducible})} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2\big]}_{B_{\text{closure}}\ (\textit{decoder bias})}. \tag{2}$$

The cross term vanishes identically, which is why $z^\star$ must be the closure's whole argument and not a part of it (proof, App. B). The split is exact with no fitting of the closure g, so the total and B_{closure}'s Jensen lower bound are fit-free. $B_{\text{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$, however, must be estimated with a conditional-variance smoother whose choice affects the verdict (§4.2).

B_{closure} is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O'Hagan,[19,20] and its dominance over B_{insuff} is the empirical content of the claim that the accuracy ceiling is set by the closure rather than by the inputs. Where model-discrepancy calibration *fits* a correction from held-out observations, here g is fixed and $z^\star$ externally observed, so the map's error is *attributed* rather than corrected—available only where an external latent oracle exists, not in gray-box hybrids (App. L). **Exactness is contingent on g being fixed**: for a *fitted* decoder the cross term does not vanish and a plug-in point split is in-valid. We therefore report *one-sided bounds*, not a point split (proved in App. B):

- $B_{\text{closure}} \geq (\mathbb{E}[m]-\mathbb{E}[g])^2$ by Jensen: a constant offset between the decoder's mean output and the target is misspecification, not input insufficiency, and this bound needs *no smoothness assumption.* Its one requirement is that m and g share the $\ln\gamma_2^\infty$ reference state, which the closure validation of §3 establishes.

- $B_{\text{insuff}} \leq \mathbb{E}[\mathrm{Var}(m \mid \mathrm{bin}(g(z^\star)))]$, a valid *upper* bound (law of total variance, since a binning of $g(z^\star)$ coarsens $z^\star$).

Bin edges. Where the equal-count edges fall is a fourth analyst choice, beside the bin count, the within-bin variance convention and the combinatorial convention. Enumerating every equal-count contiguous partition, the choice is immaterial on the broad set and material on the *corner*, whose margin changes sign inside the envelope; we accordingly read the corner's $+0.05$ as consistent with zero and carry the claim on the broad set. The Jensen constant-offset bound is convention-specific—it separates under the full combinatorial convention and inverts under the deployed residual-only default—so the separation reported in §4.2 rests on the leakage-immune (that is, model-free) law-of-total-variance (LOTV) bound. B_{insuff} is one population quantity while the two conventions coarsen two different statistics of it, so both bounds are valid on it, and so is the smaller. The binning rule and its envelope and the Jensen numbers are in App. C and App. B; §4.2(i) fixes which we headline and §4.2(iii) states why the 2002-vs-2010 contrast is not a fixed-input decoder difference.

What population, what is conditioned on, and what sets the instrument's resolution. B_{insuff} is well posed only once the population over which the conditional variance is taken and the variable conditioned on are named. The population is the *measurement superpopulation*: a draw is a nominal compound pair *at a temperature*, together with the conformer, tautomer and protonation state actually populated under those measurement conditions, and an independent experimental determination of $\ln\gamma_2^\infty$ for it. The conditioning variable is whatever the closure reads—its full argument, $(z^\star, T)$ where g reads a temperature, as on the broad IDAC set. Neither the conformational state nor the replicate determination is part of it, so $\mathrm{Var}(m \mid z^\star, T)$ is the spread of $\ln\gamma_2^\infty$ across the conformational ensemble and across replicate determinations at one reference profile and temperature, non-zero for physical reasons unrelated to sample size. Experimental label noise σ_η^2 is a *component* of it, not a term beside it: an independent zero-mean error on m enters $\mathrm{Var}(m \mid z^\star, T)$ directly, so it is inside every bound we report and never added on. What the design cannot do is *observe* it, neither set holding two rows at a common argument (§I). On these two sets B_{insuff} is therefore reached only as an *upper bound*, and B_{closure} only lower-bounded.

The bounds are conservative in one direction only—were the superpopulation variance negligible the finding would be the stronger $B_{\text{closure}} = \mathrm{MSE}$—so nothing reported can be an artifact of over-estimating B_{insuff}, and no entry is the share of the error the inputs contribute. A low P_{boot} (Table 1) is likewise loss of resolution, never evidence for the inputs. Resolution is governed by $\dim(z^\star)/n$: the VT-2005 corner supports one-sided bounds only, and the broad IDAC set is several times better resolved (§4.2(i)). What each estimator conditions on, why the identity and the coarsening bound both fail if an argument of g is left out of the conditioning variable, and the scalar-latent pKa case are in App. B, App. C and App. G.

2.3 What the instrument resolves, and what it does not

The decomposition (§2.2) is the measurement the conclusions rest on—assumption-light, needing neither a representable encoder nor a lossless bottleneck. Alongside it we report a second, purely observable quantity: the *grounding gap* $\mathcal{G} := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk; the paradox is $\mathcal{G} > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \leq \mathcal{G} \leq B_{\text{closure}}$, so an observed $\mathcal{G} > 0$ would certify $B_{\text{closure}} > 0$ (Theorem 1, App. B). A1 is, however, empirically doubtful (the encoder is transfer-limited: a linear probe drops from train $R^2 \approx 0.76$ to test $R^2 \approx 0.45$) and A2 is unverifiable and physically dubious; when A2 fails, $\mathcal{G} > 0$ conflates closure misspecification with input insufficiency—a well-specified closure with a lossy latent gives $B_{\text{closure}} = 0$ yet $\mathcal{G} > 0$ (App. K, and the pKa ortho sweep of App. G). We therefore read $\mathcal{G}$ as corroborating *evidence* and base the model-dominance conclusion on the decomposition: *the decomposition is the instrument; $\mathcal{G}$ is the indicator that motivates it.*

When a fixed prior loses; and scope. The deployed question—does the *trained* physics model outperform the *trained* black box at sample size n?—follows a sign rule with two halves. A constrained prior helps only where the estimation variance it saves exceeds its asymptotic bias Δ_∞; and a prior whose asymptotic bias is strictly positive always loses once the budget passes a critical size, the variance it saves vanishing as data accumulate while Δ_∞ does not (Proposition 1, its variance decomposition and a limiting refinement, App. B). Each physics-versus-control difference reported below is a penalty at the budget actually trained, not an asymptotic one. There is no "optimal grounding" theorem, and the decomposition does not improve out-of-distribution accuracy (§4). Everything here is measured at infinite dilution, on two sets that differ in regime (§4.2), and transfer of either to the finite-composition regime in which the paradox is observed is a conjecture (§5). We read these results against the model-discrepancy, sloppiness, identifiability, misspecification, underspecification and null-space-prior literatures (App. L).

Table 1 fixes the symbols and the coined vocabulary the Results use; the thermodynamic symbols of Eq. (1) are defined above and not repeated. Where a paper spanning two chemical domains would collide on one symbol, the table gives the renaming that resolves it.

3 Computational methods

The study is a controlled comparison: a shared graph backbone feeds either an explicit thermodynamic bottleneck (*TGNNSolv*) or a direct regressor (*DirectGNN*) that emits $\ln x_2$ with no physics. Architecture, encoder variants, the solver, the curriculum, the loss taxonomy, the COSMO-SAC constants and every hyperparameter are in §A; provenance and the audits behind the disclosures below, in §I.

3.1 Data, splits, and labels

Solubility measurements come from BigSolDB 2.0:[21] 127,930 rows over 15,665 solutes and 221 solvents between 243 and 438 K, partitioned by Bemis–Murcko scaffold into 111,724 training, 8,103 validation and 8,103 test rows so that every test solute is unseen at training; a melting-point-independent miscibility/structure filter sets row membership, and crystal T_m, ΔH_{fus}, Hansen parameters and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ are merged on as per-solute labels where available. Crystal supervision is scarce—the test and validation splits contain essentially no measured ΔH_{fus}, the quantitative basis of the identifiability argument (§2). The external single-component data used for grounding are an open crystal-property pool (∼15,000 melting points), the VT-2005 σ-profile database, and ThermoML infinite-dilution activity coefficients.[22] Two hazards the pipeline handles explicitly—melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions—are recorded in §I.

Disclosure: the scaffold guarantee is not certified for these runs. The released stream builder excludes any pool molecule whose scaffold appears in validation or test and aborts on a missing split file — but that is a property of the builder, not a certified property of the runs reported here. One σ-stream build on disk was generated against the wrong split files, 91 of its 1425 rows being held-out solutes by canonical SMILES, and per-run stream provenance was not logged, so the arti-

Table 1: Symbols and coined terms used throughout.

Symbol / term	Meaning
g (closure), h (encoder)	fixed, non-fitted physical map (COSMO-SAC; a Hammett linear free-energy relation, LFER, for pKa); learned graph network $x \mapsto z$
$z^\star$, $\hat{z}$, m	reference (oracle) vs. learned σ-profile; activity target (experimental $\ln \gamma_2^\infty$). $z^\star$ is the closure's whole argument: the concatenated profile pair, written $(z^\star, T)$ where g also reads a temperature
B_{closure}, B_{insuff}	closure misspecification $\mathbb{E}[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2]$; input insufficiency $\mathbb{E}[\text{Var}(m \mid z^\star)]$, over the measurement superpopulation of §2.2
separation margin	$\text{MSE} - 2B^{\text{up}}_{\text{insuff}}$; positive witnesses $B_{\text{closure}} > B_{\text{insuff}}$ on the sample measured, negative loses that ordering rather than reversing it
the corner / the broad IDAC set	the two activity data sets the instrument is run on: the VT-2005-matchable pairs at one temperature, and the larger set matched to the University of Delaware (UD) σ-profile database across temperatures (§4.2)
LOTV; leakage-immune	law of total variance, the model-free binning bound on B_{insuff}; *leakage-immune*, fitting no model, so no held-out row borrows strength from a near-twin
$\mathcal{G}$ (grounding gap)	$R_{\text{orc}} - R_{\text{e2e}}$: oracle risk minus best end-to-end risk (R_{free} is the free black box's)
Γ_S; σ_{H}; σ_η	COSMO-SAC segment activity coefficient (Eq. (S1)), *not* the grounding gap $\mathcal{G}$; Hammett substituent constant, renamed from σ; experimental label noise. Standing alone, σ is the charge-density profile, and $\sigma(\cdot)$ with an argument is the generated σ-algebra
Δ_∞ (physics penalty)	asymptotic risk of the physics arm minus the budget-matched control
full / res	COSMO-SAC with / without the Staverman–Guggenheim combinatorial term
compensating surrogate	a learned $\hat{z}$ that would *cancel* the closure's error instead of matching $z^\star$; the hypothesis of §4.4
P_{boot}	clustered-bootstrap frequency that the *measured* separation margin stays positive: the instrument's resolution on this design, not a posterior probability

facts cannot certify which build fed which arm. §I gives the file-order evidence, the three facts that bound what a leak could do, and the one molecule it touches in the $n{=}44$ analysis of §4.4. We therefore do not assert the scaffold guarantee for these runs.

3.2 Model, closure, and training

A *shared* message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head; two interchangeable encoder alternatives do not outperform it. A fusion head predicts T_m and ΔH_{fus}, giving the ideal term $\Phi(T)$ of Eq. (1) with the constant-ΔC_p correction omitted (audited in §E). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls: the fitted NRTL model and its symmetric one-parameter collapse, in which the identifiability analysis is cleanest (§2). The third, parameter-free, is the closure under study—a σ-profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln \gamma_2$ with an optional Staverman–Guggenheim combinatorial term, the full/residual-only distinction on which Table 2 and §4.2 turn. That closure over a network-*predicted* σ-profile, and the intervention that swaps it for the external reference, are the object of study (§4.1).

The saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$ as a damped fixed point in x_2 (§A); a bounded, gated "adaptive physics correction" perturbs the physical parameters rather than the output, so the physical decomposition survives it. Training follows a three-phase curriculum under a weighted loss. External single-component data enter through *grounding streams*—the training-time sense of grounding, and a separate operation from the evaluation-time substitution below. Each stream is a sidecar of *self-solvent* rows (solvent = solute, no solubility label, only the target factor's mask active), interleaved into training under the guard above; the streams ground the crystal factor from the melting-point pool and the activity factor from the σ-profile database, identifying from external data a factor that solubility alone leaves non-identifiable (§2) and that a black-box predictor cannot use. Training runs a three-stage curriculum: a warm-up on the auxiliary streams alone, then joint training against solubility, then a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the σ head at the end of the warm-up, so the σ stream enters the warm-up only (§C).

The in-house closures, and where their validation stops. The dispersion-off closures in the fidelity measurement (§4.2) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed σ-averaging (Mullins for 2002, Hsieh for 2010). Our 2002 layer—the *fair 2002* arm wherever the kernels are compared, because it reads all three profile channels—reproduces the NIST reference COSMO-SAC-2002[23] to RMSE 0.0035 $\ln\gamma$ on identical VT-2005 profiles, and our COSMO-SAC-2010/dsp layer,[24,25] built on the *typed* latent of three donor/acceptor-resolved σ-profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off*. The dispersion term is *not* validated to that standard, which is why every dispersion-off contrast is computed on our layers while the *2010, dispersion on* runs are quoted from `cCOSMO`. Layer definitions, validation figures and the full scope of the dispersion caveat are in §A.

3.3 Evaluation protocol, and the DirectGNN control

The *oracle substitution* replaces the predicted σ-profile $\hat{z}$ with the reference VT-2005 profile $z^\star$ before the closure, matched by canonical SMILES. It is applied three ways to rule out implementation artifacts: at evaluation only; injected at training time so the model co-adapts; and as a channel-swap control, injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase, so that only the activity branch refits against it. The last two are at seed 42 only. An analogous override substitutes reference crystal targets. The activity target for the decomposition (§4.2) is the infinite-dilution $\ln\gamma_2^\infty = \lim_{x_2\to 0}\ln\gamma_2$, on two sets that differ in their reference database and their regime (§4.2(i)–(ii)).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no heads, closure or solver. The two arms therefore share their representation but not their inputs or their tuning: the control receives an explicit temperature encoding the physics arm does not, seeing T instead through $\Phi(T)$ and the closure, and the optimisation settings were tuned per arm and differ. The TGNNSolv$-$DirectGNN gap is accordingly a comparison of two separately tuned pipelines that differ in the thermodynamic bottleneck, not an isolation of the bottleneck, and every asymptotic physics penalty comparison in §4 carries that caveat.

The controlled comparisons of §4.1 are run at three seeds and reported as mean±sd on a cross-arm row set (rows supervised and finite in every arm). Single-seed, and labelled as such where each is reported, are the training-time and channel-swap oracle controls, the output-residual arms (§4.5), the data-efficiency curve (App. F.5) and the loss-simplification ablation (App. A). The two audit runs behind §4.2 are named there with the script and the artifact file of each; the repository that would let those scripts be run is a placeholder here, and not every artifact they read is small enough to version (limitation (x), §5).

4 Results

How the claims are graded. Table 2 collects, as set, value and grade, the claims of §4.1, §4.2(i)–(iii), §4.4 and the accuracy comparison below; the downstream-correction result of §4.5 is graded where it is stated, and the second-domain result of §4.6 and the synthetic sweep of App. K carry their scope in the text instead. *Certified* means it is exactly proved, is bootstrap-certified across resamples of many clusters, or is a three-seed contrast whose per-seed values do not overlap. *Likely* means measured but short of that: a clustered-bootstrap frequency that does not clear 0.95 under every clustering, a quantity reached only through a one-sided bound, a single-seed contrast, or a result resolved only on the matched n=60/n=44 sets. *Open* means unestablished, directional, or conjectural; it is annotated with the direction of the evidence where there is one (*fails*, *negative*, *null*), and *no claim* marks a difference we decline to interpret because the arms were tuned separately. We make no accuracy claim in either direction: the two arms were tuned separately (§3), so no difference between them is attributable to the thermodynamic bottleneck, and that ΔMAE is a difference between pipelines. Bootstrap frequencies are quoted as bands—rounded to one decimal, with > 0.95 and < 0.2 closing the ends—because with 17 solvent and 41 solute clusters on the corner they do not carry two-decimal resolution (App. C). The two-decimal values are in the deposited artifacts. The rule is a rounding rule, not a fixed vocabulary, so every frequency in the paper and in Table 2 is quotable within it.

The instrument (§4.2) and the latent's departure from its reference (§4.4) are the two results the conclusions rest on; both are graded *likely* and neither is resolved. A controlled synthetic family sweeps closure fidelity directly, where the response is true by construction and so checks the instrument rather than the hypothesis (App. K); further diagnostics are in the SI.

4.1 The grounding paradox

The word "grounding" names two opposite operations on the same VT-2005 database, and they carry opposite signs. Supervising the σ head against the database *during training* improves solubility MAE from 2.043 ± 0.040 (ungrounded) to 1.846 ± 0.053 (grounded), a gain of 0.20 over three seeds (Table 4). Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the paradox; which of the two causes it fails from is not decided here but on the activity axis below. The effect erases any positive R^2 (Fig. 2, p. 13): the model predicts solubility better *from its own learned σ-profile* than when *given the external reference one* (MAE 1.85 vs 2.25, in $\ln x_2$ units). The figure's eight arms are two non-COSMO controls and six COSMO-SAC arms. The controls are DirectGNN, which shares the encoder but has no thermodynamic model, and NRTL, a more expressive fitted closure. Of the six COSMO-SAC arms, five differ only in how the σ head is trained and in what it is fed, and one (*grounded+comb.*) differs instead in the closure, which restores the Staverman–Guggenheim combinatorial term (§3). The figure's tick labels abbreviate the rest: *grounded* is the learned-σ arm; *ref. σ (eval)* is the σ-oracle, that checkpoint re-evaluated with the external VT-2005 profile substituted for the learned one at test time, and *ref. σ (train)* is the co-adaptation control, the same reference profile injected during training. The physics arms trail DirectGNN, whose optimisation settings were tuned separately from theirs, so that gap is not attributable to the bottleneck (§3); the two remaining bars are the co-adaptation controls below.

The headline +0.41 is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. The control that would remove that component injects the reference profile *at training*, so the model co-adapts; at seed 42 it costs +0.18 ($1.803 \rightarrow 1.981$) where the evaluation-only substitution costs +0.43 at that same seed ($1.803 \rightarrow 2.232$). One seed does not establish it, but the direction points to part of the headline gap going with co-adaptation; the co-adapted arm is by design the confound-free one. A channel-swap control costs +0.27, to 2.08. Both are worse than the learned-σ arm, which rules out an implementation artifact. Both controls are single-seed, so they fix a sign and not an effect size, and so does the 0.59 share they imply, which we report as a description of seed 42 rather than as a magnitude. Only the three-

Table 2: One row per load-bearing claim: what is claimed, the data set and its size, the value measured, and the grade the claim carries. Notes *a*–*l* qualify a particular row or column.

Claim	Set $(n)^a$	Value^b	Grade
The grounding paradox (§4.1)			
Reference profiles hurt solubility at evaluation (MAE)	test split (5608), 3 seeds	$1.85 \to 2.25$	certified
…and still hurt when injected at training	cross-arm rows^c, seed 42	$1.80 \to 1.98$ (own baseline $1.80 \to 2.23$)	likely
…so part of that gap is test-time distribution shift	cross-arm rows^c, seed 42	share 0.59, single seed	open
Training-time σ supervision *helps* (MAE)	cross-arm rows^c, 3 seeds	$2.04 \to 1.85$	certified
Closure against inputs on the activity axis (§4.2(i)–(ii))			
$B_{\text{closure}} > B_{\text{insuff}}$, fair 2002 (LOTV^d)	broad IDAC (477), UD	margin +0.51; $P > 0.95/{\approx}0.9/{\approx}0.9^e$	likely
… same rows, full convention^d	as above	margin +0.95; $P > 0.95$ (all three)	likely
$B_{\text{closure}} > B_{\text{insuff}}$, fair 2002 (LOTV^d)	corner (60), VT-2005	margin +0.05, $P \approx 0.7$	likely
$B_{\text{closure}} > B_{\text{insuff}}$, 2010 on its *own* typed latent	broad IDAC (477), UD	margin +1.29; $P > 0.95/{\approx}0.9/{\approx}0.9^e$	likely
$B_{\text{closure}} > B_{\text{insuff}}$, 2010 on its *own* typed latent	corner (60), UD	margin −0.77, $P \approx 0.3$	open (fails)
Whether a better closure moves the ceiling (§4.2(iii))			
The $2002 \to 2010$ step lowers the error	broad IDAC (477)	-0.99, $P \approx 0.2^f$	open (negative)
…but does on the corner's chemistry^g	57 shared pairs, UD	$1.78 \to 0.77$, $P > 0.95$	likely
A fitted kernel residual transfers across solvents	corner (60), grouped folds	$-5 \pm 16\%$	open (null)
Mechanism (§4.4)			
The latent departs from its reference along a shared direction	matched molecules (44), 3 seeds	$51 \pm 2\%^h$	likely
…and that departure's top-two mode share exceeds a phase-randomised null	one grounded checkpoint, same 44	0.40 against 0.22 ± 0.01^i	likely
…and that direction *transfers* to molecules outside the fit	matched molecules (44), 3 seeds	$4.1 \pm 0.5\times$ a zero-drift null^l	open
That departure *cancels* closure error	not measured^j	—	open
Accuracy (§5)			
Physics arm vs. DirectGNN accuracy (ΔMAE)	test split (5608)^k	+0.14	no claim

[a] Sets and profile databases: Table 1; *corner*, the n=60 VT-2005-matchable pairs at 298 K, *broad IDAC*, the 477 IDAC ∩ UD measurements over 185 distinct pairs at all temperatures, UD the University of Delaware σ-profile database. Resolution $\dim(z^\star, T)/n$ is given per set, per row and per distinct pair in §4.2(i)–(ii) and App. I.
[b] *Margin* is $\text{MSE} - 2B^{\text{up}}_{\text{insuff}}$, positive when B_{closure} exceeds B_{insuff}; P is a two-way (solute × solvent) cluster-bootstrap frequency, reported as a band, by the rounding rule at the head of this section. [c] The cross-arm row set common to every arm of Table 4. [d] LOTV rows use eight equal-count bins, the unbiased within-bin variance, and the *deployed residual-only* combinatorial convention on both sets and both kernels (§4.2(i)); rows naming the full convention are comparisons. [e] Bootstrap band over pair / two-way / source clusters. [f] Deployed convention, dispersion off; under the full convention −0.07, $P \approx 0.6$, and +0.06, $P \approx 0.6$ with dispersion on (Table 8). [g] At fixed profile database and temperature, full convention. [h] Magnitude not stable across analysis configurations (§4.4). [i] ± is the null's spread; the fine-tuning drift has no matched null of its own (§5, limitation (v)). [j] The direct check is under-powered and runs against it (§4.4). [k] Separately tuned arms (§3). [l] Zero-drift null, with no constant-offset control on these seeds; the ± spans seeds at one partition (§4.4).

seed arms can be set against one another, and there the substitution costs about twice what supervision gains (+0.41 against −0.20); the co-adapted arm's +0.18 cannot enter that comparison. The between-seed arithmetic that excludes it, and the row set the three oracle controls are computed on, are in App. C. That the fault is a wrong *map* rather than a missing input is measured on the activity axis, where the reference input is clean (§4.2). There the fixed reference-input closure sits well above the bound on what its inputs can fail to resolve, under-predicting $\ln \gamma_2^\infty$ with a systematic bias (SI, Fig. 6).

Why the solubility paradox does not by itself carry the attribution. The solubility effect is real and certified (three seeds on the evaluation-only substitution, plus two single-seed controls), but it cannot carry the closure-versus-input attribution, for a data reason. Drug-like solutes are essentially absent from the VT-2005 reference set, so the reference substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers only ≈7 solutes—too few to separate the cause either way. Enlarging that set would mean computing new reference profiles rather than looking them up. We therefore carry the causal claim on the activity axis, where the reference input is clean and matched on both channels: the decomposition (§4.2) and the latent's departure from its reference (n=44, §4.4). The title names that axis for this reason.

4.2 The closure-fidelity axis: locating the ceiling, and moving it

The externally-referenced decomposition of §2.2 is an *instrument*: it separates a fixed closure's error into a part in the map and a part in its inputs. This subsection runs it on two data sets that must not be conflated—*(i)* the *broad IDAC set* (n=477, UD profiles, all temperatures) and *(ii)* the *corner* (n=60 at 298 K, where the paradox's own VT-2005 database supplies both profiles; Table 1)—and *(iii)* then asks whether a better closure moves the ceiling. A better-resolved counterpart in another domain is the scalar-latent pKa closure (§4.6), where the same instrument recovers the a-priori channel assignment (there, insufficiency)—evidence the instrument works, not a re-confirmation of this verdict.

(i) The decomposition on the broad IDAC set (n=477). The closure reads a temperature as well as a profile pair, and is evaluated here at each row's own temperature, so the conditioning variable is the closure's full argument $(z^\star, T)$—the concatenated UD solute/solvent profile pair together with T, 103 dimensions against n=477 ($\dim/n \approx 0.22$)—as §2.2 requires. The target m is the experimental $\ln \gamma_2^\infty$, so floor and closure error are measured on one data set and one profile database. Its chemistry is small. No solute exceeds 12 heavy atoms (median 7), against a median of 14 and a maximum of 122 for the drug-like solids of this paper's own solubility corpus, and only 0.46% of corpus rows carry a solute from this set. *Broad* therefore names the reach of the γ^∞ corpus the kernel is calibrated and benchmarked on, not coverage of the chemistry a solubility model is deployed against: the set covers the solvent side of deployment and not the solute side. It is matched to UD by InChIKey with a first-block fallback, a looser rule than the corner's; a wrong $z^\star$ inflates B_{closure}, so the looser rule runs against the ordering we claim, and the verdict is unchanged on the 465 exactly matched rows. The construction ladder, composition, provenance audit and deposited table are in §I.

One convention rule, applied to both sets. The margin depends on the combinatorial convention, so we fix the rule before the numbers: headline the *deployed* residual-only convention, pairing MSE and the LOTV bound computed under that same convention, on the broad IDAC set and on the corner alike. Headlining the deployed convention on the corner and the full convention here would be the favourable choice on each set taken separately; one rule applied to both costs the broad set's margin about half its value.

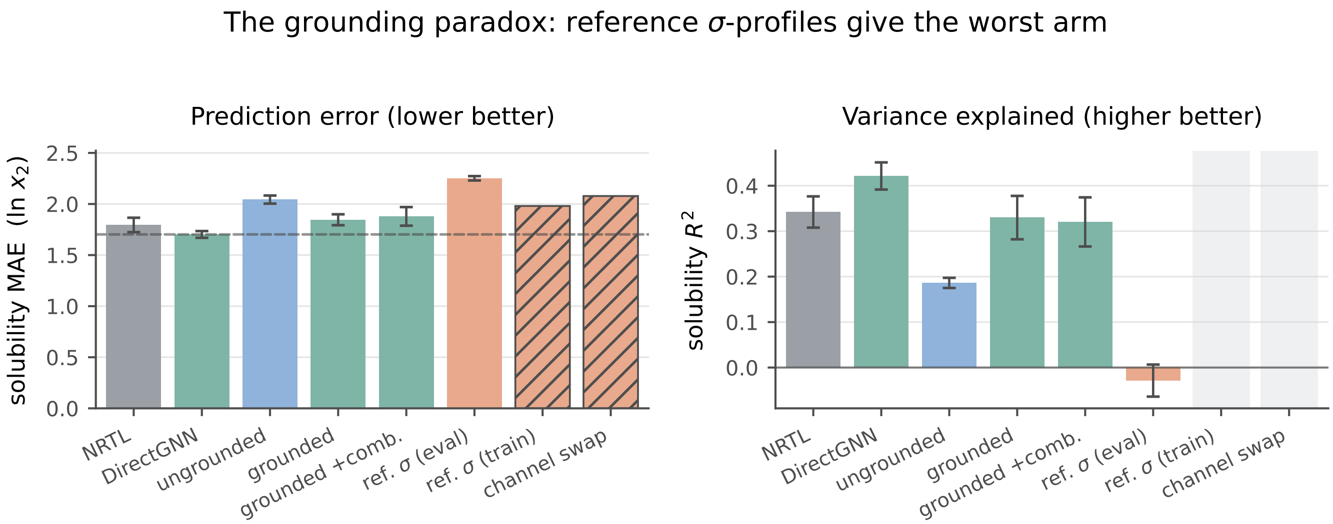

Figure 2: Controlled comparison on the scaffold split (n=5608). Solid bars: three seeds, mean ± a population standard deviation (`ddof`= 0). The two hatched bars are the co-adaptation controls, seed 42 only, so they carry no error bar and no R^2, and keep an empty slot in the right panel. Dashed in the left panel only, the DirectGNN reference.

Under that rule the fair 2002 arm scores MSE = 1.902 against Var(m) = 4.85, the LOTV bound gives $B_{\text{insuff}} \lesssim 0.70$ and hence $B_{\text{closure}} \gtrsim 1.21$, at eight equal-count bins under the *unbiased* within-bin variance: a separation margin of **+0.51**. The full convention on the same rows gives +0.95, reported throughout as the comparison rather than the headline. Two of the three bootstrap clusterings extend just below zero, so on the deployed convention the broad set's ordering is *likely*, not resolved—the grade Table 2 carries for it; under the full convention all three clear. The margin is positive in every cell of the bin-count × variance × convention grid, and collapsing to one row per distinct pair leaves the sign standing at a smaller margin and a weaker band, so the verdict is not an artifact of counting repeated measurements of one pair as independent draws. Both sweeps, and the tighter-of-the-two-conventions reading we do not headline, are in App. C.

Two caveats qualify the corroborating estimators, not the bound. Under random folds they leak, and under folds blocked on molecules they are uninformative on the corner and split on the broad set, so on neither set do they support the separation. The verdict rests on the LOTV bound, which fits no model and so has no fold design to leak through, and not on the agreement of three estimators. Nor is the binning-free pair bound promoted, tighter than the binning bound though it is, because its multi-row pairs each come from a *single* publication and it therefore samples no between-laboratory component. Read together, the two say the binning bound is slack rather than that B_{insuff} is negligible. The leak's mechanism, and both bounds in every fold design the audit records and against each set's MSE/2 threshold, are in App. C (Fig. 5, Table 9).

The broad set is stratifiable where the corner's 60 pairs were not: the chromatographic-only cut that *inverts* the margin on the enlarged 298 K corner (§5, limitation (i)) does not invert here under either convention. On the chemistry the deployed kernel has been most widely exercised on, the ceiling is in the map—within the scope set above. Method and source composition, the per-stratum values, the squared-error composition audit and the DOI- and pair-clustered bootstraps are in §I.

(ii) The decomposition on the corner (n=60). The corner carries the paper's other reference-matched analyses, and it is far weaker: all 60 rows sit at 298 K, so $z^\star$ is the profile pair alone and $\dim(z^\star)/n \approx 102/60$ leaves the split behind one-sided bounds. It is also strongly unbalanced—methanol occupies 25 of the 60 rows, and the two worst-fit pairs, each resting on a single record, carry 34% of the squared error, a concentration we do not read as established chemistry. The separation does not survive their deletion at the headline cell (margin -0.002): a corner margin of $+0.05$ cannot absorb 34% of the squared error leaving the sample, and we report the deletion as a failure of the headline cell rather than as a robustness pass. Its full composition audit, and the four neighbouring cells that do survive the deletion, are in App. C. These are not 60 exchangeable observations, and the folds below hinge on the same molecules. Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are not independent samples—the corner is the sub-design on which VT-2005, and therefore the paradox's own oracle, is defined.

The separation margin, and the choices it depends on. On the corner, scored against VT-2005 profiles under the deployed residual-only convention and the *unbiased* within-bin variance, the LOTV bound gives $B_{\text{insuff}} \lesssim 0.713$ and a separation margin (Table 1) of $+\mathbf{0.05}$. We headline that rather than the maximum-likelihood variant, which biases the within-bin variance downward and so the margin upward to $+0.24$; on the corner the ordering holds by a margin the design cannot resolve. Four analyst choices set it (§2.2), and the separation rests in addition on the maximum-leverage solute: dropping methanol loses it. Blocked out-of-fold estimators do not support it either, so on the corner "the larger" is estimator-conditional and the ordering holds by the grade *likely* rather than certainly. The verdict rests on the broad IDAC set, whose several times more numerous clusters hold the margin above zero in a larger fraction of resamples; that margin is not resolved either, being of the order of its own 90% bootstrap half-width. What the corner adds is that the ordering is not overturned on the sub-design where VT-2005 is defined. Zero-mean label noise can understate the separation but not create it (§2.2); what the margin does bound is a label systematic tracking $z^{\star}$, implausible on the maximum-likelihood binning and *not excluded* on the Bessel-corrected one. The four choices, the leave-one-solvent, leave-one-pair, leave-one-solute and leverage-deletion folds, the cell grids, the cluster bootstraps, the label-noise arithmetic and the convention, leakage and γ-stratification stress-tests are in App. C.

Whether the closure defect is correctable, and where it fails chemically. That B_{closure} dominates locates the ceiling in the map; it does not say the map is *fixably* wrong. A small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel, fit on the reference VT-2005 profiles, removes $46 \pm 5\%$ of the reference-input error under random folds but $-5 \pm 16\%$ once folds are grouped so that no solvent is shared. Only the grouped design is free of leakage, the closure error being predominantly a between-solvent systematic. Stated as a bound rather than a null, that design excludes a cross-solvent reduction above about 30%. B_{closure} is therefore a defect a handful of kernel parameters can absorb *within* a solvent and that we have not shown to be correctable across solvents—weaker than a general repair. Yet 90% of the squared error falls on strong hydrogen-bond-donor solvents and is near-exact on acceptor-only ones, placing B_{closure} in COSMO-SAC's documented single-parameter H-bond kernel—the specific defect the donor/acceptor-typed 2010 closure was built to fix. The residual's fold design, rank ordering and the scope of its standard errors, its failure to gain over the free 2002 $\rightarrow$ 2010 step below, the NIST reproduction that rules out an implementation artifact, and the Staverman–Guggenheim counter-test in which the physically *more* complete combinatorial term *increases* error on size-asymmetric pairs are in App. C.

(iii) The closure-fidelity lever: present on the corner's chemistry, bounded on the broad set's chemistry. If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 $\rightarrow$ 2010 step delivers one on the corner (n=60) and not on the broad IDAC set (n=477); the block split below identifies the factor that carries that reversal.

The two kernels have no *prescribed* common input, σ-profile averaging being part of each model's definition, so evaluating each on its native profile is the specified use of the model. An input-fixed *control*—profile database, cavity area, volume and averaging held fixed, kernel alone varied—gives $2.047 \rightarrow 0.765$ on the corner ($P > 0.95$), so the $2002 \rightarrow 2010$ step survives with the input held fixed. Three limits sit on it: it cannot separate the donor/acceptor typing, which the 2002 kernel has no channel to consume and which is therefore charged to the kernel by construction; it is favourable to the conclusion it supports, its step being larger than the native-input one; and its re-averaging term is itself unresolved. The arithmetic, and the NIST implementation detail that fixes the *fair 2002* arm at 1.757, are in App. C.

The three arms—*fair 2002*, *2010, dispersion off* and *2010, dispersion on*—are Table 8 of App. C, on both sets under both conventions. The standard $2002 \rightarrow 2010$ step more than halves the corner's error under the full convention ($P > 0.95$) and cuts it 42% under the deployed one, so "more than halves" is a full-convention statement. On the broad IDAC set the same step is not an improvement under either rule, and the 3% the dispersion term recovers there is a null whose interval is an order of magnitude wider than the effect, not a measured equality. That arm's corner run is on n=57, three halogenated pairs having no tabulated dispersion value—a coverage limit of the model offered as the lever; no claim rests on the 0.765-versus-0.785 ordering, which App. C shows is not an artifact of the differing n. The comparison runs under the full convention, where the two kernels' published error figures are reported, with the deployed-convention reading recorded beside it rather than leaving the rules mixed.

The reversal is the *chemistry* and neither the profile database nor the temperature range. With UD profiles, one extraction and one convention throughout, the *2010, dispersion off* arm wins on the 57 pairs shared with the corner, does not win on the complementary 420, and still does not win on the 80 complementary rows that are *also* at 298 K. That last block carries $P \approx 0.3$ and no interval, the three blocks not having been tested against one another (Table 8). The corner is a low-γ, hydrogen-bond-dominated sub-design, which is exactly the regime the donor/acceptor-typed 2010 kernel was built for, and where the error localises above. The lever is therefore bounded rather than absent: on the corner's chemistry a free $2002 \rightarrow 2010$ step removes most of B_{closure}, and on the broad set's chemistry it does not.

The floor checks are same-database, same-latent *and* same-convention, each stated under the deployed rule of (i) with the full-convention reading beside it. The corner's *UD-native* floor check, the one commensurable with the broad set, is at zero under the rule we deploy and clears at +0.69 under the full convention. Under the 2010 kernel's own typed latent, which is 306-dimensional and so several times worse resolved, the ordering is *lost* on the corner (margin −0.77) and holds on the broad set (+1.29; *likely*, not certified). That loss is loss of resolution, not a certified flip to the inputs, which would need a *lower* bound on B_{insuff} (§2.2). The full arithmetic, under both conventions and all three clusterings, is in App. C.

We know of no large-scale infinite-dilution benchmark of the $2002 \rightarrow 2010$ step, and our search was not exhaustive, so this is an absence of evidence rather than evidence of absence. The broad-set null lies within the range of magnitudes the comprehensive assessment [26] reports (search scope and arithmetic, App. C). We claim only that **the fidelity lever we can demonstrate is confined to the corner's chemistry**; the broad set's null is additionally a small-molecule and glycol-heavy one (composition audit, §I), so on drug-like solids the lever is untested rather than absent.

The reference is itself imperfect, and what that makes B_{closure} a property of. The external ground truth used to charge B_{closure} against B_{insuff} is quantum chemistry, not experiment: the VT-2005 profiles are computed for a single reference conformer and protonation state (§I), and the successor Delaware database revises those conformations for about a third of its compounds.[26] That is the gap between $z^\star$ and the state actually populated under the measurement, and the mismatch does not load on B_{insuff} alone: if the reference is displaced, even a perfectly specified g carries a term in B_{closure}, whose share we do not bound. That scopes what the attribution is *about*: B_{closure} as measured here is the misspecification of the composite object, the kernel applied to one fixed computed conformer, not of the kernel at the true conformational ensemble. The deployed pipeline is that same composite, so this is the quantity a practitioner faces. Why the superpopulation has a non-degenerate B_{insuff} at all, and the conformer term's own arithmetic, are in §I. The corner measurement is additionally a low-γ/298 K one, and neither set transfers to the finite-composition regime the paradox lives in (§5).

4.3 The status of the closure's known defects

That COSMO-SAC-2002 is a low-fidelity closure is the contribution rather than a qualification of the result. The 2002 kernel is what a practitioner adopts by default, and a closure's fidelity is *not knowable in advance* on a new system. The field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. This paper reports what follows when it is not: the learned latent silently departs from the quantity it is named after (§4.4), and neither that departure nor its cause is visible without an external oracle.

4.4 The end-to-end latent departs from its reference along a shared direction

A candidate *mechanism* for the low-fidelity, end-to-end regime of the map (§5) is this: when a σ-profile head is trained *end-to-end* through a fixed, misspecified closure—with no σ supervision in that phase—it is under pressure to *cancel* the closure's error rather than to be physically faithful. The hypothesis has two parts and this section establishes only the first. Its antecedent—that the latent leaves the reference, and does so along a direction shared across molecules rather than as per-molecule noise—is measured. Its consequent—that the departure cancels part of B_{closure}—is not, and the one direct check we can run on retained checkpoints argues against it, on the wrong axis and with little power. We name the hypothesis a *compensating surrogate* and grade it open. The external reference that fixes the ceiling (§4.2) is what makes the first part testable: we compare the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile σ molecule by molecule (n=44).

The drift under fine-tuning is systematic and low-rank, and generalises against a zero-drift null only—confound-free. The comparison is within a single grounded model, checkpointed before and after solubility fine-tuning (the σ head unfrozen), so architecture and initialisation are held fixed and only the objective differs; we repeat it over *three seeds* and report mean±sd. Under σ supervision the head reproduces the reference profile to within 36±1%, so the intermediate *can* carry the physical shape, though only as an in-sample fit: all 44 probe molecules lie inside the σ-grounding pool (§E). Solubility fine-tuning then drives the profile 41 ± 3% off the grounded one, for a total departure from the reference of 51 ± 2%. What this section carries is the sign of the departure and the presence of a shared direction, not the magnitudes: that figure, and the two reported below it, are not stable across analysis configurations (§E; §5(v)). The drift is not per-molecule noise, and its structure is not a constant offset: the first two principal components of the *mean-centred* deviation carry 72 ± 1% of the variance. That share is read against a *phase-randomised* null, which matches each deviation's own smoothness and destroys only the between-molecule alignment at issue (§E). The observed share clears it (0.399 against 0.221, $p < 0.001$), at roughly 1.8×; a wrong-molecule null built from real reference profiles does *not* clear (0.772, p = 1.00), and if the case for preferring the phase-randomised one is not accepted the spectrum is uninformative rather than supporting. That case, the calibration behind it, and the fact that the isolation drift has no matched null of its own are in §E. To test whether the structure transfers across molecules we fit one distortion operator D on half the 44 molecules and score it on the other half. There D predicts the drift $4.1 \pm 0.5\times$ better than the zero-drift (identity) null out of sample. That null is deliberately weak, nothing controls it on these three seeds, whose checkpoints were not retained, and a molecule-independent constant offset reaches $2.1\times$ on the one companion run where that control can be computed, so the ratio stands against the identity null alone. The ± 0.5 spans seeds at one partition, and the held-out half is homologue-saturated. Structure *beyond* an offset therefore rests on the mean-centred spectrum, not on the ratio—and that spectrum does not exclude a shrinkage of every profile toward the corpus-mean shape (§E). $\hat{\sigma}$ therefore departs from its reference systematically and at low rank, and the departure generalises across molecules—against a zero-drift null only, on a set whose halves are chemically close—driven by the fitting objective rather than by architecture or grounding. Whether that departure is compensation is the separate question below. A reference ladder calibrating the 51% against uninformative baselines is in §E; on it the drift consumes essentially all the molecular specificity grounding had provided, as a collapse toward the corpus-mean shape would also look.

Dominant mode, and the direct cancellation test. The dominant drift mode (Fig. 3) transfers area *out of* the nonpolar bins ($\sigma \approx 0$) and *into* the polar wings: the network represents molecules as more polar than they are. Whether that shift *cancels* part of B_{closure}—the input compensating the map, as the non-identifiability of the split allows (Lemma 2)—is the comparison $\text{MSE}(m, g(\hat{\sigma}))$ against $\text{MSE}(m, g(z^\star))$ on the same pairs. It does not support the cancellation reading. Holding the cavity area at its reference value so that only the profile shape varies, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint gives 18.75 against 1.47: on the activity axis its drifted profile is an order of magnitude *worse* through the same closure, not better. Three limits sit on that number. The checkpoint is the uncontrolled one, whose departure (82%) is half again the three-seed $51 \pm 2\%$, so it may not be representative of the three seeds. The axis is wrong: the drift is produced by the finite-composition solubility objective and scored here at infinite dilution. The check also has almost no power in the direction that matters, so it can separate "no cancellation" from "large cancellation" but not from "some cancellation" (§E). It is the only direct evidence available, and it is evidence against rather than a clean pass or fail.

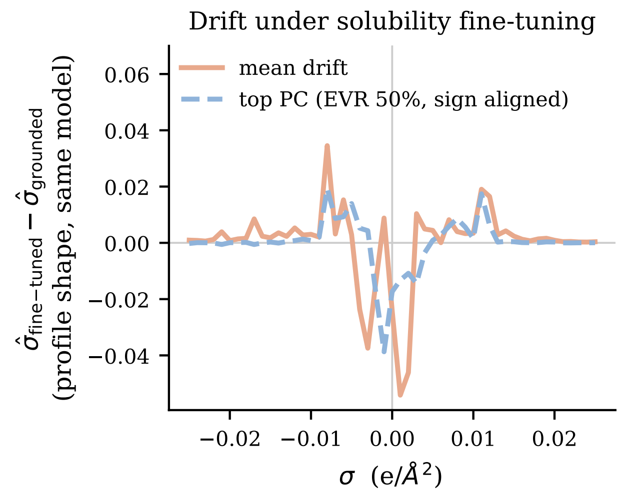

Figure 3: The drift solubility fine-tuning imposes on the learned σ-profile (n=44 VT-2005-matched molecules): $\hat{\sigma}$ after fine-tuning minus the *same model's grounded profile*, not minus the external reference. Shown are the mean drift and its dominant PCA mode, defined only up to sign and drawn in the mean drift's direction; the legend's EVR is that mode's explained variance ratio.

A second, independent check on the closure's own axis points the same way: a *more* polar profile moves g further from m (§E). Both checks sit on that same infinite-dilution axis, so neither refutes cancellation in the regime that generates the drift. They remove the one place where it could have been measured, leaving the cancellation reading inferred from the paradox itself and graded open. What is established is weaker and still consequential: the model's "physical" intermediate is not the molecule's charge distribution, consistent with cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification,[20,27] and that adapting a representation can itself collapse an otherwise physically-faithful latent.[28] Its *structure*—low-rank rather than diffuse—excludes a random collapse but not the structured shrinkage above; the diagnostic that would separate them is under-resolved here (App. F.6).

Why σ is under-determined by activity. The freedom that would make a compensating surrogate possible is the closure's geometry: at infinite dilution the activity map concentrates on the one or two hydrogen-bonding directions where the closure is least faithful (§F.4). The end-to-end head can therefore move σ along the remaining, low-sensitivity directions at little cost to the fit—the σ-profile form of Lemma 2. The Fisher measurement behind that statement, and the two respects in which it does not settle the question, are in §E; nothing here shows recovery of the physical profile to be impossible in principle.

The contrast that fixes the sign, and how the axes interact. The sign of grounding turns on *both* axes, and on which sense of grounding is meant. Freezing the σ head at its supervised warm-up state instead of letting it drift end to end cuts the reference-input penalty by 65% (ΔMAE from +0.40 to +0.14 over n=5571 matched test rows). Those are a separate matched pair of runs, one base resumed twice with the freeze off and on, and not the +0.41 headline arms of §4.1 (§C). A near-physical intermediate makes the pipeline far more robust to substituting the reference σ. Supervision alone does not, however, *flip* the sign here—the penalty stays positive, the closure still being the low-fidelity COSMO-SAC-2002 and the warmup σ itself only partly physical. The full flip needs supervision *and* fidelity together, which is what TeNNet-SAC[7] has and this pipeline does not (§1). Supervision moves the system toward the regime in which grounding helps, and closure fidelity carries it across (§5).

4.5 One downstream correction: a gated output residual restores calibration, not accuracy

If the closure binding were merely an inaccurate point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not: an additive bounded residual with a learned confidence gate leaves the gate effectively shut, and forcing it open removes the systematic *bias* (+0.45 $\rightarrow$ +0.06) but recovers little *accuracy* (R^2 0.236 $\rightarrow$ 0.274 on the same rows), as expected if the closure, not the estimate, is the binding constraint. Both arms are seed 42 only, so by the grading rule opening this section this is *likely*, and it is a statement about the one correction tested: an additive bounded residual on the output, gated, on this closure and this corpus. Other downstream corrections, and this one at other seeds, are untested here.

4.6 A second chemical closure: pKa through a fixed Hammett relation

The instrument is not specific to COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. There the fixed closure is a Hammett linear free-energy relationship in a substituent constant, the learned intermediate is that constant predicted from structure, the external reference is the Hansch–Leo table, and the data are the OPERA experimental compilation, split within scaffold K=20 times. On the meta- and para-substituted aromatics the relation was built for (n=158), substituting the tabulated constant for the learned one *improves* accuracy—MAE 0.48 against 0.315, the reference better in 19 of 20 splits and in three of the four reaction series; on the ortho-substituted and heteroaromatic ones it does not describe (n=846), the same substitution *degrades* accuracy—0.83 against 1.62, the reference worse in 20 of 20 splits and in all four series. The sign of grounding therefore reverses inside one closure, in the direction the penalty rule (Prop. 1) gives a priori and on the channel it names: there the ortho and steric information the tabulated constant omits, which is input insufficiency, rather than the closure-error cancellation of §4.4. Only G=4 Hammett reaction series exist, so the accompanying bootstrap resolves nothing finer than series-level sign agreement, read as a sign test on four units (p=1/16 on the pole where grounding hurts); the construction, the estimator validation, the measured fidelity sweep and the crossover law that predicts both signs are in App. G.

5 Discussion

The result is a cautionary measurement for physics-informed modelling, about one of the two operations the word "grounding" names. Supervising a learned physical latent on reference values during training helps; *supplying* those reference values as an input at inference hurts, and it hurts because the fixed map that consumes them is misspecified. When that map is of unknown fidelity—and the standard route does not improve it—better inputs surface its bias rather than help. Both sets are scoped by the reference database that defines them, and neither is the deployment regime (composition audit, §I). The fidelity null is therefore a null on small-molecule chemistry, and the lever remains untested on drug-like solids.

The unconstrained control does not carry the thermodynamics either. If the physics arm's intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is not. A ridge probe of the control's pair representation recovers the model's own prediction at $R^2 = 0.98$ but not the measured ideal-solubility term—an upper bound at $R^2 \lesssim 0.30$ rather than an exclusion (limitation (vi)). What the representation is organised by instead is scaffold identity, which is consistent with the transfer limitation of §2.3: a test molecule on an unseen scaffold is not a small extrapolation but a point off the axis the representation is organised along. The seed-by-seed scores, the permutation nulls, the within-scaffold measure and its weakness, and the temperature probe are in App. J.

What the contribution is. The unit of contribution is the *instrument*, not any single measurement. Its two ingredients are individually standard—the closure–insufficiency split is a law-of-total-variance identity, and the sign of grounding's effect follows the model-discrepancy principle[19,20]—and what is new is their combination into a usable measurement. The split is an identity on a *fixed* operator, with the insufficiency side bounded rather than point-identified (§2.2), and the resulting sign rule can be tested *a priori* and *bidirectionally.* The pKa domain supplies that test as an in-distribution sign flip (§4.6), at the resolution four series allow and no better—an oracle-on-a-fixed-operator affordance the concept-leakage setting lacks, its intermediate having no ground truth.[17,18] The solubility results are the instrument's first application, graded in Table 2; their transfer out of the matched regime is not established.

Why we make no accuracy claim, and what remains after that. The physics and control arms were tuned separately—the settings differ by up to an order of magnitude and are listed in §A—and no hyperparameter-matched control was run. A global ΔMAE of +0.14 between pipelines whose learning rates differ by an order of magnitude carries no information about the thermodynamic bottleneck, and neither do the per-solvent-class map, the per-solute-class map, the novelty inversion or the data-efficiency trend, all of which resolve the *same* difference along chemical axes. We therefore claim no accuracy result in either direction. Those strata are reported in the SI

(§F.4, §F.5) as hypothesis-generating descriptions of where two particular pipelines differ, each carrying that caveat, and no conclusion in this paper rests on them. Running the matched control is the direct repair and we have not run it. What survives the confound is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning: §4.1, §4.2 (which uses no trained model at all), §4.4 and §4.5. Separately, and not as an accuracy claim about the bottleneck: we did not find a configuration in which the physics path attains a lower error than the control. External state-of-the-art models (FastSolv, SolProp) do no better on a by-solute split, which bounds the task's difficulty and does not rank the models against one another (App. D), and grounding the crystal factor on external labels did not move the ceiling (§F.3). The one place the physical *structure* helps is temperature extrapolation, as a hypothesis-class property the trained models do not exploit (§F.1).

When a fixed-closure prior helps versus hurts. The penalty sign rule (Prop. 1) predicts *when* to expect these negatives: a prior is predicted to hurt precisely in the mature-data, low-noise, well-covered regime. The synthetic sweep (Fig. 11; App. K) is true by construction there, so it verifies the instrument rather than the hypothesis, while the pKa closure (§4.6) exercises the sufficiency channel on real data. Whether grounding helps is shaped by *two* axes, closure fidelity and how the latent is trained; characterising that plane on real systems is future work. The chemical strata form a pattern in the direction the sign rule would give; separately tuned pipelines can produce it for reasons unrelated to the prior and there is no multiplicity control, so §F.4 records it as the experiment a matched control should run, not as evidence, and the diagnostic is *proposed* rather than validated.

Limitations. (i) The phenomenon and its causal attribution lie in *different regimes*: the paradox is measured where the model operates (finite composition, all γ, n=5608), the closure–insufficiency attribution and the latent-departure result only on infinite-dilution activity data (n=477 across temperatures, and the 298 K, low-γ, VT-2005-matched n=60 and n=44 sets). Transfer back to the paradox's own regime, argued from the in-regime signs, is a conjecture rather than a measured fact and is the principal open question here. The matched n=60 set is also narrowly sourced: all by gas chromatography and all from three publications, 33 pairs from one, so neither a method-stratified nor a source-clustered re-estimate is available on it. Its margin is also extraction-sensitive: a broader ThermoML pull at 298 K widens it but *inverts* it on the gas-chromatography subset, a sensitivity we leave unresolved and do not read as a verdict on the extraction. The broad IDAC set does not move the same way under the same cut (§4.2(i)); the arithmetic, the enlarged-pull values and the matching-rule alternatives are in §I. Because few drug-like solutes carry a VT-2005 profile, the solubility substitution also falls almost entirely on the solvent channel; the clean both-channel evidence is on the activity axis. Under the deployed convention the margin is resolved on neither set, and only the literature-standard convention resolves it, while the headline convention is the deployed one (Table 2). The broad IDAC set is in any case not the set on which the paradox's own oracle is defined. (ii) B_{insuff} is upper-bounded, not point-identified (no replicates at fixed $z^\star$); the smoothness-free Jensen bound assumes m and g share the $\ln \gamma_2^\infty$ reference state, established by the closure validation of §3, so the mean offset is genuine decoder bias, not a units artifact. (iii) The constant-offset margin is convention-specific and inverts under the deployed residual-only default, on both sets; there the separation rests on the law-of-total-variance bound and the more-physics-worse effect. That bound is not itself convention-independent—only the estimand B_{insuff} is (§2.2)—so the deployed claim rests on the law-of-total-variance bound *computed under the deployed convention*, not on a quantity insensitive to the choice. (iv) Absolute scaffold accuracy is encoder-bound but well *above* the inter-lab noise floor (0.15–0.31

in $\ln x_2$; §F.1), so label noise does not explain the ceiling. (v) The latent-departure result is three-seed mean±sd (n=44; §4.4); what it establishes is the *sign* and the presence of a transferable shared direction, with the magnitudes (51 ± 2%/72 ± 1%/4.1 ± 0.5×) reported but not stable across analysis configurations. The two top-two shares in play are distinct quantities: the learned-versus-reference deviation clears a smoothness-matched phase-randomised null but not a wrong-molecule one, and is measured on one grounded checkpoint rather than three seeds, whereas the isolation drift has no matched null of its own (§4.4, Table 2). (vi) The probe result on the unconstrained control is bounded by the same reference scarcity: its crystal-term and temperature tests rest on 13 held-out solutes and cannot be enlarged, so the negative is an upper bound at $R^2 \approx 0.30$ rather than a demonstration that no crystal information is present, and the crystal-slope test is additionally limited by the corpus, whose empirical slope carries $-\Delta H_{\mathrm{fus}}/R$ at $r = +0.24$. The scaffold-organisation result needs no external reference, but its two partitions nearly coincide by construction, so a ratio near one is weak evidence of within-scaffold indistinguishability (App. J). (vii) The closure-fidelity contrast compares *oracle-input* activity error across kernels, so $\mathcal{G}$ itself is not measured under a modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested. Both require a typed (donor/acceptor) σ-profile head trained end to end through the 2010 kernel, and the solubility test a UD$\,\cap\,$BigSolDB matched set; we have neither. (viii) No difference in accuracy between the physics and control arms is attributable to the thermodynamic bottleneck; the confound and what survives it are stated above. (ix) The scaffold-leak guard is a property of the released stream builder and not a certified property of these runs: one σ-stream build on disk carries 91 held-out solutes, and per-run stream provenance was not logged (§I). The paradox contrast is one checkpoint evaluated two ways and so cannot be generated by such a leak, and DirectGNN uses no stream, but auditability is not restored by those arguments. (x) The repository URL, archive DOI, licence and commit hash are placeholders, so the reproduction commands the Data availability statement describes cannot be run, and several of the numbers the verdict rests on are checkable only through code we have not yet been able to release (§I names which). The objection stands in full for those artifacts, and it reaches much of the verdict, which rests on evaluations that cannot be opened without the code. One partial remedy is in place: the two decomposition sets are deposited as row-level tables (Tables S1 and S2), so the sets and their margins can be recomputed without code. That does not deliver the provenance of the closure evaluations in them, or of the trained arms anywhere else.

6 Conclusion

Grounding a learned physical intermediate in reference values is two operations, not one, and only one of them helped here. Training the intermediate against reference values improved the model. Supplying those values at prediction time did not. On the activity data where the two terms can be separated, the bound on the fixed thermodynamic model's own error lies above the bound on the contribution of its inputs, which is the condition under which more accurate inputs expose a model's bias rather than remove it. The intermediate trained through that model stopped reporting the physical quantity it is named after, on the molecules where it can be held against its reference, which is the price of reading a grounded latent as a measurement.

The general statement is a design rule with a test attached. Wherever a predictor composes a fixed physical model over a learned intermediate for which an external reference exists, that reference measures which of the two limits accuracy, and the answer decides the remedy: a more faithful physical model, or better inputs. Here the answer was the model, and the standard route to a more faithful one did not clear the bar on the broader set measured. Whether a latent trained through an imperfect model becomes, in general, a surrogate for that model's error is the prediction this work leaves testable.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The analysis code, and those intermediate artifacts small enough to version, are available at `https://github.com/doctawho42/tgnn-solv` (commit hash to be supplied at submission) under the MIT licence; the model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts—among them the two audit files §I names—will be deposited in a Zenodo archive, whose DOI is registered at submission. **The repository URL, the archive DOI and the commit hash are therefore not yet the final ones.** An anonymised repository and an archived snapshot at the commit the reported numbers were produced from will be supplied with the manuscript; until then the reproduction commands cannot be run (limitation (x)). The data sources are public: BigSolDB 2.0[21] for solubility, the VT-2005 database for the reference σ-profiles, Ref.[26] as redistributed with Ref.[23] for the UD profiles, ThermoML for the infinite-dilution activity coefficients, and the OPERA compilation[29] for the second-domain pKa data. Most analysis scripts carry their run commands in their docstrings, and the two audits behind §4.2 are named with the script and the artifact file of each in the compute ledger (App. I). Four deposited artifacts carry disclosed defects, recorded in App. I and App. C: the seed-44 oracle per-row file is truncated, the three-seed surrogate run's per-run split provenance was not logged, one σ-stream build on disk was written against the wrong split files, and the two corner audit artifacts disagree on one out-of-fold estimate.

Acknowledgement This research received no external funding. Computations were carried out on commercial cloud services (Google Colab Pro+, Modal, and Google Cloud Platform) using NVIDIA T4 and L4 GPUs. During the preparation of this work the authors used generative artificial-intelligence tools (large language models, including Anthropic's Claude) to assist with literature search, drafting and editing of the text, writing and debugging of analysis code, and generating figures. All content was reviewed, verified, and edited by the authors, who take full responsibility for the manuscript.

Supporting Information Available

The Supporting Information contains supplementary methods; proofs; per-arm tables and decomposition robustness; external baselines and ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa analysis; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; and the positioning against adjacent literatures. (The notation table, Table 1, is in the main text.) Two machine-readable tables are deposited with it: the 477-row broad IDAC set (Table S1, `broad_idac_set_477.csv`) and the 60-row corner (Table S2, `corner_set_60.csv`); their columns are listed in App. I.

Supplementary Information

A Supplementary methods

COSMO-SAC constants. The differentiable COSMO-SAC layer uses the standard VT-2005 / Lin–Sandler constants: effective segment area $a_{\rm eff} = 7.5\,\text{Å}^2$, electrostatic-misfit constant $\alpha' = 16466.72\,\text{kcal}\,\text{Å}^4\,\text{mol}^{-1}\,\text{e}^{-2}$, hydrogen-bond coefficient $c_{\rm hb} = 85580$ with cutoff $\sigma_{\rm hb} = 0.0084\,\text{e}\,\text{Å}^{-2}$, coordination number $z = 10$, and Staverman–Guggenheim normalisers $r_0 = 66.69\,\text{Å}^3$, $q_0 = 79.53\,\text{Å}^2$; the σ-profile grid is 51 bins on $[-0.025, 0.025]\,\text{e}\,\text{Å}^{-2}$. The segment-activity fixed point is damped ($\lambda = 0.7$) and run for 16 iterations at train time and 30 at evaluation.

A usage pitfall in `cCOSMO` (`COSMO1` on typed profiles). Supplying the 2002 kernel with a typed three-profile base produces a silently-wrong result that the library returns without warning. The reference implementation[23] pairs its 2002 model (`COSMO1`) with the single-profile Virginia Tech database; the demonstrated input is one untyped, Mullins-averaged profile. When `COSMO1` is instead supplied with a typed three-profile (Hsieh, `sigma3`) base — the configuration that arises when a 2002 and a 2010 arm share one profile source — it silently reads the nonpolar channel only, discards the donor and acceptor channels, keeps the total cavity area as prefactor, and returns a number without error. The prescribed way to recover a single 2002 profile from the typed set is instead their *sum*, $p = p_{\rm nhb} + p_{\rm OH} + p_{\rm OT}$ (Eq. 9 of Ref.[23]); the nonpolar-only read is neither that sum nor a documented mode. This single-channel arm scores MSE 1.263 on the matched corner, whereas the *fair 2002* arm under the full convention (the same molecule supplied either to our differentiable layer or to `cCOSMO` with the full profile placed in the single nonpolar channel it actually reads) scores 1.757/1.758. This is an issue for the library maintainers (a silently-wrong return where an error would be safer), not a defect of any published benchmark. The kernel arms of §4.2 are deposited across two artifacts, and which file holds a value does not follow the set. The dispersion-*off* values (1.757 and 0.765 on the corner, 1.911 and 1.978 on the broad IDAC set), all under the full convention in which the kernel comparison is run, are our differentiable layers on each kernel's native profiles (`fidelity_lever_fair2002.json`). Both dispersion values are `cCOSMO`'s own `COSMO3` call: the corner's 0.785 from `closure_flip_ci.json`, where the reference returns no dispersion for the three bromine/iodine pairs and that arm therefore scores on n=57; the broad IDAC set's 1.855 from the `mse_2010dsp` field of `fidelity_lever_fair2002.json`, whose corner counterpart in the same file is instead our own layer's 7.43. That 7.43 is the measure of the gap: our transcribed dispersion term is not validated against the reference, which is why the dsp arm is deferred to `cCOSMO` throughout. The input-fixed control of §C(a′) is a third file, `fidelity_lever_inputfixed.json`: it runs the 2002 kernel on the summed typed profile and re-derives the 1.757 and 0.765 arms of the first file as its own gate.

The differentiable COSMO-SAC-2010/dsp layer, and the scope of its dispersion caveat. The 2010 layer implements the four mechanisms of:[24,25] the donor/acceptor-typed three-profile representation, the temperature-dependent electrostatic misfit $c_{\rm ES}(T) = A_{\rm ES} + B_{\rm ES}/T^2$, the typed hydrogen-bonding kernel, and an optional dispersion term in the tabulated parameter $\epsilon/k_{\rm B}$. With dispersion *off* it reproduces the NIST COSMO-SAC-2010 reference to RMSE $1.4 \times 10^{-4}\ \ln\gamma$. That figure is quoted from the validation run's console output: unlike its 2002 counterpart, whose comparison is deposited as `closure_reference_validation.json`, the 2010 validator wrote no artifact, so this one figure cannot be checked against the deposited record. The dispersion term is not validated to that standard, for two reasons we state rather than absorb. It carries the tabulated coefficient at a single sign, the class-dependent sign reversal of Ref.[25] being unimplemented and unexercised on the matched set, which holds no water or carboxylic-acid pair; and no ϵ is tabulated for the three matched pairs containing bromine or iodine, which our layer scores at ϵ=0. That alone accounts for the MSE 7.43 our layer records over all 60 corner pairs; on the remaining 57 it returns the reference 0.785, showing no divergence where both are defined. Every dispersion-off contrast in the paper is therefore computed on our layers and cross-checked against the NIST `cCOSMO` reference, and the *2010, dispersion on* runs are quoted from `cCOSMO` itself (§4.2).

Solver, curriculum and loss. The SLE fixed point is iterated as the damped map $x_2^{(k+1)} = \lambda \exp(-\Phi(T) - \ln\gamma_2(x_2^{(k)}, T)) + (1-\lambda)x_2^{(k)}$ with closure-specific kernels. Gradients use the implicit-function theorem at the converged $x_2^\star$ (with $\eta = \partial \ln\gamma_2/\partial x_2$) rather than backpropagation through the iterations: exact at the fixed point, memory-constant in the iteration count, and well conditioned unless $1 + x_2^\star\eta \to 0$. The iteration counts above follow a convergence audit: a shorter n=8 schedule fails a 5.86-$\ln x_2$ convergence test on strongly non-ideal pairs. The three-phase curriculum is (P1) auxiliary-property pretraining on crystal, Hansen and $\ln\gamma_2^\infty$ targets with no SLE loss; (P2) full SLE training, with the adaptive correction unfrozen partway and an optional crystal→pair weight ramp; (P3) low-learning-rate stabilisation with monotonicity and van't Hoff regularisers. Of the weighted loss, few terms are load-bearing for this paper: a bin-weighted Huber on $\ln x_2$; auxiliary losses on the σ-profile (an earth-mover distance), on crystal $T_m/\Delta H_{\text{fus}}$, on $\ln\gamma_2^\infty$ and on Hansen parameters; and structural regularisers for temperature monotonicity and the van't Hoff slope. The remainder — channel orthogonality, contrastive, mixture-of-experts balance, priors — is auxiliary. The grounded arms begin with a warm-up on the σ pool of up to 40 epochs, early-stopped on a held-out tenth, and the pinned recipe freezes the σ head for P2/P3, so that stream enters the warm-up and P1 only.

The closure itself, and the remaining terms of the 2002 layer. The COSMO-SAC layer of §3 maps a profile pair to $\ln\gamma_2$ through a damped fixed point for the segment activity coefficient Γ_S,

$$\Gamma_S(\sigma_m) = \Big[\sum_n p_{\text{norm}}(\sigma_n)\,\Gamma_S(\sigma_n)\,\exp\big(-\Delta W(\sigma_m,\sigma_n)/RT\big)\Big]^{-1}, \tag{S1}$$

with a precomputed exchange-energy matrix ΔW (electrostatic misfit plus a hydrogen-bonding term), a restoring contribution assembled from those segment activities, and an optional Staverman–Guggenheim combinatorial term. The restoring contribution is $\ln\gamma_2^{\text{res}} = (A_2/a_{\text{eff}})\sum_m p_2^{\text{pure}}(\sigma_m)[\ln\Gamma_{S,\text{mix}}(\sigma_m) - \ln\Gamma_{S,2}^{\text{pure}}(\sigma_m)]$, and the optional Staverman–Guggenheim combinatorial term takes its size factor $r_i = V_i/r_0$ from the *reference* COSMO cavity volume so that it shares the cavity basis of the area A_i. Small hydride solvents such as water carry explicit hydrogens so that the polar and hydrogen-bond channels activate. The pair representation is passed through an MLP and may be augmented with RDKit descriptors, Morgan fingerprints and an ionic-context feature; the crystal head additionally offers Walden-entropy and group-contribution T_m modes. The two NRTL control closures of §3 are the Non-Random Two-Liquid model,[30] the most expressive closure here, which fits interaction parameters per pair — predicting two interaction energies with a $1/T$ law and a non-randomness factor, optionally regularised by a UNIFAC group-contribution prior[31] — and its symmetric one-parameter collapse $\ln\gamma_2(x_2,T) = \ln\gamma_2^\infty(T)(1-x_2)^2$, in which the identifiability analysis of §2 is cleanest. The crystal head's bounded parametrisations are below.

Crystal head. $T_m = 100 + 600\,\text{logistic}(\cdot) \in [100, 700]$ K, $\Delta H_{\text{fus}} = S_H\,\text{softplus}(\cdot)$; the group-contribution T_m prior is affine- calibrated on the train split before entering the head.

Architecture. The headline COSMO-SAC model uses a message-passing encoder (a shared backbone with a 2-layer residual role-specific branch) of hidden width 64 over 3 message-passing layers, on 35-dimensional node and 8-dimensional edge features; a set2set readout (3 processing steps) pools each graph, solute and solvent interact through a 3-layer cross-attention block, and the 512-dimensional pair representation $[\,g_s, g_v, g_s \odot g_v, |g_s - g_v|\,]$ (dropout 0.012) feeds the crystal head, the 51-bin σ-profile head, and the parameter-free COSMO-SAC closure. The full model has 5.40M trainable parameters—capacity that the in-sample linear probe ($R^2 \approx 0.76$, §F.1) indicates is not the binding constraint. The DirectGNN control shares this encoder, interaction, and readout verbatim, and adds a thermometer encoding of T in place of the temperature dependence the physics arm receives through $\Phi(T)$ and the closure. Figure 4 draws the two arms and marks where each of the two error terms enters.

Encoder backbones. The shared encoder has three interchangeable backbones: the headline message-passing network (MPNN); a hybrid local-message/global-attention graph transformer (GPS[32]); and a physics-channelled variant, Thermodynamically-Informed Message Passing (TIMP), which splits messages into dispersive, polar, and hydrogen-bonding channels supervised toward the Hansen parameters to prevent channel collapse. TIMP does not outperform the plain MPNN across our runs, consistent with the encoder being transfer-limited rather than expressivity-limited (linear probe train $R^2 \approx 0.76$, test $R^2 \approx 0.45$).

Optimisation, and the two arms' settings. The physics arm uses Adam (weight decay 5×10^{-4}, gradient clipping at 2.0, batch 64) under the three-phase curriculum of §3: 30 epochs of auxiliary pretraining (P1), 70 of full SLE training (P2), and 10 of low-rate stabilisation (P3), at learning rates 2.6×10^{-4}, 8.5×10^{-5}, and 8.5×10^{-6} respectively. The DirectGNN control was tuned separately and its settings differ: dropout 0.174 against 0.012, P2 learning rate 8.6×10^{-4} against 8.5×10^{-5}, weight decay 4.2×10^{-5} against 5×10^{-4}, gradient clipping 2.54 against 2.0, over 110 P2 epochs equal to the physics arm's total budget at the same width (64) and depth (3). Width, depth, batch size and epoch budget are therefore matched and the optimisation is not, so Δ_∞ and the physics-versus-control ΔMAE are differences between two separately tuned pipelines and are not attributable to the thermodynamic bottleneck alone (§2). A hyperparameter-matched control was not run.

Loss. The objective is a weighted sum. Its registry holds 32 terms, of which 17 are hard-wired to 0 and two more (bridge, solubility variance) default to 0 through the configuration—disabled experimental scaffolding, released with the code for completeness. Table 3 lists the 13 carrying a non-zero default, together with the solubility-variance term and the σ-profile earth-mover term, which is applied outside the registry when the grounding stream is on; only the top two groups are load-bearing for the claims, and the *Loss-simplification ablation* paragraph below tests which.

Table 3: The 13 registry terms carrying a non-zero default weight, with the solubility-variance term (zero through the configuration) and the σ-profile earth-mover term, which is not a registry entry.[a] A further 17 registered terms are hard-wired to 0, and one more (bridge) is zero through the configuration.

role	term	default wt.
task	solubility (bin-weighted Huber on $\ln x_2$)	1.0
crystal grounding	melting point T_m	0.3
	fusion enthalpy ΔH_{fus}	0.3
σ/act. ground.	infinite-dilution $\ln \gamma_2^\infty$	0.5
	finite-composition $\ln \gamma_2$	0.5
	Hansen parameters	0.2
	σ-profile EMD (grounding stream on)	†
regulariser	solubility monotonicity in T	0.1
	physics preference	0.1
	adaptive-correction residual	0.01
	NRTL τ	0.01
	solubility variance	cfg
control coupling	DirectGNN NLL	0.2
	DirectGNN regulariser	0.05
architecture	mixture-of-experts balance	0.02

[a] † marks the σ-profile earth-mover term, active only when the grounding stream is on. Per-phase weight overrides are pinned in each run's resolved configuration.

Loss-simplification ablation. For a diagnostic result the elaborate objective is a potential confound: the paradox could be an artifact of loss tuning. The test registered in advance of the run trains the end-to-end grounded arm with only the task ($\ln x_2$ Huber), the crystal grounding ($T_m, \Delta H_{\text{fus}}$), and the σ-profile warmup, *dropping* the activity-grounding auxiliaries ($\ln \gamma_2^\infty$, $\ln \gamma_2$, Hansen) and every regulariser, and then reads the sign of the paradox (oracle vs. learned) and the surrogate drift.

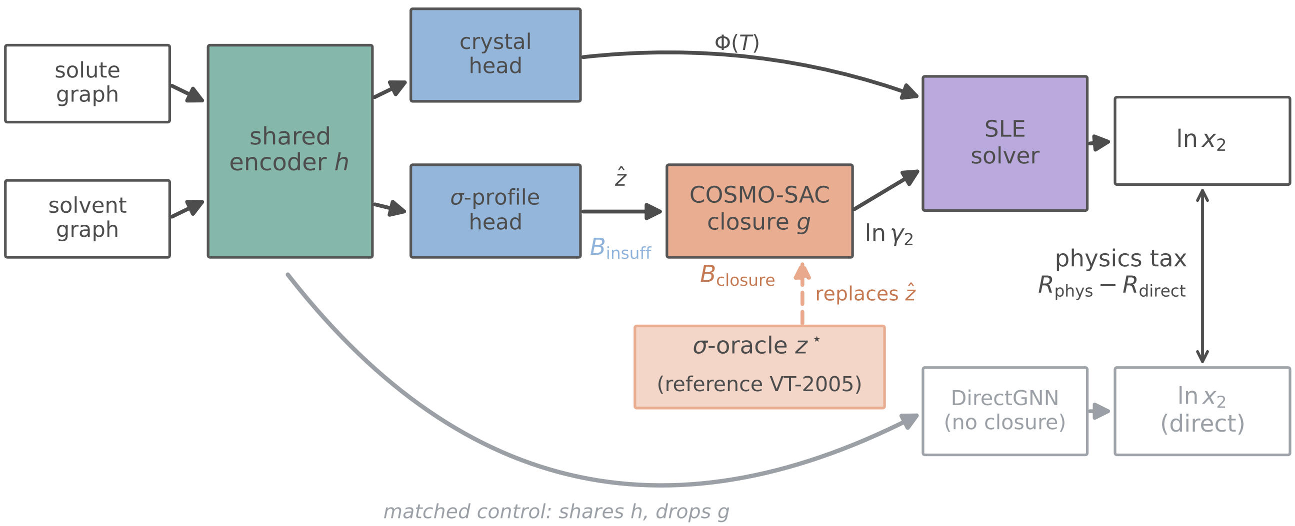

Figure 4: The model and the oracle substitution. Input insufficiency B_{insuff} enters at $\hat{z}$ and closure misspecification B_{closure} at the fixed map g; the σ-oracle (salmon) replaces $\hat{z}$ with the reference VT-2005 profile $z^\star$. DirectGNN (grey) is the capacity- and budget-matched control, separately tuned (§2), and the double-headed arrow between the two $\ln x_2$ outputs is the physics tax $R_{\text{phys}} - R_{\text{direct}}$.

The registered prediction was that both persist, the mechanism being end-to-end training through a misspecified closure rather than the auxiliary terms, and both do at the level the ablation measures. The ablation re-measures the solubility *paradox* (reference minus learned $\ln x_2$ MAE), the observable effect, not the profile drift directly: on the full test set it is $+1.13$ (95% CI $[0.88, 1.38]$) under the minimal objective, *larger* than the $+0.30$ ($[0.20, 0.39]$) under the full one. This is the *overall*, solvent-dominated paradox, and it strengthens once the activity-grounding auxiliaries—the only pressure toward physics—are removed; on the both-reference subset (n=297 rows over the $\approx$7 VT-covered solutes) the sign reverses (-0.23 minimal vs $+0.22$ full), so the ablation is evidence that the overall observable effect is loss-robust, not subset-level confirmation of the mechanism. It is *consistent with* the drift being intrinsic to end-to-end training through a misspecified closure rather than a regularisation artifact. It is an implication of the amplified paradox, not a direct re-measurement of $\|\hat{\sigma} - \sigma\|$ under the minimal run (the minimal-loss checkpoint was not retained, and the larger reference-arm error is consistent with more profile drift but does not isolate it from closure-path recalibration). The minimal model trains normally (validation $\ln x_2$ MAE 1.87, within 0.1 of the full-loss base), ruling out a degenerate-training reading (seed 42; `scripts/analysis/run_loss_ablation_analysis.py`).

B Proofs

Lemma 2 (Structural non-identifiability of the SLE split). *Consider the infinite-dilution SLE observable $\mu(T) = -\Phi(T) - \ln \gamma_2^\infty(T)$ with $\Delta C_p = 0$, so that $\Phi(T) = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$ is affine in $u = 1/T$, and a symmetric two-suffix activity class ($\ln \gamma_2 = \ln \gamma_2^\infty (1 - x_2)^2$) in which $\ln \gamma_2^\infty(T)$ is affine in u. Then, for any design with at least two distinct temperatures, μ depends on the parameters only through the two-dimensional statistic (its slope and intercept in u), and the activity intercept and slope already span that plane; hence the crystal parameters $(\Delta H_{\text{fus}}, T_m)$ are unidentified, the profiled Fisher information for ΔH_{fus} is exactly zero, and the* *indeterminacy set is two-dimensional. The deficiency is closed only by a rank-completing external constraint (a direct crystal label, or a fixed activity slope). At finite dilution the composition factor $(1-x_2)^2$ breaks the exact affinity, leaving a non-zero but near-degenerate information of order x_2^2.*

Proof of Lemma 2 (non-identifiability). Write $u = 1/T$ and $h = \Delta H_{\mathrm{fus}}/R$, $u_m = 1/T_m$. At infinite dilution and with $\Delta C_p = 0$ the observable is

$$\begin{aligned}\mu(u) &= -\Phi(u) - \ln\gamma_2^\infty(u)\\ &= -h(u-u_m) - (c + d\,u)\\ &= -(h+d)\,u + (hu_m - c),\end{aligned}$$

using $\ln\gamma_2^\infty(u) = c + d\,u$ (affine in u by hypothesis). Thus μ depends on the parameters $\theta = (\Delta H_{\mathrm{fus}}, T_m, c, d)$ only through the pair $(S,K) = (-(h+d),\ hu_m - c) \in \mathbb{R}^2$. The activity block spans this plane, $\partial(S,K)/\partial c = (0,-1)$ and $\partial(S,K)/\partial d = (-1,0)$; hence for *any* crystal value (h, u_m) and any target (S,K) there is an activity pair $d = -S-h$, $c = hu_m - K$ reproducing it. The likelihood level set therefore contains the full two-dimensional family $\{(h,u_m)\}$, so $(\Delta H_{\mathrm{fus}}, T_m)$ are unidentified. Equivalently, the crystal score $\partial(S,K)/\partial h = (-1, u_m)$ lies in the span of the activity columns, so profiling out (c,d) annihilates it and $\mathcal{I}_{\mathrm{eff}}(\Delta H_{\mathrm{fus}}) = 0$. At finite dilution $\ln\gamma_2(u) = \ln\gamma_2^\infty(u)\,(1-x_2)^2$ adds $\ln\gamma_2^\infty(u)\,(2x_2 - x_2^2) = O(x_2)$, which is not affine in u because $x_2 = x_2(u)$; the crystal score then has a component outside the activity span of order x_2, so $\mathcal{I}_{\mathrm{eff}}(\Delta H_{\mathrm{fus}}) = O(x_2^2) > 0$. $\square$

Lemma 3 (Class-dependent efficient information)**.** *Let the activity nuisance lie in a class $\mathcal{H}$ with tangent projection $\Pi_{\mathcal{H}}$, and let $v = \partial\mu/\partial\Delta H_{\mathrm{fus}}$ be the crystal score. The efficient information for ΔH_{fus} is $\mathcal{I}_{\mathrm{eff}}(\Delta H_{\mathrm{fus}}) = \sigma^{-2}\,\|v - \Pi_{\mathcal{H}}v\|^2$. If $\mathcal{H}$ contains the crystal score's span $\{1, 1/T\}$ (a free intercept and slope) then $\Pi_{\mathcal{H}}v = v$ and $\mathcal{I}_{\mathrm{eff}} = 0$, recovering Lemma 2; if $\mathcal{H}$ is restricted (the slope is known or constrained) then $\mathcal{I}_{\mathrm{eff}} > 0$ and the Cramér–Rao lower bound (CRLB) on ΔH_{fus} is finite. (The melting point T_m is likewise an unknown crystal parameter, but is profiled jointly: its score $\partial\mu/\partial T_m$ lies in* span$\{1\} \subset$ span$\{1, 1/T\}$, *already inside the nuisance tangent whenever the slope is free, so the dichotomy is unaffected by treating T_m as free.)*

Proof of Lemma 3 (class-dependent information). This is the semiparametric efficient-score construction[33] (here finite-dimensional and Gaussian, so equivalently the partitioned-information / Frisch–Waugh–Lovell identity). With Gaussian noise of variance σ^2, parameter-of-interest score $v = \partial\mu/\partial\Delta H_{\mathrm{fus}}$, and nuisance tangent space $\mathcal{T}_{\mathcal{H}}$ (the L^2-closure of the span of nuisance scores), the efficient score is the residual $\tilde{v} = v - \Pi_{\mathcal{H}}v$ and the efficient information is $\mathcal{I}_{\mathrm{eff}} = \sigma^{-2}\|v - \Pi_{\mathcal{H}}v\|^2$. Here $v = \partial\mu/\partial\Delta H_{\mathrm{fus}} = -(u-u_m)/R \in \mathrm{span}\{1,u\}$. If $\mathcal{T}_{\mathcal{H}} \supseteq \mathrm{span}\{1,u\}$ then $\Pi_{\mathcal{H}}v = v$ and $\mathcal{I}_{\mathrm{eff}} = 0$, recovering Lemma 2; if $\mathcal{T}_{\mathcal{H}}$ is a proper subspace not containing v (e.g. a fixed slope, $\mathcal{T}_{\mathcal{H}} = \mathrm{span}\{1\}$) then $\Pi_{\mathcal{H}}v \neq v$ and $\mathcal{I}_{\mathrm{eff}} > 0$, with a finite Cramér–Rao bound. $\square$

Proof of Lemma 1 (decomposition and bounds). Throughout, $z^\star$ is the closure's *entire* argument, so that $g(z^\star)$ is $\sigma(z^\star)$-measurable by construction. Every step below uses only that measurability, and each fails if any argument of g is left out of the conditioning variable: with $g = g(z^\star, T)$ and T excluded, the cross term in (a) need not vanish and the binning in (c) is no longer a coarsening of the conditioning σ-algebra, so neither the identity nor the upper bound survives. Where g reads a temperature—the broad IDAC set of §4.2, dim = 103—we accordingly take $z^\star = (\text{profile pair}, T)$; on the corner all rows sit at 298 K, where T is constant and the two readings coincide. *(a) Identity.* Split the residual as $m - g(z^\star) = [m - \mathbb{E}(m \mid z^\star)] + [\mathbb{E}(m \mid z^\star) - g(z^\star)]$, square, and take expectations. The squared terms give $\mathbb{E}[\mathrm{Var}(m \mid z^\star)] = B_{\mathrm{insuff}}$ and $\mathbb{E}[(\mathbb{E}(m \mid z^\star) - g(z^\star))^2] = B_{\mathrm{closure}}$. The cross term vanishes: since $\mathbb{E}(m \mid z^\star) - g(z^\star)$ is $\sigma(z^\star)$-measurable, the tower rule gives $\mathbb{E}\big[(m - \mathbb{E}(m \mid z^\star))(\mathbb{E}(m \mid z^\star) - g(z^\star))\big] = \mathbb{E}\big[(\mathbb{E}(m \mid z^\star) - g(z^\star))\,\mathbb{E}(m - \mathbb{E}(m \mid z^\star) \mid z^\star)\big] = 0$. *(b) Jensen lower bound.* With $\Delta(z^\star) = \mathbb{E}(m \mid z^\star) - g(z^\star)$, $B_{\mathrm{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m] - \mathbb{E}[g])^2$ by Jensen, using $\mathbb{E}[\mathbb{E}(m \mid z^\star)] = \mathbb{E}[m]$; it requires only that m and g share a reference state, so that $\mathbb{E}[m] - \mathbb{E}[g]$ is a genuine mean discrepancy. *(c) Law-of-total-variance upper bound.* For any binning $B = \mathrm{bin}(g(z^\star))$, $\sigma(B) \subseteq \sigma(z^\star)$, so conditioning on the coarser σ-algebra cannot reduce expected conditional variance: $\mathbb{E}[\mathrm{Var}(m \mid B)] \geq \mathbb{E}[\mathrm{Var}(m \mid z^\star)] = B_{\mathrm{insuff}}$. *(d) Convention-independence of the estimand, but not of the bound.* $B_{\mathrm{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$ is a functional of the law of $(m, z^\star)$ alone and does not involve g; hence it is identical across closure conventions, and any $\mathrm{MSE}(g_1) - \mathrm{MSE}(g_2)$ is entirely $B_{\mathrm{closure}}(g_1) - B_{\mathrm{closure}}(g_2)$. The *bound* of (c) is not convention-independent, and no contradiction arises: $\mathrm{bin}(g_{\mathrm{full}}(z^\star))$ and $\mathrm{bin}(g_{\mathrm{res}}(z^\star))$ coarsen the same $\sigma(z^\star)$ along two different statistics, so (c) yields two different valid upper bounds on one quantity—which is why Table 7 carries two numbers per set and why their minimum is again a valid population upper bound, admissible against the MSE of *either* convention. In finite samples the minimum of two estimates is biased low even when each is valid, so we report it and do not headline it (§4.2(i)). □

Two consequences of this one-sidedness reconcile the bounding apparatus with the injectivity of the design, and are stated in the main text at §2.2. First, because every reported B_{insuff} is an *upper* bound and every B_{closure} the matching *lower* one, the bounds are conservative in one direction only: were the superpopulation variance negligible, the finding would be the stronger $B_{\mathrm{closure}} = \mathrm{MSE}$. Nothing reported can therefore be an artifact of having over-estimated B_{insuff}, and no entry in any table is the share of the error the inputs contribute. Second, the resampling frequencies are statements about the *instrument* and not about the superpopulation: the clustered-bootstrap frequency P_{boot} is the fraction of cluster resamples in which the *measured bound* still falls below half the total error, MSE/2 being the threshold that orders the two terms. It measures whether this design resolves the ordering, not the probability that an inequality between two population quantities holds, so a low P_{boot} is loss of resolution and never evidence for the inputs.

Theorem 1 (Grounding-gap sandwich)**.** *Assume all risks are finite ($m, g(z^\star) \in L^2$; g and each $h \in \mathcal{E}$ measurable), (A1) the true encoder is representable, $\phi^\star \in \mathcal{E}$ with $z^\star = \phi^\star(x)$, and (A2) the intermediate is a lossless bottleneck, $\mathbb{E}[m \mid x] = \mathbb{E}[m \mid z^\star]$. Then*

$$0 \;\leq\; \mathcal{G} \;\leq\; B_{\mathrm{closure}}, \qquad \mathcal{G} = B_{\mathrm{closure}} - \big(R_{\mathrm{e2e}} - R_{\mathrm{free}}\big), \tag{S2}$$

with $\mathcal{G} = B_{\mathrm{closure}}$ iff $R_{\mathrm{e2e}} = R_{\mathrm{free}}$ (the composed class realizes the Bayes predictor by distorting the intermediate). In particular a well-specified closure ($B_{\mathrm{closure}} = 0$) gives $\mathcal{G} = 0$; hence under A1–A2, *$\mathcal{G} > 0$ certifies $B_{\mathrm{closure}} > 0$.*

Proof of Theorem 1 (grounding-gap sandwich). Throughout assume $m, g(z^\star) \in L^2$ and g, every $h \in \mathcal{E}$ measurable. *(a)* $\mathcal{G} \geq 0$. By A1, $\phi^\star \in \mathcal{E}$ is a feasible encoder with $g(\phi^\star(x)) = g(z^\star)$, so $R_{\mathrm{e2e}} = \inf_h \mathbb{E}[(m - g(h(x)))^2] \leq \mathbb{E}[(m - g(z^\star))^2] = R_{\mathrm{orc}}$; the infimum need not be attained. Hence $\mathcal{G} = R_{\mathrm{orc}} - R_{\mathrm{e2e}} \geq 0$. *(b)* $R_{\mathrm{e2e}} \geq R_{\mathrm{free}}$. Every $g(h(x))$ is $\sigma(x)$-measurable, and $R_{\mathrm{free}} = \mathbb{E}[(m - \mathbb{E}(m \mid x))^2]$ is the infimum of $\mathbb{E}[(m-p)^2]$ over *all* $\sigma(x)$-measurable p; an infimum over the subclass $\{g(h(\cdot)) : h \in \mathcal{E}\}$ is no smaller. *(c) $R_{\mathrm{free}} = B_{\mathrm{insuff}}$ under A2.* From $\mathrm{Var}(m) = \mathbb{E}[\mathrm{Var}(m \mid x)] + \mathrm{Var}(\mathbb{E}(m \mid x)) = \mathbb{E}[\mathrm{Var}(m \mid z^\star)] + \mathrm{Var}(\mathbb{E}(m \mid z^\star))$ and A2 ($\mathbb{E}(m \mid x) = \mathbb{E}(m \mid z^\star)$, so the two $\mathrm{Var}(\mathbb{E}(\cdot))$ terms agree), the two expected conditional variances agree: $R_{\mathrm{free}} = \mathbb{E}[\mathrm{Var}(m \mid x)] = B_{\mathrm{insuff}}$. *(d) Sandwich and identity.* By Lemma 1, $R_{\mathrm{orc}} = B_{\mathrm{insuff}} + B_{\mathrm{closure}}$. With (c), $\mathcal{G} = R_{\mathrm{orc}} - R_{\mathrm{e2e}} = B_{\mathrm{closure}} - (R_{\mathrm{e2e}} - R_{\mathrm{free}})$, and (b) gives $R_{\mathrm{e2e}} - R_{\mathrm{free}} \geq 0$, so $0 \leq \mathcal{G} \leq B_{\mathrm{closure}}$; equality $\mathcal{G} = B_{\mathrm{closure}}$ holds iff $R_{\mathrm{e2e}} = R_{\mathrm{free}}$. Finally $B_{\mathrm{closure}} = 0$ forces $0 \leq \mathcal{G} \leq 0$, i.e. $\mathcal{G} = 0$; contrapositively, under A1–A2, $\mathcal{G} > 0 \Rightarrow B_{\mathrm{closure}} > 0$. *(e) A2 is necessary for the certificate.* If A2 fails, take $x \in \{a, b\}$ equiprobable, $z^\star \equiv 0$ (so A1 holds), $m(a) = +1, m(b) = -1$, and g with $g(0) = 0 = \mathbb{E}(m \mid z^\star)$ so $B_{\mathrm{closure}} = 0$; an encoder $h^\dagger(a) = 1, h^\dagger(b) = -1$ with $g(\pm 1) = \pm 1$ gives $R_{\mathrm{e2e}} = 0$ while $R_{\mathrm{orc}} = 1$, hence $\mathcal{G} = 1 > 0 = B_{\mathrm{closure}}$. Thus $\mathcal{G} > 0$ alone does not certify $B_{\mathrm{closure}} > 0$; the decomposition (Lemma 1) does. □

The penalty-sign result is stated and proved here; §2.3 gives its two halves in prose and reads the consequences off them. Writing each trained risk as an asymptote plus estimation variance, $\mathbb{E}[\hat{R}_{\mathrm{phys}}(n)] \approx R_{\mathrm{e2e}} + V_{\mathrm{phys}}(n)$ and $\mathbb{E}[\hat{R}_{\mathrm{direct}}(n)] \approx R_{\mathrm{direct},\infty} + V_{\mathrm{direct}}(n)$ with $V(n) \downarrow 0$, the *physics penalty at budget* n is $\mathcal{T}(n) \approx \Delta_\infty + [V_{\mathrm{phys}}(n) - V_{\mathrm{direct}}(n)]$, with $\Delta_\infty := R_{\mathrm{e2e}} - R_{\mathrm{direct},\infty}$ its *asymptotic* part.

Proposition 1 (Sign of the physics penalty)**.** *(i)* $\mathcal{T}(n) < 0$ *(physics helps) iff* $V_{\mathrm{direct}}(n) - V_{\mathrm{phys}}(n) > \Delta_\infty$*: the variance a constrained prior saves must exceed its asymptotic bias. (ii) If* $\Delta_\infty > 0$ *and the variance gap vanishes (*$V_{\mathrm{direct}} - V_{\mathrm{phys}} \to 0$*), there is a critical budget* $n^\star$ *beyond which* $\mathcal{T}(n) > 0$ *for all* $n > n^\star$*.*

Proof of Proposition 1 (sign of the physics penalty). Write $D(n) := V_{\mathrm{direct}}(n) - V_{\mathrm{phys}}(n)$ and $\Delta_\infty := R_{\mathrm{e2e}} - R_{\mathrm{direct},\infty}$, so that $\mathcal{T}(n) = \Delta_\infty - D(n)$. *(i)* $\mathcal{T}(n) < 0$ exactly when $D(n) > \Delta_\infty$, immediately. *(ii)* If $\Delta_\infty > 0$ and $D(n) \to 0$, choose $n^\star$ with $|D(n)| < \Delta_\infty$ for all $n > n^\star$ (possible since $D(n) \to 0$); then $\mathcal{T}(n) = \Delta_\infty - D(n) > 0$. Monotonicity of D is *not* required for existence of $n^\star$; it is needed only for the stronger claim "$\mathcal{T}(n) > 0$ at every n," which we treat as an empirical observation. *(iii) A limiting refinement.* Since $V_{\mathrm{phys}}(n) \geq 0$, $D(n) \leq V_{\mathrm{direct}}(n)$: the variance a prior can save is bounded by the control's own estimation variance. As the black box approaches the aleatoric floor $V_{\mathrm{direct}}(n) \to 0$, so $\limsup_n D(n) \leq 0$; with both estimators consistent, so $V_{\mathrm{phys}}, V_{\mathrm{direct}} \to 0$, one has $D(n) \to 0$ and $\mathcal{T}(n) \to \Delta_\infty$. That limit is ≥ 0 when the control is asymptotically Bayes, so $R_{\mathrm{direct},\infty} = R_{\mathrm{free}}$ (the risk of an unconstrained predictor), and not merely from $V_{\mathrm{direct}} \to 0$; under that condition and A2, $\Delta_\infty = R_{\mathrm{e2e}} - R_{\mathrm{free}} = B_{\mathrm{closure}} - \mathcal{G}$ by (S2). □

The variance term $D(n)$ of Proposition 1 is what carries the semiparametric phenomenon §5 appeals to, in which an estimated propensity outperforms the known-true one:[34,35] a fitted nuisance can be the more *efficient* estimator even when its bias is no smaller, so a true-valued input need not give the lower total risk at finite n.

C Per-arm tables and decomposition robustness

Table 4 gives the exact per-arm metrics behind the six three-seed bars of Fig. 2: three seeds on the cross-arm intersection lock (n=5608, rows supervised and finite in every arm); the two single-seed co-adaptation bars are valued in the paragraph below. Per-seed values are printed to three decimals; every mean, difference and ± quoted anywhere is computed on the unrounded values, so the printed triples reproduce them only to that precision. Every ± that spans *seeds* in this paper is a population standard deviation over them (`ddof`$=0$); a sample standard deviation (`ddof`$=1$) would be larger by $\sqrt{3/2}$, and the error bars of Fig. 2 follow the same `ddof`$=0$ convention. Three kinds of ± in this paper are *not* that, and each says so where it is printed: the kernel-residual figures of the correctability test ($46 \pm 5\%$, $-5 \pm 16\%$, $+2 \pm 15\%$) carry a standard *error* over five fit seeds; a ± on a permutation or phase-randomised null (Table 2; §4.4) is that null's own spread over draws, not a spread over seeds at all; and the pKa arm errors of App. H span the 20 within-scaffold splits. Reading the table off the rounded means mis-states differences by up to 0.01: the learned-σ and DirectGNN arms differ by 0.1435 from their unrounded means, which we quote as +0.14, while their rounded values 1.85 and 1.70 differ by 0.15. Both closure families—NRTL and COSMO-SAC over a learned σ-profile—trail the DirectGNN control (the physics penalty), though the arms are separately tuned and that gap is not attributable to the bottleneck (§4). The σ-oracle (reference VT-2005 profile) is the worst arm. Restoring the Staverman–Guggenheim combinatorial term makes the *grounded learned-*σ arm slightly worse (1.8457 → 1.8788)—the signature of a misspecified map (§4.2); no oracle-plus-combinatorial arm was run, so the combinatorial contrast is against the grounded learned-σ row and not against the oracle row. The unrounded means are 1.7022, 1.7950, 1.8457, 1.8788, 2.0434 and 2.2517, with population standard deviations 0.033, 0.071, 0.053,

Table 4: Per-arm scaffold-split metrics (3 seeds, n=5608 cross-arm lock), with per-seed MAE to three decimals. Ordered by MAE. The $\pm$ is a population standard deviation over the three seeds, computed on the unrounded values.

Arm	MAE	R^2	per-seed MAE (42/43/44)
DirectGNN (no closure)	1.70 ± 0.03	$+0.42 \pm 0.03$	1.749/1.674/1.684
NRTL closure	1.80 ± 0.07	$+0.34 \pm 0.03$	1.734/1.758/1.894
COSMO-SAC, learned σ (grounded)	1.85 ± 0.05	$+0.33 \pm 0.05$	1.803/1.921/1.813
+ Staverman–Guggenheim combinatorial	1.88 ± 0.09	$+0.32 \pm 0.05$	1.805/2.007/1.824
COSMO-SAC, ungrounded σ	2.04 ± 0.04	$+0.19 \pm 0.01$	2.040/1.996/2.093
COSMO-SAC, reference σ (σ-oracle)	2.25 ± 0.02	-0.03 ± 0.04	2.232/2.281/2.242

0.091, 0.040 and 0.021. The two senses of grounding are the second and the last of these: training-time supervision moves the arm from 2.0434 to 1.8457 (helps, -0.20), evaluation-time substitution from 1.8457 to 2.2517 (hurts, $+0.41$).

σ-head schedule of these arms. The four COSMO-SAC rows come from three training runs under one pinned configuration — 30 P1, 70 P2 and 10 P3 epochs with `freeze_sigma_head_during_sle` on, so the σ head's weights are held from P2 onward. The two grounded rows precede that with a σ warm-up on the grounding pool (up to 40 epochs, early-stopped on a held-out tenth of the pool) and interleave the pool through P1; under the freeze the stream is inactive in P2/P3. The ungrounded row sets warm-up and stream budget to zero, so its σ head is shaped in P1 by the corpus-internal $\ln \gamma_2^\infty$ auxiliary loss alone, never by an external σ label, and is frozen from P2 in the same way. The σ-oracle row is the grounded checkpoint re-evaluated with $z^\star$ injected, so it shares the grounded schedule exactly. Because the freeze holds the head and not the encoder that feeds it, the predicted profile is not pinned at its warm-up value in these arms. We have not run the surrogate isolation (§4.4) on these checkpoints, so how far their latent drifted is not measured. The supervision contrast quoted there ($+0.40 \to +0.14$) is a separate matched pair of runs — one warm-up-plus-short-P1 base resumed twice for SLE, with the freeze off and on, on n=5571 matched rows — and not these arms.

Oracle-application controls: two of the three are single-seed. All three oracle applications (evaluation-only, training-time injection, channel-swap; §3) are computed on the same three-seed cross-arm row set as Table 4, and their per-arm predictions are released with the code. Only the evaluation-only substitution is a three-seed result. Training-time injection (MAE 1.98) and the channel swap (MAE 2.08) were run at seed 42 alone, so their matched comparator is the seed-42 learned-σ value 1.803 on the same n=5608 lock, not the three-seed mean 1.8457 of Table 4. Both exceed the worst of the three learned-σ seeds (1.921), so the direction of both survives a worst-case seed comparison. The channel-swap margin ($+0.27$) is 2.3 times the learned arm's full between-seed range; the training-time margin ($+0.18$) is one and a half times it, so that control fixes a sign rather than an effect size. This matters for how the paradox's magnitude should be read. The evaluation-only substitution ($+0.41$ against the three-seed mean) mixes closure misspecification with the ordinary degradation any pipeline suffers when an input distribution is swapped at test time; the training-time injection, in which the model co-adapts to the reference profile, is the arm that removes that component and it costs $+0.18$. The co-adapted arm is therefore the confound-free one and the evaluation-only $+0.41$ is not. How large a share of the gap co-adaptation carries off can be read only on the seed the two arms share: at seed 42 the injection costs $1.981 - 1.803 = 0.177$ where the evaluation-only substitution costs $2.232 - 1.803 = 0.428$, a share of 0.59. We report that share as a description of seed 42 and not as a magnitude, because it is not resolved. Its numerator was measured once, so the injected arm's own between-seed spread is unmeasured, and a spread the size of the learned arm's 0.118 would move the share by about ± 0.27; holding the injected arm at 1.981 and reading it against the seed-43 and seed-44 learned baselines instead gives 0.83 and 0.61. What survives that latitude is the sign—at the matched seed part of the evaluation-only gap does go with co-adaptation, which is what makes the co-adapted arm the confound-free one—and the sign, not the share, is what the claim rests on. The ratio $(0.406 - 0.177)/0.406 = 0.56$ sets the one-seed injection against the three-seed mean gap; it is a share of nothing and is reported nowhere. Neither $+0.18$ nor $+0.27$ can be ordered against the -0.20 supervision gain either, whose own per-seed values are 0.237/0.075/0.281 and straddle $+0.18$.

The estimator grid. Three analyst choices set the separation margin on the corner, and Table 5 reports every combination of them rather than one setting of each. The margin is positive in 66 of the 72 cells. The six exceptions are the coarsest conditionings—three bins under either combinatorial convention and four bins residual-only—where the law-of-total-variance bound is valid but weakest. The grid is not a set of interchangeable settings, because two opposing biases run across it. Coarser bins condition on less, so the bound is weaker; finer bins condition on more but leave few points per bin, and the within-bin variance is then downward-biased in finite samples, which inflates the margin favourably. Neither end is unbiased. We headline the eight-bin cell under the *unbiased* within-bin variance on an occupancy rule fixed in advance of the grid—the coarsest binning that still leaves fewer than eight rows per bin, here 7.5 at n=60—and we carry the same *bin count* to the broad IDAC set, so the two are conditioned at the same granularity. On 477 rows that is not the same occupancy rule (which would call for about 60 bins) and is deliberately conservative. The broad set's margin grows with bin count over the grid as a whole, though not cell by cell: in the deployed convention under the unbiased variance it falls from $+0.10$ at three bins to $+0.01$ at four, from $+0.57$ at seven to $+0.51$ at eight, and from $+0.83$ at twenty to $+0.82$ at twenty-four. The endpoints are what the choice turns on: $+0.51$ at eight bins against $+0.96$ at forty-eight. Eight bins therefore gives a looser bound and a smaller margin there than the occupancy rule would. It is *not* the least favourable cell of the middle range: seven bins residual-only under the unbiased variance gives $+0.01$ against the eight-bin $+0.05$, and twenty bins gives $+0.08$. Nothing turns on the choice—the whole middle range of the deployed convention lies between $+0.01$ and $+0.47$, all positive—but the reason for it is occupancy and not conservatism, and taking the least favourable middle cell instead gives $+0.01$ and the same verdict of an unresolved margin. The per-cell P holds that cell's bin count fixed across resamples. The main text instead quotes the deposited adaptive rule $\max(3, \lfloor n_{\text{resample}}/10 \rfloor)$, which conditions on less on small resamples and is therefore more conservative: it gives ≈ 0.78 (maximum-likelihood) and ≈ 0.72 (unbiased) residual-only, against ≈ 0.89 and ≈ 0.86 full. The two schemes measure different things and we report both rather than the higher.

A fourth analyst choice: what "equal-count" means when n is not divisible by the bin count. The rule this paper deploys, and which §2.2 names as the fourth analyst choice without restating, is interpolated (`numpy` default `linear`) quantile edges of g at $j/8$, $j = 1, \ldots, 7$, with a value falling exactly on an edge assigned to the upper bin. Neither n divides by eight ($60/8 = 7.5$, $477/8 = 59.625$), and that alone is what makes standard implementations of the same phrase disagree; there are no exact ties in g on either set under either convention, so each of them is a genuine coarsening of $\sigma(g)$ and the bound is valid under all of them. On the corner at eight bins, the deployed convention and the unbiased variance, $B^{\text{up}}_{\text{insuff}}$ is 0.713 under our rule, under the same edges with ties sent to the lower bin, and under `pandas.qcut` alike; 0.624 under `numpy.array_split` of the

g-sorted order (margin $+0.225$); 0.669 under $\lfloor k\,\mathrm{rank}/n \rfloor$ ($+0.134$); and 0.736 under order-statistic quantile edges ($+0.001$). Exact enumeration by dynamic programming over *every* partition of the g-ordered rows into eight contiguous bins of sizes in $\{7, 8\}$ puts the corner bound in $[0.620, 0.742]$ and its margin in $[-0.012, +0.232]$; at seven bins the margin envelope is $[-0.134, +0.157]$, and one such rule realises -0.089 there. The sign of the corner margin is therefore set by the binning rule and not by the data, which is a further reason to read its $+0.05$ as consistent with zero. The broad IDAC set absorbs the same choice: the six strictly equal-count implementations give 0.694 to 0.700, a spread of 0.006, and the exact envelope over all sizes-$\{59, 60\}$ layouts is $[0.687, 0.701]$, margin $[+0.499, +0.528]$ (at seven bins, $[+0.564, +0.570]$). Our 0.697 sits mid-range in both enumerations. Nothing load-bearing moves, since the paper already grades the corner unresolved and carries the claim on the broad set; we name the rule because the phrase "equal-count" does not by itself fix it.

Table 5: Separation margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ on the corner (n=60), over bin count, within-bin variance (ML, Be) and combinatorial convention.[a] Negative margins in bold.

bins	full/ML	full/Be	res/ML	res/Be
3	**−0.06** (.64)	**−0.16** (.59)	**−0.43** (.43)	**−0.53** (.38)
4	+0.18 (.82)	+0.07 (.78)	**−0.14** (.63)	**−0.26** (.57)
5	+0.50 (.90)	+0.38 (.87)	+0.21 (.77)	+0.10 (.70)
6	+0.66 (.94)	+0.54 (.92)	+0.35 (.84)	+0.22 (.77)
7	+0.76 (.96)	+0.62 (.94)	+0.18 (.89)	+0.01 (.83)
8	+0.74 (.97)	+0.58 (.96)	+0.24 (.91)	+0.05 (.86)
9	+0.90 (.98)	+0.75 (.97)	+0.42 (.93)	+0.24 (.88)
10	+1.12 (.98)	+0.99 (.97)	+0.42 (.94)	+0.21 (.89)
11	+1.11 (.99)	+0.95 (.98)	+0.38 (.96)	+0.14 (.91)
12	+0.96 (.99)	+0.75 (.98)	+0.49 (.96)	+0.24 (.93)
13	+1.01 (.99)	+0.80 (.98)	+0.59 (.97)	+0.34 (.94)
14	+1.05 (.99)	+0.81 (.98)	+0.50 (.97)	+0.21 (.94)
15	+1.00 (.99)	+0.73 (.99)	+0.52 (.98)	+0.20 (.95)
16	+1.03 (1.0)	+0.77 (.99)	+0.56 (.98)	+0.22 (.96)
17	+1.25 (1.0)	+1.05 (.99)	+0.56 (.99)	+0.20 (.96)
18	+1.08 (1.0)	+0.76 (.99)	+0.69 (.99)	+0.39 (.97)
19	+1.11 (1.0)	+0.77 (1.0)	+0.79 (.99)	+0.47 (.97)
20	+1.23 (1.0)	+0.94 (.99)	+0.55 (.99)	+0.08 (.97)

[a] ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive.

The same grid on the broad IDAC set is Table 6, now with its own per-cell bootstrap frequency rather than a bare range of point margins. None of its 56 cells is negative. Within the *deployed* residual-only convention the margin runs from $+0.01$ (four bins) and $+0.10$ (three bins) up to $+0.96$, with the eight-bin unbiased headline at $+0.51$; the frequencies at the two coarsest cells, 0.65 and 0.42, say that those conditionings do not resolve the ordering, and they are the reason the broad set's claim is graded *likely* and not certified. Under the full convention the same grid runs from $+0.23$ to $+1.25$ and reaches $P > 0.95$ from five bins upward.

Table 6: Separation margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ on the broad IDAC set (n=477), over bin count, within-bin variance (ML, Be) and combinatorial convention.[a] No cell is negative; the headline is the 8/res/Be cell.

bins	full/ML	full/Be	res/ML	res/Be
3	+0.24 (.60)	+0.23 (.59)	+0.11 (.42)	+0.10 (.42)
4	+0.50 (.93)	+0.49 (.93)	+0.02 (.67)	+0.01 (.65)
5	+0.84 (.98)	+0.82 (.98)	+0.50 (.77)	+0.48 (.76)
6	+0.84 (.99)	+0.82 (.99)	+0.51 (.86)	+0.50 (.85)
7	+0.94 (.99)	+0.93 (.99)	+0.59 (.89)	+0.57 (.88)
8	+0.96 (1.0)	+0.95 (1.0)	+0.53 (.93)	+0.51 (.92)
10	+1.06 (1.0)	+1.04 (1.0)	+0.67 (.96)	+0.64 (.94)
12	+1.05 (1.0)	+1.03 (1.0)	+0.74 (.97)	+0.71 (.97)
16	+1.11 (1.0)	+1.09 (1.0)	+0.76 (.99)	+0.72 (.99)
20	+1.17 (1.0)	+1.14 (1.0)	+0.88 (1.0)	+0.83 (.99)
24	+1.19 (1.0)	+1.15 (1.0)	+0.87 (1.0)	+0.82 (1.0)
32	+1.21 (1.0)	+1.16 (1.0)	+0.93 (1.0)	+0.86 (1.0)
40	+1.30 (1.0)	+1.25 (1.0)	+1.03 (1.0)	+0.95 (1.0)
48	+1.29 (1.0)	+1.22 (1.0)	+1.06 (1.0)	+0.96 (1.0)

[a] ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive. Regenerates from `results/b_insuff/convention_audit.json`.

The decomposition on both sets, and the three kernel arms. Table 7 is the closure–insufficiency decomposition of §4.2(i)–(ii) on both sets under both conventions, with the maximum-likelihood variance variant, the constant-offset (Jensen) row and the bootstrap bands. Table 8 is the three COSMO-SAC arms of §4.2(iii) on both sets under both conventions, with the bootstrap frequencies, the 90% margins and the block split of the 2002 → 2010 contrast into the 57 pairs shared with the corner, the complementary 420 and the 80 of those at 298 K. Every claim either table carries has a row in Table 2 with its set, value and grade.

The bounds Fig. 5 draws, named and valued. Table 9 lists one row per mark in that figure, together with the corroborating estimators Table 7 points here for instead of listing them row by row. The inclusion rule is note *a* of the table, and it is a rule for that figure and not a claim of completeness over the paper: the LOTV half is a selection from two bin grids far too large to draw, and the $\ln \gamma_2^{\infty}$-stratified bounds of Table 10 are computed on subsets of the corner rather than on either whole set and are not marks in the figure. It regenerates from `make_paradox_figures.py` (in `scripts/analysis/`) run with `--dump-bounds`, which prints the inventory the figure draws, so the table can be re-checked against the marks. The regeneration's guard against an artifact changing under the inventory is *not* the same on the two sets, and we state which is which rather than claim one rule. On the corner it stops both ways: on an estimator the artifact records and the inventory does not name, and on a named one the artifact no longer records. It additionally checks the entries that set's two audit artifacts share against each other. On the broad set it stops only the first way, so a named estimator absent from that set's audit artifact would be dropped rather than raised. In both panels of Fig. 5 the bar's two blocks are contiguous shares of one total, so what separates the terms is the margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ and not a gap on the axis. Two rendering conventions in that figure: the dashed threshold MSE/2 prints three decimals when a bound would print the same two-decimal string, and a mark prints its value when its disc reaches that line on the page. Two rows of the table carry the whole risk on each set. On the broad set the coarsest and the least favourable cell differ—three bins gives 0.901, four bins 0.948 against a threshold of 0.951, a margin of +0.007 at bootstrap frequency 0.65 and the tightest anywhere on that set. On the corner the two coincide at three bins (1.000), and both it and the four-bin cell (0.866) sit *above* that panel's threshold of 0.736: the negative-margin cells Table 5 sets in bold, and both are drawn in Fig. 5.

The constant-offset (Jensen) bound on both sets. §2.2 promises these here. On the broad IDAC set $\mathbb{E}[m] = 3.424$, $\mathbb{E}[g_{\mathrm{res}}] = 2.998$ and $\mathbb{E}[g_{\mathrm{full}}] = 2.579$, so the constant-offset bound is $B_{\mathrm{closure}} \geq 0.182$ under the deployed residual-only convention and $B_{\mathrm{closure}} \geq 0.715$ under the full one. Against the law-of-total-variance ceiling on B_{insuff} (0.697 residual-only, 0.482 full) the assumption-free bound therefore separates under the full convention and inverts under the deployed one, exactly as it does on the corner (0.426 against 0.713) and by a wider margin—the deployed ratio is 0.26 here against 0.60 there. On this set it is in addition strictly slacker than the bound the decomposition already supplies, $B_{\mathrm{closure}} \geq \mathrm{MSE} - B^{\mathrm{up}}_{\mathrm{insuff}}$ (0.182 against 1.205 residual-only, 0.715 against 1.429 full), so it adds nothing to the ordering under either convention; the deployed-convention separation rests on the law-of-total-variance bound alone, which is what §2.2 says.

The broad set's matching rule, and two bounds we decline to headline. Three details of §4.2(i) follow. *(a) The matching asymmetry.* The corner is matched to VT-2005 by canonical SMILES, a rule we defend there on the ground that a profile imported across a tautomer or protonation variant is the wrong $z^{\star}$. The broad IDAC set is matched to UD by InChIKey with a first-block fallback, which is looser, and a wrong $z^{\star}$ inflates MSE and so B_{closure}—that is, it runs against the ordering we claim. The exposure is small and we measure it rather than argue it: of the 90 distinct molecules there, 86 resolve by full InChIKey and four only through the first block, touching 12 of the 477 rows. On the 465 rows matched exactly the verdict is unchanged—deployed convention MSE 1.930, $B_{\mathrm{insuff}} \lesssim 0.70$, margin +0.53 at $P \approx 0.9$; full convention 1.926, 0.49, +0.94 at $P > 0.95$—so the looser rule is not carrying the result. *(b) The tighter of the two conventions' bounds.* B_{insuff} is the same population quantity under both conventions

Table 7: Closure–insufficiency decomposition on the reference-input activity path, as bounds rather than a point split. **The headline columns are the deployed residual-only ones on *both* sets**, one rule applied throughout; the full-convention columns are the comparison.

	broad IDAC (477)[a]		corner (60)[a]	
Quantity	**res**[b]	full[b]	**res**[b]	full[b]
$\mathrm{MSE}_{\mathrm{total}} = B_{\mathrm{closure}} + B_{\mathrm{insuff}}$ (exact)	1.90	1.91	1.47	1.80
B_{insuff} upper bound (LOTV)[c]	0.70	0.48	0.71	0.61
B_{closure} lower bound[d]	1.21	1.43	0.76	1.19
B_{closure} lower, constant-offset (Jensen)[e]	0.18	0.71	0.43	0.72
separation margin[f]	**+0.51**	+0.95	**+0.05**	+0.58
… bootstrap band[g]	$>$0.95/≈0.9/≈0.9	$>$0.95 (all three)	≈0.7	≈0.9
… maximum-likelihood variance[h]	0.68/+0.53	0.47/+0.96	0.62/+0.24	0.53/+0.74
… tighter of the two conventions[i]	0.48/+0.94		0.61/+0.26	
Further B_{insuff} upper bounds[j]	Table 9			
Label noise σ_η^2 (IDAC)	already inside every row above[k]			

[a] Broad IDAC, n=477 on UD profiles; corner, n=60 on VT-2005 profiles throughout (§4.2(iii) scores the same pairs on UD). [b] *full*, with the Staverman–Guggenheim combinatorial term; *res*, residual-only. [c] Law of total variance, at eight equal-count bins of $g(z^\star, T)$ under the unbiased (Bessel-corrected) within-bin variance. It fits no model to the data. [d] $\mathrm{MSE} - B^{\mathrm{up}}_{\mathrm{insuff}}$; this row decides the ordering. [e] $(\mathbb{E}[m] - \mathbb{E}[g])^2$. [f] $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$, the quantity that orders the two terms. [g] Pair / two-way / source clusters. The corner supports the two-way clustering only, so it carries one entry per column. [h] Same rows and bins, maximum-likelihood within-bin variance, as B_{insuff} upper / margin. [i] Charged against the deployed MSE, as B_{insuff} upper / margin. Not headlined. [j] The marks of Fig. 5: out-of-fold RF and ridge (random and blocked folds), kNN-in-$z^\star$, and the binning-free $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})]$. Not a paper-wide inventory: the bin grids behind the LOTV selection are Tables 5 and 6, and the $\ln\gamma_2^\infty$-stratified bounds, computed on corner subsets, are Table 10. [k] A *component* of B_{insuff}, not a term beside it (§2.2).

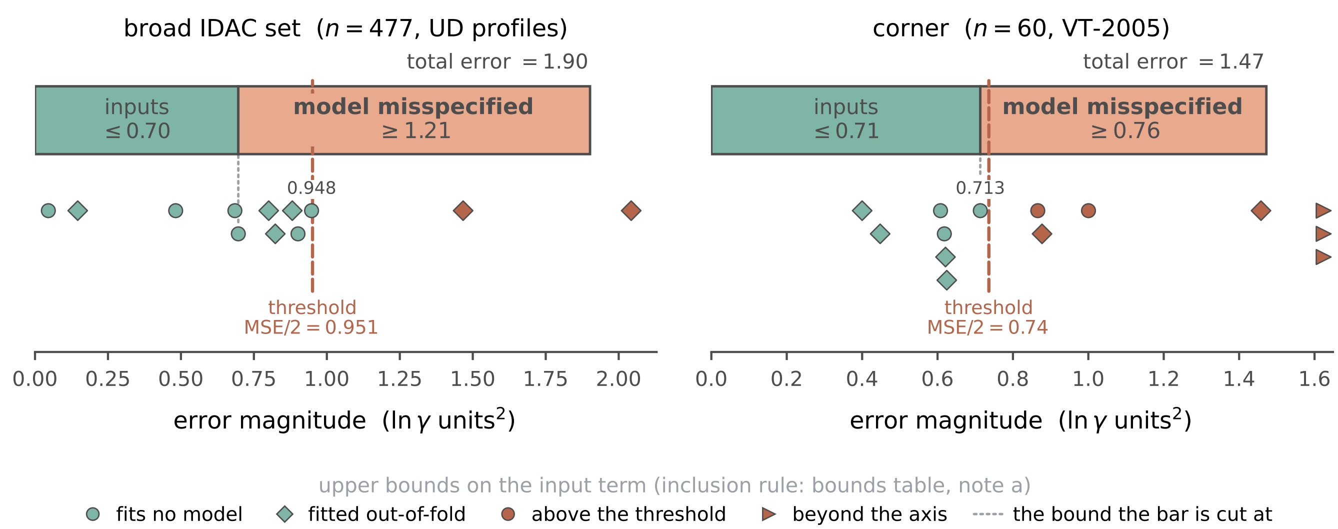

Figure 5: One-sided bounds on the closure–insufficiency split; each bar, threshold and mark uses the deployed residual-only convention except the one full-convention bound per panel that Table 9 keys, and each bar's two blocks are contiguous shares of its total. Inventory, rule and the two rendering conventions: Table 9 and the text above it.

(step (d) of the proof of Lemma 1, App. B); what differs is the statistic the coarsening bins on, $g_{\mathrm{full}}(z^\star, T)$ or $g_{\mathrm{res}}(z^\star, T)$, each a deterministic function of the same argument and each therefore giving a separately valid upper bound. The smaller of the two is consequently also a valid upper bound in population, and charging it against the deployed error gives 1.902

Table 8: Activity error (MSE, $\ln \gamma_2^\infty$ units) of the three COSMO-SAC arms, and the block split that reads the 2002 → 2010 reversal against profile database and temperature held fixed.

	corner (60), UD[b]		broad IDAC (477), UD[b]	
(a) Arm[a]	full[b]	res[b]	full[b]	res[b]
fair 2002	1.757	1.436	1.911	1.902
2010, dispersion off	0.765	0.831	1.978	2.893
2010, dispersion on	0.785[e]	—[d]	1.855	—
P (2010 better)[c], dispersion off	>0.95	—	≈ 0.6	≈ 0.2
90% margin[c]	[+0.06, +2.18]	—	[−1.95, +0.93]	[−3.06, +0.70]
P (2010 better)[c], dispersion on	—	—	≈ 0.6	—
90% margin[c]	—	—	[−2.24, +1.31]	—
(b) Block (UD profiles, full convention)	2002 → 2010, dispersion off		P[c]	90% margin[c]
shared with the corner (57)	1.783 → 0.767		>0.95	[+0.06, +2.36]
complementary (420)	1.928 → 2.143		≈ 0.5	[−2.41, +0.87]
… of these, at 298 K (80)	0.796 → 1.284		≈ 0.3	—

[a] The three arms, named here and in the text: *fair 2002*, COSMO-SAC-2002 in our NIST-validated layer reading all three profile channels; *2010, dispersion off* and *2010, dispersion on*, COSMO-SAC-2010 without and with the dispersion term (§4.2(iii)). [b] *full* / *res* as in Table 7. *Both* sets are scored on UD profiles here, the corner included—Table 7 scores the corner on VT-2005 instead, and that database difference is the whole difference between its corner/full 1.80 and the 1.757 here (§4.2(iii)). Within a set, each arm uses the σ-profile averaging its own model prescribes (Mullins for 2002, Hsieh for 2010). [c] P is the two-way cluster-bootstrap frequency that the 2010 arm is the better one, with the 90% margin on the paired squared-error difference beneath it. [d] A dash marks a cell the deposited artifacts do not carry. [e] This one cell is n=57, not 60: `cCOSMO` tabulates no dispersion ϵ for the three bromine- or iodine-containing corner pairs. On those common 57 the *dispersion off* arm scores 0.767, so the 0.765-versus-0.785 ordering is not an artifact of the differing n, and no claim rests on that ordering (§4.2(iii)).

against 2×0.48, a margin of +0.94 (on the corner, 1.472 against 2×0.61, +0.26). We report this but do not headline it: a minimum over two finite-sample estimates is anti-conservative even when each is valid in population, and the same-convention pairing is the conservative reading. *(c) The binning-free pair bound.* $\mathbb{E}[\mathrm{Var}(m \mid \text{pair})] = 0.046$ is thirteen times tighter than the binning bound on the same 359 rows (0.61), and would put the margin *there* at +2.16 against the +1.02 the binning bound gives on those rows. It is not promoted because all 67 multi-row pairs draw their rows from a single publication, so it samples no between-laboratory spread (§4.2(i)).

Whether the closure defect is correctable: within a solvent, not across one. That B_{closure} dominates locates the ceiling in the map; it does not say the map is *fixably* wrong. We tested this with a small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel ($\Delta = B\,\mathrm{diag}(a)\,B^\top$ over the σ-grid, symmetric, zero-initialised), fit on the *reference* VT-2005 profiles and cross-validated five-fold over five fit seeds. The answer depends on the fold design, and only one design is free of leakage. With random folds over the 60 matched pairs, a rank-one residual (52 parameters) removes 46 ± 5% of the reference-input error out of sample (activity MSE 1.47 → 0.79). The same solvent, however, then appears in training and test, and the closure error is predominantly a between-solvent systematic. Under folds grouped so that no solvent is shared, the reduction is −5 ± 16% at rank one and +2 ± 15% at rank two. Each ± is a standard error over the five fit seeds on one fixed 60-pair set and prices optimiser noise only; the held-out spread across folds is an order of magnitude larger (sd 1.02 in MSE under the grouped design, 0.54 under random folds) and is not propagated into it. Stated as a bound rather than a null, the grouped design excludes a cross-solvent reduction above about 30%, and to that extent the apparent 46% is a within-solvent ef-

Table 9: **Upper bounds on the input-insufficiency term** B_{insuff}, on both sets—one row per mark in Fig. 5, in the order it draws them, plus one estimator that set cannot support, with the value, what each fits and what qualifies it. What is included, and what is not: note *a*.

Estimator[a]	$B^{\mathrm{up}}_{\mathrm{insuff}}$	fits[b]	note[c]
Broad IDAC set (n=477, UD profiles). MSE = 1.902; threshold MSE/2 = 0.951; headline margin +0.51.			
LOTV, 8 bins, unbiased variance **[headline]**	0.697	no model (○)	
LOTV, 8 bins, maximum-likelihood variance	0.685	no model (○)	
LOTV, 3 bins (coarsest)	0.901	no model (○)	
LOTV, 4 bins (least favourable)	0.948	no model (○)	tightest margin on this set
LOTV, 8 bins, full convention (also valid)[d]	0.482	no model (○)	
RF out-of-fold, folds blocked on the molecule pair	0.882	out-of-fold (⋄)	
ridge out-of-fold, folds blocked on the molecule pair	1.467	out-of-fold (⋄)	above thr.
RF out-of-fold, folds blocked on the pair, conditioning on $(z^\star, T)$	0.823	out-of-fold (⋄)	
RF out-of-fold, folds blocked on the solute	2.043	out-of-fold (⋄)	vacuous
RF out-of-fold, random folds	0.146	out-of-fold (⋄)	leaky
ridge out-of-fold, random folds	0.801	out-of-fold (⋄)	leaky
$\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})]$, binning-free, on the 359 rows that support it	0.046	no model (○)	single-source
Corner (n=60, VT-2005 profiles). MSE = 1.472; threshold MSE/2 = 0.736; headline margin +0.05.			
LOTV, 8 bins, unbiased variance **[headline]**	0.713	no model (○)	
LOTV, 8 bins, maximum-likelihood variance	0.618	no model (○)	
LOTV, 3 bins (coarsest, least favourable)	1.000	no model (○)	above thr.
LOTV, 4 bins	0.866	no model (○)	above thr.
LOTV, 8 bins, full convention (also valid)[d]	0.608	no model (○)	
kNN-in-$z^\star$ (convention-independent; least conservative)	0.400	out-of-fold (⋄)	deflated
kNN, neighbour from a distinct solute	0.448	out-of-fold (⋄)	
kNN, neighbour from a distinct solvent	1.458	out-of-fold (⋄)	above thr.
kNN, neighbour from a distinct pair	1.831	out-of-fold (⋄)	vacuous
RF out-of-fold, random folds	0.621	out-of-fold (⋄)	deflated
ridge out-of-fold, random folds	0.625	out-of-fold (⋄)	
RF out-of-fold, folds blocked on both molecules	3.653	out-of-fold (⋄)	vacuous
ridge out-of-fold, folds blocked on both molecules	divergent	out-of-fold (⋄)	
ridge out-of-fold, deliberately aggressive	0.877	out-of-fold (⋄)	above thr.
$\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})]$[e]	—	—	no repeated pairs

[a] One inclusion rule governs both sets, in two halves that differ in kind. The LOTV half is a *selection* from bin grids too large to draw (56 cells on the broad set, 72 on the corner; Tables 6 and 5): *within the deployed convention and under the unbiased within-bin variance*, the headline cell, the *coarsest*, the *least favourable* and every such cell above the threshold, with the headline cell's maximum-likelihood variant and its full-convention counterpart listed beside it. Cells above the threshold under the other variance or the other convention are not drawn; they are in those two grids, where Table 5 sets the negative-margin ones in bold. The estimator half is every out-of-fold estimator that set's audit artifact records, in every fold design it records, vacuous designs included—the per-set guards on that are stated with the regeneration command above. [b] What the estimator fits to the data; matches the figure's marker. *no model* (○) for the law-of-total-variance (LOTV) cells and for $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})]$, *out-of-fold* (⋄) for a fitted estimator. The LOTV bound, fitting nothing, is unaffected by any fold design. [c] *above thr.*, the bound exceeds that set's separation threshold MSE/2, so that cell does not separate; *vacuous*, it exceeds the total MSE—the three corner marks at that panel's right edge are its two vacuous ones and the *divergent* ridge, while the broad set's vacuous solute-blocked forest fits on its own axis; *leaky*, random folds leave a held-out row's own molecule pair in the training fold at a neighbouring temperature; *deflated*, biased low by the crossed design, in which two pairs sharing a molecule agree on 51 of the 102 $z^\star$ coordinates; *single-source*, every multi-row pair supporting the bound comes from one publication, so it is valid on those rows but samples no between-laboratory component and is not promoted. [d] Computed under the other combinatorial rule and drawn against the deployed-convention threshold beside it. It is valid there, and it is the only mark on either panel that is not residual-only. [e] The estimator that set cannot support, not a mark.

fect.

That fold spread also exceeds the gaps between ranks (≤ 0.06) by an order of magnitude, so these data do not resolve a rank ordering; and the 60 pairs form a single connected component under shared molecule identity, so a molecule-disjoint holdout does not exist at any fold design. B_{closure} is therefore a defect that a handful of kernel parameters can absorb *within* a solvent, and that we have not shown to be correctable across solvents—weaker than a general repair. The fitted correction also gains nothing over a free one: the parameter-free 2010 kernel reaches 0.765 on the same molecule pairs (§4.2(iii)) against the 52-parameter residual's random-fold 0.79. The residual's leakage-free number, however, is the grouped-fold one, a $-5 \pm 16\%$ reduction, i.e. no reduction at all, so the comparison that holds is a free 2002 → 2010 step against a fitted correction that does not transfer. The two also run on different profile sources and conventions (residual-only on VT-2005 here, full on the University of Delaware (UD) profiles there[26]), so this is indicative rather than a head-to-head; but each stays above its own convention's insufficiency ceiling (0.62–0.71 residual-only, 0.53–0.61 full), so neither erases B_{closure}.

Where the residual error sits chemically. The composition audit behind §4.2: 90% of the corner's squared error falls on the strong hydrogen-bond-donor solvents, 84% once the two doubted amine/methanol pairs are deleted, and the acceptor-only solvents are near-exact. That localisation is what places B_{closure} in COSMO-SAC-2002's single-parameter hydrogen-bonding kernel rather than spreading it over the profile. A counter-test on the same closure points the same way, and is the one §4.2 cites. The Staverman–Guggenheim combinatorial term is the physically *more* complete treatment and acts only where the two molecules differ in size; restoring it *increases* the corner's reference-input error, MSE = 1.47 under the deployed residual-only convention against 1.80 with the term in place (Table 7), the gap between the two being carried by the size-asymmetric pairs. A term that adds physics and costs accuracy is what a misspecified map does and not what missing information about the inputs would do. The same signature appears on the solubility axis, where restoring the term makes the grounded learned-σ arm slightly worse (Table 4).

The closure-fidelity lever: the arithmetic behind §4.2(iii). *(a′) The input-fixed kernel control.* The prescribed reduction of the typed Hsieh profile to a single 2002 profile is its sum $p = p_{\text{nhb}} + p_{\text{OH}} + p_{\text{OT}}$ (Eq. 9 of Ref.[23]), which holds database, cavity area, volume and averaging fixed and varies only the kernel. On the corner that contrast is 2.047 → 0.765, two-way cluster bootstrap $P > 0.95$ with 90% margin $[+0.05, +2.90]$; the averaging step taken alone moves 2002 by +0.29 in the opposite direction and is not resolved ($P < 0.2$, $[-0.79, +0.11]$). That averaging term is why the control's 2002 endpoint is 2.047 and not the 1.757 of the *fair 2002* arm, and why the input-fixed step (1.28) is *larger* than the native-input one (0.99): the control is favourable to the conclusion it supports. The control cannot separate the donor/acceptor typing, which the 2002 kernel has no channel to consume and which is therefore charged to the kernel by construction. What fixes the *fair 2002* arm at 1.757 rather than lower is an implementation detail: the NIST reference implementation, handed a typed three-profile database, silently reads the nonpolar channel only, an arm scoring 1.26 on the matched corner (§A gives it unrounded), which is not the reference closure, so the arm is taken from our own NIST-validated layer instead. *(a) The* 2010, dispersion on *arm's coverage.* `cCOSMO` returns no dispersion value for the three bromine- or iodine-containing corner pairs—bromobenzene and iodobenzene in 1, 2-ethanediol, and 2-bromopropane in pyridine—so that arm's corner run scores on n=57. On those common 57 the *2010, dispersion off* arm scores 0.767, so the 0.765-versus-0.785 ordering is not an artifact of the differing n. They are the *same* three pairs the broad set's construction excludes, its ladder also requiring a tabulated ϵ, so the dispersion-defined 57 and the 57 shared between the two sets are one set, not two. *(b) A rounding coincidence.* The 1.783

quoted for the shared 57 and the 1.783 quoted for the 334 chromatographic rows are distinct numbers that round alike (unrounded 1.78257 against 1.78245), the smaller set lying wholly inside the larger. *(c) The 2010 kernel on its own typed latent.* That latent is three 51-bin profiles per molecule, 153 in all, so the conditioning variable is 306-dimensional and the resolution ratios are $307/477 \approx 0.64$ per row and $307/185 \approx 1.66$ per distinct pair on the broad IDAC set, and $306/60 \approx 5.1$ on the corner, where all rows sit at 298 K and T adds no dimension. Under the deployed residual-only convention the corner gives 0.831 against $B_{\mathrm{insuff}} \lesssim 0.80$, hence $B_{\mathrm{closure}} \gtrsim 0.03$ and a margin of -0.77 at $P \approx 0.3$; the broad IDAC set gives 2.893 against $B_{\mathrm{insuff}} \lesssim 0.80$, hence $B_{\mathrm{closure}} \gtrsim 2.09$ and a margin of $+1.29$ at $P \approx 0.9$. Under the literature-standard full convention the same rows give 1.978 against $B_{\mathrm{insuff}} \lesssim 0.60$, margin $+0.78$ at $P \approx 0.8$, and the corner 0.765 against 0.69, margin -0.61. The switch to the deployed convention raises the 2010 kernel's broad-set error from 1.978 to 2.893 and its floor bound from 0.60 to 0.80, so the wider margin comes from a worse closure and not from a tighter floor, and the grade does not move. On the broad set the binding clustering is the source-DOI bootstrap on 15 clusters, whose 90% interval $[-0.41, +4.53]$ still admits zero, as does the two-way one $[-0.24, +4.71]$; only the pair clustering is strictly positive, and it was so before the convention switch as well. *(d) The published comparison.* The comprehensive assessment[26] covers 29,173 γ^{∞} points for the 2010 and dsp kernels only. Its largest documented infinite-dilution step—adding the whole dispersion term—yields a ∼10-point mean-absolute-deviation gain (95% → 86%) with a *reversal* on aqueous systems (203% vs. 194%); 2010's gains are otherwise concentrated in liquid–liquid equilibria and in the temperature dependence of the electrostatic term, where an infinite-dilution design has little leverage. *The scope of the search behind the null of §4.2(iii)*: the papers introducing each kernel,[5,24,25] the benchmark implementation,[23] this comprehensive assessment[26] and the works citing them. It was not exhaustive, which is why the main text reads the result as an absence of evidence rather than as evidence of absence. *(e) The floor checks.* Each is same-database, same-latent and same-convention, stated under the deployed residual-only rule with the full-convention reading beside it. *(1)* Broad IDAC set: the fair 2002 error 1.902 against $2B_{\mathrm{insuff}} \lesssim 1.39$ on the *same* UD profiles, margin $+0.51$ at $P \approx 0.9$ (1.911 against 0.96, $+0.95$, $P > 0.95$ full). *(2)* Corner, VT-2005 against VT-2005: 1.472 against $2B_{\mathrm{insuff}} \lesssim 1.427$, so $B_{\mathrm{closure}} \gtrsim 0.759$ against $B_{\mathrm{insuff}} \lesssim 0.713$ and a margin of $+0.05$ (1.80 against 1.22, $+0.58$ full; Table 7). The 1.757 of (a′) is those same 60 pairs scored on *UD* profiles, which is the whole difference between 1.80 and 1.757. Charged against a UD-native floor recomputed for those pairs, the deployed convention gives 1.436 against $B_{\mathrm{insuff}} \lesssim 0.71$, a margin of $+0.01$ at $P \approx 0.8$, against 1.757 versus a UD-native $B_{\mathrm{insuff}} \lesssim 0.54$ and a margin of $+0.69$ full—so on UD profiles the corner's own floor check is, under the rule we deploy, at zero.

Robustness of the closure/insufficiency separation. We stress-test the deployed-convention verdict $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$ (§4.2)—equivalently $B_{\mathrm{insuff}} < \mathrm{MSE}/2 = 0.74$ at $\mathrm{MSE}_{\mathrm{res}} = 1.47$—four ways on the n=60 set.

(i) Estimator sensitivity. The four random-fold B_{insuff} upper bounds of Table 9 (the LOTV one is also the headline row of Table 7)—0.40 (k-NN), 0.71 (binning/LOTV at the headline unbiased variance; 0.62 at the maximum-likelihood one), 0.62 (random forest), 0.62 (ridge out-of-fold)—all fall below the 0.74 threshold, the binning bound only just. The forest is the one estimator our two corner audit artifacts disagree on: the standalone decomposition run's own forest gives 0.565 on these rows against the audit's 0.621, a spread of the size a forest's own randomisation produces. The audit value is the one Table 9 and Fig. 5 carry, because the audit artifact is what the inclusion rule names; both are below the threshold and neither is the headline. The k-NN value is the smallest and least conservative input to $B_{\mathrm{closure}} \geq \mathrm{MSE} - B^{\mathrm{up}}_{\mathrm{insuff}}$, so we take the leakage-immune LOTV bound as the headline instead. The LOTV bound bins

$g(z^\star)$ and is therefore finite-sample downward-biased when points per bin are few, which would *inflate* B_{closure} favourably. The two non-binning bounds (RF 0.62, ridge 0.62) are immune to that particular per-bin sparsity, but they do not provide independent corroboration: under random folds every held-out row shares its solvent or its solute with a training row, so both can borrow strength from a near-twin, anti-conservatively and in the same direction as the binning (§4.2). Blocking folds on both molecules leaves no informative estimator on this set. The broad IDAC set behaves the same way once its own leak—a held-out row's molecule pair recurring at a neighbouring temperature—is blocked: the random forest goes from 0.15 to 0.88 and the ridge from 0.80 to 1.47, past its threshold. The headline should therefore be read as resting on the leakage-immune LOTV bound alone, on either set, not on agreement between three estimators. The sole non-certifying configuration is a deliberately aggressive ridge, which lifts B_{insuff} to 0.88 (then $B_{\text{closure}} \geq 0.59 < 0.88$): a sensitivity edge, not a refutation.

(0) The corner's design. The 60 pairs span 41 solutes and 17 solvents; two formamide pairs are excluded for a canonical-SMILES tautomer mismatch. Methanol occupies 25 of the 60 rows (12 as solute, 13 as solvent) and pyridine is the largest solvent stratum (20 rows); $\text{Var}(m) = 2.47$ across the set. The two worst-fit pairs, 1-hexanamine and cyclohexylamine in methanol, carry 34% of the squared error; both rest on a single record, and 1-hexanamine sits far off its own homologous series ($\ln \gamma_2^\infty$=2.62 against −0.78 for 1-butanamine in the same solvent), which is why we do not read that concentration as established chemistry. Deleting both leaves MSE 1.01; charged against this the headline cell of the deployed convention—eight equal-count bins under the *unbiased* within-bin variance—gives $B_{\text{insuff}}^{\text{up}}$ 0.5056 and margin −0.0015, so the separation does *not* survive the deletion where the paper headlines it. The four neighbouring cells do: 0.4370 (margin +0.14) at eight bins under the maximum-likelihood variance, 0.4717 (+0.07) at six bins unbiased, 0.4241 (+0.16) at six bins maximum-likelihood, and, under the full convention on the same 58 rows, MSE 1.20 against 0.3309 (+0.53) at eight bins unbiased. The reading we take from the cell that fails is that the corner's +0.05 headline margin has no room to absorb the loss of 34% of the squared error, which is the conclusion the leverage folds below reach by a different route.

(i bis) Leverage folds on the corner. The separation survives all 17 leave-one-solvent and all 60 leave-one-pair folds and deletion of the eight highest-error pairs, but only 40 of 41 leave-one-solute folds (six equal-count bins): the exception drops methanol, the maximum-leverage solute (12 pairs), and loses the separation (−0.26), as does dropping methanol as solute with pyridine as solvent (n=28; −0.36). Deleting methanol from both roles (n=35) holds only marginally—the margin falls from 0.35 to 0.11 at six bins and from 0.24 to 0.003 at eight, and inverts under the Bessel-corrected variance (−0.03, −0.21). Those fold counts and the bootstraps use the maximum-likelihood variance at six bins. Repeated at the headline cell—eight bins, unbiased variance—the leave-one-solvent count is unchanged at 17/17 (minimum margin +0.03), the leave-one-pair count falls to 56/60 (minimum −0.03) and the leave-one-solute count to 36/41 (minimum −0.22); deletion of the eight highest-error pairs still holds there (+0.04).

(ii) Leave-one-solvent-out. Dropping each of the 17 solvents in turn, the verdict holds in all 17 folds under the LOTV bound. Under the random-forest estimator it holds in only 2/17, because an RF over-fits conditional variance in 102 dimensions at n=60 (its out-of-fold MSE approaches $\text{Var}(m)$), so its leave-one-out estimate is unstable—an estimator artifact, not a sign flip.

(iii) Label noise. Injecting zero-mean Gaussian noise $\sigma_\eta \in \{0.2, \ldots, 0.8\}$ into m leaves the B_{closure} lower bound stable (0.84 → 0.70), as expected since $B_{\text{closure}} = \text{MSE} - \mathbb{E}[\text{Var}(m \mid z^\star)]$ is invariant to zero-mean noise.

(iv) Clustered-bootstrap interval. A solvent-clustered bootstrap (3000 resamples of the 17 solvents, LOTV bound) makes the confidence explicit: the B_{closure} lower bound has median 0.93 (CI_{90} [0.40, 1.62]), and the mar-

gin $B_{\text{closure}} - B_{\text{insuff}}$ has median 0.36 (CI_{90} $[-0.22, +0.97]$) with $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.89$ (single-axis; the conservative two-way solute×solvent bootstrap, which also charges for repeated solutes, gives ≈ 0.78, the value reported in the main text). With $G = 17$ solvent and 41 solute clusters this small-G regime has optimistic interval coverage, and the finite-cluster correction acts in the direction of less confidence and non-certification. We therefore treat the P values as a coarse band, not two-decimal quantities, and the main text quotes them as bands (rounded to one decimal, with > 0.95 and < 0.2 closing the ends) throughout; the two-decimal values are in the deposited artifacts and in the tables here, where the estimator that produced each is named. The separation on the corner is *likely* but not certified at the 90% level under either clustering, which is why the main text treats it as supporting rather than decisive. On the broad IDAC set, with 78 solute and 36 solvent clusters and a third clustering over 15 source publications, the bands mean something; but they do not all clear. Under the deployed residual-only convention the margin bootstraps to $P > 0.95$ over the 185 molecule pairs (90% margin $[+0.23, +1.16]$) and only $\approx$0.9 two-way over the 78×36 solute/solvent clusters ($[-0.08, +2.16]$) and over the 15 sources ($[-0.07, +1.74]$), with two of the three 90% intervals extending just below zero, so that set is graded *likely* as well. Read as a ratio, the broad set's +0.51 is 1.1 times its own 90% bootstrap half-width under pair clustering and 0.46 times it under the two-way one, which is the sense in which §4.2 calls that margin unresolved. It is only under the full convention that all three clear ($P > 0.95$), in the same order: $[+0.71, +1.27]$ over pairs, $[+0.48, +1.85]$ two-way, $[+0.50, +1.57]$ over sources.

(v) A label systematic tracking $z^\star$, and what the corner's fold designs leave. Zero-mean noise cannot manufacture the separation: it raises MSE and $B^{\text{up}}_{\text{insuff}}$ alike, leaves the B_{closure} lower bound unchanged, and shrinks the margin by σ^2_η, so it can understate the separation but not create it. What the margin does bound is a *systematic* label error correlated with $z^\star$, which enters B_{closure} rather than B_{insuff}: on the corner it would close the separation at root-mean-square 0.49 $\ln\gamma$ under the maximum-likelihood binning but at only $\approx$0.21 under the Bessel-corrected one. The matched IDAC is essentially single-method (158/165 gas chromatography), which makes a within-method systematic of 0.5 implausible; one of 0.21 is not, so on the corrected binning that threat is *not* excluded. The corner's out-of-fold estimators cannot police it either: under random folds every held-out row shares its solvent (47/60 = 78% of rows) or its solute (27/60 = 45%) with a training row, and under folds blocked on both molecules three of the four estimates (3.65, 1.83, and the divergent ridge) exceed the total MSE = 1.47, beyond which an upper bound on B_{insuff} constrains B_{closure} vacuously.

In-regime (γ-stratified) separation. The n=60 measurement is matched at low γ, while the paradox spans all γ. Stratifying the n=60 set by $\ln\gamma_2^\infty$ tests whether closure-dominance is a regime artifact. It is not: the verdict $B_{\text{closure}} > B_{\text{insuff}}$ (leakage-immune LOTV binning bound) holds in every stratum and *strengthens* toward higher γ (Table 10)—in the highest stratum ($\overline{\ln\gamma_2^\infty} = 2.94$, approaching the non-ideal regime) $B_{\text{closure}} \geq 2.68$ exceeds $B_{\text{insuff}} \leq 0.67$ several-fold. The high-γ strata are small (n = 17–30 in 102 dimensions): the law-of-total-variance bound can be pushed toward zero by too-few-points-per-bin, which mechanically widens the margin exactly where data is thinnest, so it is the several-fold separations rather than the precise values that carry the qualitative reading. The bound here bins each stratum into three equal-count bins (against eight in the main text) and uses the maximum-likelihood within-bin variance, as in the headline row of Table 7; recomputed with the unbiased (Bessel) one it rises to 0.08/0.95/0.81 across the three strata, so each still separates, but the middle stratum only marginally (0.95 against its own MSE/2 = 0.96). The $m >$ med. stratum separates through its upper (high-γ) half, and the *isolated* moderate band (med.$< m \leq 1.5$, n=13) does *not* separate. This is not a counter-example. The total error there is only

≈0.06, so $B_{\text{closure}} \leq \text{MSE} \approx 0.06$ assumption-free (since $B_{\text{insuff}} \geq 0$), and there is nothing to attribute—near ideality the target is small and the split is uninformative. We read no closure-quality claim from it in either direction. This supports—but does not prove—the conjecture that the same closure misspecification drives the all-γ paradox.

Table 10: Closure/insufficiency verdict across $\ln\gamma_2^\infty$ strata of the matched set (deployed residual-only convention; $B_{\text{insuff}}^{\text{up}}$ is the leakage-immune LOTV binning bound throughout, at the maximum-likelihood within-bin variance).

Stratum	n	$\overline{\ln\gamma_2^\infty}$	$B_{\text{insuff}} \leq$	$B_{\text{closure}} \geq$
lower γ ($m \leq$ med.)	30	-0.40	0.07	0.95
higher γ ($m >$ med.)	30	$+1.90$	0.85	1.07
high γ ($m > 1.5$)	17	$+2.94$	0.67	2.68

The closure/insufficiency decomposition (full/res conventions; law-of-total-variance (LOTV), random-forest (RF), ridge, and kNN estimators; solvent-clustered bootstrap; the pyridine stratum) is tabulated in §4.2. Figure 6 shows the shape of that error on the corner: the reference-σ closure's parity plot against the measured $\ln\gamma_2^\infty$.

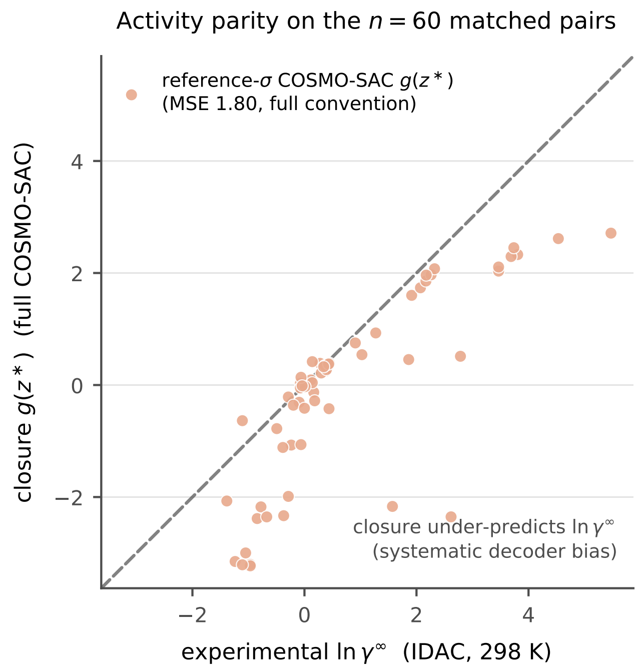

Figure 6: Activity-coefficient parity on the n=60 matched pairs. The reference-σ COSMO-SAC closure $g(z^\star)$ (salmon) falls below the diagonal at larger $\ln\gamma_2^\infty$, under-predicting the activity coefficient (MSE 1.80, full convention).

Four qualifications on the decomposition (full). Four qualifications bear on the decomposition of §4.2. None of them inverts the verdict; each sharpens its scope.

1. The deployed combinatorial default is residual-only. Under it the *constant-offset* (Jensen) lower bound inverts ($B_{\text{closure}} \geq 0.43 < B_{\text{insuff}} \leq 0.71$; it separates only under the full convention, $B_{\text{closure}} \geq 0.72$ against $B_{\text{insuff}} \leq 0.61$), so that assumption-free bound does not separate under deployment; Table 7 carries that row for both sets, and the full-convention margin (the sharpened MSE$-B_{\text{insuff}}$ bound, 1.19 vs 0.61) is a full-COSMO-SAC-only result. What separates under the deployed convention is the leakage-immune bound $B_{\text{closure}} \geq \text{MSE}_{\text{res}} - B_{\text{insuff}}^{\text{LOTV}} \approx 0.76 > 0.71$, which carries the headline claim, together with the combinatorial argument above. Every B_{insuff} ceiling quoted in this item is the unbiased within-bin variance, the headline setting of Table 7; the maximum-likelihood pair (0.85 against 0.62) is the demoted variant and is not quoted here.

2. The closure error is predominantly a *between-solvent* systematic bias (∼64% of total error); within the largest single-solvent stratum (pyridine, n=20) the closure is nearly correct (stratum MSE 0.07). Because solvent σ is part of $z^\star$, these offsets are still decoder error and give an independent solvent-conditional lower bound $B_{\text{closure}} \geq 0.89$ (full)/0.56 (res).

3. Confidence on the full-convention constant-offset separation is the solvent-clustered bootstrap $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$ (0.83 vs the random-forest upper bound, 0.89 vs the law-of-total-variance bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent depen-

dence dominates. The leakage-immune margin (0.76 vs 0.71) is a point estimate. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).

4. The 60 matchable pairs are the low-to-moderate γ regime (matched mean $\ln \gamma_2^\infty = 0.75$ vs 3.56 for VT-2005-unmatchable pairs, $t = -7.44$; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is $\ln \gamma_2^\infty$ at 298 K, so the numeric ceiling does not necessarily transfer unchanged to the finite-composition solubility regime.

D External baselines and ablations

We evaluated the direct state-of-the-art model (FastSolv[1]) and the physics-informed state-of-the-art model (SolProp[2]) on a by-solute test split (`test_solute.csv`, $n \approx$ 10–12k). The two arms are not obtained the same way, and the difference matters below: FastSolv is retrained on our corpus, whereas SolProp is the released pretrained thermodynamic cycle—the arm that instantiates the physics—mapped onto our target by a three-parameter linear recalibration fitted on the training rows, not by retraining. That cycle is run at 298 K on every row, so the temperature of a measurement reaches the SolProp arm only through the calibrator's temperature term. In $\ln x_2$ they reach MAE 1.94 (FastSolv, R^2 0.23) and 2.34 (SolProp, calibrated, R^2 −0.18); in $\log_{10} S$, 0.87 and 1.04. These are on a *by-solute* split that is *not comparable* to our scaffold numbers (Table 4): they bound the task's difficulty, not a ranking of the models against one another, so this does *not* establish that our scaffold-split DirectGNN (1.70) outperforms them. A like-for-like scaffold evaluation of the external models was not run, the deposited external runs being by-solute and pair-random only. The internal ordering—the physics-informed SolProp trailing the direct FastSolv, the same ordering we find between our physics closures and DirectGNN—is therefore a comparison of the *cycle* against a direct model. It reverses when SolProp's encoder is instead retrained directly on $\ln x_2$ with the cycle discarded: that arm reaches MAE 1.63 (R^2 0.39) in $\ln x_2$ and 0.76 in $\log_{10} S$ on the n=10,287 rows FastSolv is scored on, against FastSolv's 1.94 and 0.87. We quote the cycle as the physics-informed arm because only it assembles the separately learned components; the retrained arm measures what discarding the closure achieves on this corpus, not a better physics-informed model. The regime-level reading is that on this corpus no black box, ours or external, escapes the accuracy ceiling—consistent with the structural account of §5. Per-arm ablations (NRTL vs COSMO-SAC, combinatorial on/off) are in Table 4.

E Supplementary mechanism results

The activity map's Fisher geometry, and why σ is under-determined. At infinite dilution the aggregate activity Fisher $J^\top J$ ($J = \partial \ln \gamma_2^\infty / \partial \sigma$) is rank-one (participation ratio 1.0 by autograd; finite differences decorrelate the redundant flat-profile coordinates and inflate it, App. F.6, so we read PR against a permuted control and not absolutely). The per-molecule Fisher that governs the drift has participation ratio ≈1.4 of 51 even at full solvent coverage. The participation ratio is a concentration statistic and not an exact count. It is measured at infinite dilution, so its transfer to the finite-composition regime that generates the drift is not established, the same scope gap that qualifies B_{closure} (§5). The external reference is itself *computed* (VT-2005 quantum chemistry) rather than experimental, so none of this establishes that recovery of the physical profile is impossible in principle.

Why the top-two share is read against a phase-randomised null, and how unstable

the magnitudes are. A top-two explained-variance share is not interpretable against an isotropic null, because smoothness alone concentrates a spectrum: smoothing isotropic noise reaches a top-two share of 0.25 to 0.83 as the correlation length grows from one bin to eight, spanning the observed value without any between-molecule alignment at all. The *phase-randomised* null of §4.4 preserves each deviation's amplitude spectrum exactly—and so its smoothness—and destroys only the alignment across molecules that a shared compensating direction asserts; the observed 0.399 against its 0.221 ± 0.011 is therefore a statement about alignment and not about smoothness. The wrong-molecule null (0.772 ± 0.023), which the observed share does *not* clear, is a difference between two points on the reference manifold and inherits that manifold's own concentration, so it demands more alignment than a shared direction implies. That is the argument for preferring the phase-randomised one, and §4.4 states that if it is not accepted the spectrum is uninformative rather than supporting. In each case the $\pm$ is the null's spread over resamples, not an uncertainty on the observed value. The three magnitudes §4.4 reports but does not carry move in both directions across analysis configurations: a shortened run gives only ~16% departure and a transfer of 1.7×, below what a constant offset alone reaches on the one run where that control can be computed, while uncontrolled two-model comparisons over-state both the departure (~82%) and the transfer (16–26×). With n=3 the $\pm$ on the three-seed figures is indicative and not a precision. Nor does mean-centred low-rank structure exclude a shrinkage of each profile toward the corpus-mean shape: the reference profiles themselves sit at 0.77 mean-centred, so any drift linear in the profile inherits that concentration.

The drift spectrum, and the null it lacks. Figure 7 is the spectrum of the drift whose mean and dominant mode are Fig. 3. Its top two modes carry 72.9% in the companion run plotted there and $71.8 \pm 1.3\%$ over the three seeds (quoted as $72 \pm 1\%$ in §4.4): the same statistic on different runs. The companion run's spectrum is `results/compensation/isolation_gpu.json` and the three-seed aggregate `results/sur/surrogate_seeds/surrogate_seeds.json`, whose `top2_evr` has mean 0.71819 and standard deviation 0.01265 over seeds 0–2; both this figure and Fig. 3 are drawn by `scripts/analysis/make_surrogate_figure.py`. That share has no matched null, so the panel is titled by what it plots and not by an inference. The phase-randomised and wrong-molecule nulls of §4.4 are run on the $\hat{\sigma}$-versus-reference deviation instead, a different quantity; the isolation drift has not been re-tested that way, because the two locally available checkpoints share a σ head and so have identically zero drift between them.

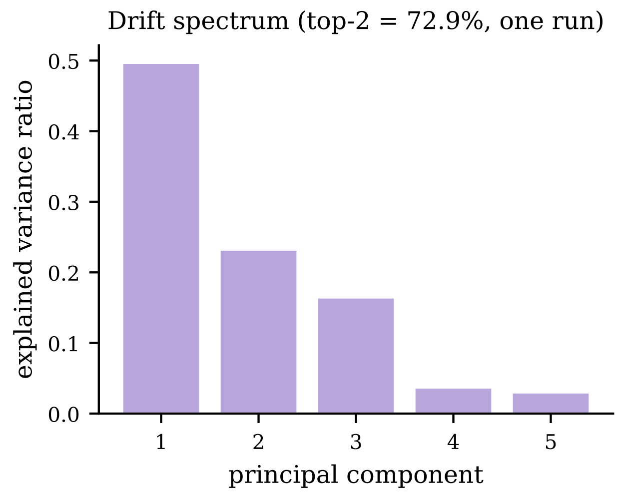

Figure 7: Explained-variance spectrum of the fine-tuning drift on the 44 matched molecules (companion run). *Caveat:* this share carries no matched null (text).

The grounded warm-up checkpoint's 0.73, and the power of the two-MSE check. The direct cancellation check of §4.4 has almost no power in the direction that matters. On the corner, 18.75 against $\mathrm{Var}(m) = 2.47$ is 7.6× worse than predicting the mean, the signature of a broken profile rather than of a partly compensating one, so the comparison separates "no cancellation" from "large cancellation" and not from "some cancellation". The same comparison run on a grounded warm-up checkpoint, whose profile sits 28% from the reference, scores 0.73 where the end-to-end SLE-trained checkpoint scores 18.75. That is a statement about the closure's sensitivity to profile roughness and about an in-sample fit to $z^\star$ on molecules inside the grounding pool, not evidence that the encoder carries information the reference lacks, and we do not read it as a both-channel reproduction of the paradox.

The closure's own-axis check on polarity. The second check of §4.4: on the n=60 corner the reference-input closure *under*-predicts, its mean output being +0.10 under the residual-only convention and −0.10 under the full one against a mean target $m = +0.75$. Sweeping its hydrogen-bond coefficient at zero, one and two times the standard value gives mean squared errors 2.42/2.83/10.15 on the strong-donor pairs: raising the coefficient makes them worse and zeroing it makes them better, so a *more* polar profile moves g further from m, which is the direction opposite to a compensating drift.

The reference ladder. A relative deviation carries no absolute scale, which is what the ladder supplies. Its rungs are built from the reference profiles the grounded arm was supervised on, so it calibrates that deviation rather than competing with it. The rungs calibrating the 51% departure of §4.4, all on the same 44 molecules and the same relative-L_2 metric: a leave-one-out corpus-mean profile (one fixed shape reused for every molecule, carrying no molecular information) 0.493; each molecule assigned another molecule's reference profile, 0.651–0.684 over five derangements; a uniform profile 0.889. Against these, the grounded base sits at 0.364 and the fine-tuned latent at 0.507. The grounded base is the σ-supervised recovery §4.4 reports as reproducing the reference profile far more closely: it is 36 ± 1% over the three seeds in the norm ratio $\|\hat{\sigma}_{\text{grounded}} - \sigma\|/\|\sigma\|$, and it is an *in-sample* fit—all 44 probe molecules lie in the σ-grounding pool and the warm-up early-stops on it—so no out-of-sample recovery estimate exists on this set.

The distortion operator D. The transfer test of §4.4 fits a dense affine map on the 51-bin grid: a 51 × 51 linear part plus a 51-bin offset, 2652 parameters, each output bin a ridge regression of 52 coefficients (ℓ_2 penalty 10^{-3}) on 22 of the 44 molecules. The constant-offset control replaces D by a single molecule-independent shift of the whole profile, fitted on the same 22 molecules and applied unchanged to the held-out 22. It can be computed on one companion run only—the two-model comparison whose two checkpoints are both still on disk—and there it is 2.1× better than the identity null on both of that run's arms. That run is the uncontrolled one, whose own transfer of 16–26× §4.4 reports as over-stated, so its residual factor is not read across. The three seed checkpoints behind the $4.1 \pm 0.5\times$ were not retained, and the remaining companion run records only an ephemeral checkpoint path, so no offset control exists for either and the 4.1× is a ratio against the identity null with no constant-offset control of its own. The homologue saturation that makes "held out" largely within-cluster interpolation is six n-alkanes, four halobenzenes, five pyridines and four butyl and hexyl amines among the 44; the median held-out molecule sits 0.048 in L_2 from its nearest training-half neighbour, about 16% of its own profile norm.

Non-identifiability, empirically (the corrupted twin). Training a twin model on data whose crystal labels are corrupted (every T_m shifted by +273 K) leaves the *solubility* fit essentially unchanged—$\ln x_2$ MAE 1.835 versus 1.812 on the corrected data—while the recovered crystal grounding differs by hundreds of kelvin (T_m MAE 258 K versus 45 K). Two very different crystal/activity splits produce the same solubility, which is Lemma 2 made concrete.

Crystal grounding on physical labels. When the crystal head is supervised on measured melting points it predicts T_m to 45 K MAE, about 4× better than a Joback group-contribution reference—so the flat downstream result of §F.3 is a dose/identifiability effect, not an inability to learn crystal properties.

The $\Delta C_p = 0$ approximation. The ideal term omits the constant-heat- capacity correction; a group-contribution audit finds a nonzero ΔC_p for 1441 of 1525 solutes (108,287 rows), so this is a bounded model-form approximation. It cannot produce either headline result. The closure/insufficiency decomposition is measured on the activity target ($\ln \gamma_2^\infty$, at 298 K on the corner and at each row's own temperature over 273–403 K on the broad IDAC set), where ΔC_p does not enter at all on either set: the ideal term is not evaluated in the decomposition, so no temperature range of it is at issue. The grounding paradox is an arm-to-arm difference (learned-σ vs. oracle-σ) whose two arms share the *identical* crystal term, so any ΔC_p error cancels exactly in the comparison. It is thus orthogonal to the activity closure, not a source of the paradox.

Fusion-supervision scarcity. Direct ΔH_{fus} labels are extremely sparse: 1279 training rows across 31 solutes, and essentially none in validation or test. This is the quantitative basis of the identifiability argument—there is almost no signal with which to pin the crystal term from data.

The NRTL branch trains (a trainability check). The asymptotic physics penalty is not an under-training artifact of the activity branch. In the headline NRTL arm the head produces solvent-varying interaction parameters—τ_{12} over the n=8103 test pairs has mean 0.97 and std 2.5 (range $[-2.6, 22]$), far from collapsed—yet the arm still trails the closure-free DirectGNN (1.80 vs 1.70; Table 4). A trained, non-degenerate activity branch that the control still outperforms is evidence that the penalty is structural, not a failure to train the branch. (The core closure-misspecification result—the σ-oracle and the decomposition—is in any case training-*independent*: it feeds the reference VT-2005 profiles through the fixed closure with no encoder optimisation, so it cannot be a training pathology.)

Structured temperature extrapolation. Withholding *temperature* rather than structure (train at $T \leq 309.75$ K, test at higher T; $n = 3343$), the affine-in-$1/T$ hypothesis classes extrapolate far better than an unstructured descriptor regressor (Table 11); the van't Hoff class also predicts the *direction* of the high-T shift correctly 99% of the time versus 40% for the random forest. This is a property of the structured hypothesis class—the trained neural models are weaker than the van't Hoff baseline—not an accuracy claim for our model.

Table 11: Same-pair high-temperature extrapolation ($n = 3343$, train $T \leq 309.75$ K). "dir." is the fraction of pairs whose high-T shift direction is predicted correctly.

Model	MAE	R^2	dir.
per-pair van't Hoff ($1/T$)	0.37	0.89	0.99
per-pair linear in T	0.41	0.85	0.99
per-pair last-low-T hold	1.20	0.69	0.01
RF (Morgan $+T$)	1.29	0.66	0.40
per-pair mean	1.46	0.56	0.01

F Supplementary diagnostics and broad negatives

F.1 Supporting diagnostics

The interpretive claims of the preceding sections are based on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that are unfavourable to the physics-informed hypothesis.

Aleatoric noise floor. Repeated solubility measurements of the same solute in the same solvent disagree across laboratories, and that disagreement sets a floor below which no model can be expected to fit. Measuring inter-source spread on,[21] the mean absolute disagreement is approximately 0.15 $\ln x_2$ units for pairs backed by two sources ($n = 3948$ groups), rising to 0.31 for groups backed by three or more sources ($n = 349$); the corresponding medians are smaller still (0.03 and 0.09). The scaffold-split MAE we report ($\approx$ 1.7 to 1.9) sits well above this band, so label noise does not account for the accuracy ceiling. The floor is small; the ceiling is not a data-quality artifact.

The encoder is transfer-limited, not expressivity-limited. To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a linear probe from the encoder representation onto RDKit descriptors and compare in-distribution and held-out fit. The median train R^2 is 0.758 against a test R^2 of 0.448 ($n = 700$ solutes in each partition), a gap of 0.31. Individual descriptors split sharply: FractionCSP3 transfers well (test $R^2 = 0.96$) while HeavyAtomCount does not (0.23). The embedding is expressive enough in sample; what degrades out of distribution is the generalisation of that expressivity across scaffolds. This evidence is suggestive rather than decisive: the sharp per-descriptor spread (FractionCSP3 0.96 vs HeavyAtomCount 0.23) means the train–test gap partly reflects covariate shift in the descriptor *targets* across scaffolds rather than an encoder property alone, and the probe measures descriptor-representability, not task-relevant expressivity directly. It therefore *suggests*—rather than establishes—that a new graph encoder is unlikely to be the main lever for out-of-distribution accuracy, with pretraining and coverage the more likely ones.

Temperature extrapolation. A structured van't Hoff activity class (log-solubility affine in $1/T$) encodes the correct temperature dependence of $\ln \gamma_2$ by construction, and this is apparent when the split withholds temperature rather than structure. Training on measurements at $T \leq 309.75$ K and testing at $T \geq 330$ K ($n_{\text{test}} = 3343$), the structured class attains a high-temperature MAE of 0.37 against 1.29 for a random forest over Morgan fingerprints plus temperature (in $\ln x_2$ units), consistent with the temperature-aware $\ln \gamma_2^\infty$ modelling of.[4] This is a property of the structured hypothesis class, not a performance claim for the trained model: the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not.

Table 12: Solvent-ranking quality of the physics path on the scaffold test ($n = 826$ solute groups).

Metric	Value
Spearman ρ	0.511
Top-1 accuracy	0.477
Kendall τ	0.436
nDCG@3	0.732

Ranking and calibration. Absolute error and decision quality diverge. Despite its high MAE, the physics path orders candidate solvents usefully (Table 12; nDCG@3 $\approx$ 0.73 across $n = 826$ solute groups). Uncertainty, by contrast, is badly miscalibrated in the native model—a nominal-90% interval covers only PICP@90 = 0.13 of held-out points—but split-conformal wrapping repairs it to 0.92 empirical coverage at the same nominal level. The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

F.2 Identifiability and compensation

Lemma 2 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher information matrix I of the observation model at a reference operating point and compute the conditioning $\text{cond}(I)$ and the Cramér–Rao lower bound on the standard deviation of ΔH_{fus} under three supervision regimes.

Every entry below is conditional on a design that must be stated, because a Cramér–Rao bound in J/mol scales directly with the assumed measurement noise. The operating point is $\Delta H_{\text{fus}} = 25{,}000\,\text{J/mol}$, $T_m = 400\,\text{K}$, and an activity term $\ln \gamma_2^\infty(T) = a + b\,(T_{\text{ref}}/T - 1)$ with $a = 1$, $b = 200\,\text{K}$ and $T_{\text{ref}} = 298.15\,\text{K}$;

Table 13: Fisher audit of the SLE design at $\Delta T = 40\,\mathrm{K}$ (synthetic, closure-independent).

Supervision regime	cond(I)	CRLB sd(ΔH_{fus}) [J/mol]
Solubility only	2.3×10^5	19,895
+ T_m label	5.4×10^3	3,151
+ $T_m + \Delta H_{\mathrm{fus}}$	1.6×10^3	1,689

x_2 is not a free input but the SLE solution at each temperature, running from 3×10^{-8} to 0.15 across the window. The design is $n = 9$ equally spaced temperatures spanning $\Delta T = 40\,\mathrm{K}$ from 280 K, one measurement each. The noise levels are $\sigma_{\ln x_2} = 0.3$ on each log-solubility observation, $\sigma_{T_m} = 10\,\mathrm{K}$ on a melting-point label and $\sigma_{\Delta H_{\mathrm{fus}}} = 2{,}000\,\mathrm{J/mol}$ on an enthalpy label. Parameters are non-dimensionalised by the scales used in the model's own configuration (5,000 J/mol, 50 K, 1, 300 K), so the condition number reports rank structure rather than unit choice. Table 13 reports the result at that window. With solubility measurements only, the design is severely ill-conditioned and the CRLB on ΔH_{fus} is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct T_m label reduces the conditioning by nearly two orders of magnitude and lowers the bound to about 3,200 J/mol; supplying T_m and ΔH_{fus} together lowers it to about 1,700 J/mol. The ill-conditioning is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which the grounding streams supply.

Widening or narrowing the temperature window does not remove the ill-conditioning of the solubility-only design. Varying ΔT over $\{120, 80, 40, 20, 10\}$ K moves the solubility-only conditioning through 1.7×10^3, 5.6×10^3, 2.3×10^5, 6.5×10^9 and 1.9×10^{16}, and the CRLB on ΔH_{fus} through 1,348, 2,249, 19,895, 3.2×10^6 and 8.0×10^6 J/mol. The point value quoted in the main text is the $\Delta T = 40$ K entry of that sweep and is not a design-independent constant. Multi-temperature data alone therefore does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps only partially: it reduces sd(ΔH_{fus}) to 8,251 J/mol, still well above the label-supervised regimes. A second, model-side diagnostic is not part of the primary argument. On the supervised test set, with a group-contribution (Joback) crystal reference, the crystal and activity residuals are strongly anti-correlated ($\mathrm{corr}(\delta\Phi, \delta \ln \gamma_2) = -0.962$). This is an *accounting artifact*, not evidence of a learned split: the two residuals sum to the (negative) $\ln x_2$ fit residual, so their correlation tends to -1 as the fit improves, a permutation null that destroys any learned coupling still reproduces most of it, and the model's own factors barely co-vary. The large apparent crystal offset is likewise not a usable misgrounding metric — it is dominated by reference error, since the Joback reference is physically implausible on these drug-like polycyclics, and capping predicted T_m at a physical bound collapses the offset from ≈ -7.1 to ≈ -1.5.[1] We therefore base the non-identifiability claim on the reference-free Fisher/CRLB audit above, and present the compensation figures in the SI (§E) as a cautionary diagnostic rather than a decomposition-quality measurement.

F.3 Grounding the crystal factor: a dose-limited null

Whether the ceiling lies in the inputs rather than the closure can be tested by grounding a *different* factor—the crystal term—on abundant external data. We trained the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points, added through the grounding-stream mechanism after excluding any pool solute whose Bemis–Murcko scaffold appears in the validation or test split. The pool did not improve prediction of the crystal properties it supplies (melting-point MAE 45.0 → 46.6 K, a negligible change in the unfavourable direction). Downstream solubility did not improve either ($\ln x_2$ MAE 1.81 → 1.88; two nominally favourable but slight shifts, in $\ln x_2$ R^2 and the crystal offset $|\delta\Phi|$, do not

[1] Permutation null (learned coupling destroyed) -0.90 to -0.93; model-factor $\mathrm{corr}(\hat{\Phi}, \ln \gamma_2) = -0.19$; the Joback reference predicts $T_m > 1000\,\mathrm{K}$ for a substantial fraction of these polycyclics.

amount to a gain). This result is *dose-limited and inconclusive rather than an unambiguous null*. The crystal grounding stream was applied here at a low dose, a small fraction of optimiser steps at low loss weight, so at the selected checkpoint the model had completed less than one full pass over the external pool. A stream that does not lower the MAE of the property it directly supervises cannot justify a strong conclusion. The result shows that *this low-dose intervention* does not move the ceiling, not that external crystal labels cannot. The activity-path $B_{\text{closure}}/B_{\text{insuff}}$ decomposition is uninformative here: the crystal "closure" is the near-exact ideal-solubility term $\Phi = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$, not a misspecified map, so the flat T_m recovery is more consistent with the crystal/activity split's structural non-identifiability (Lemma 2) and the encoder's transfer limitation (§F.1) than with closure dominance. A properly dosed crystal-grounding test is left to future work.

F.4 Chemistry-resolved structure of the asymptotic physics penalty

Scope. The two arms compared in this section differ from each other in capacity and in the per-arm optimisation settings of §2, and no hyperparameter-matched control was run, so nothing here is an accuracy claim about the thermodynamic bottleneck: Table 2 grades that ΔMAE *no claim*, and what follows is a hypothesis-generating description of where two particular pipelines differ (§5). "Physics penalty" names that measured difference throughout, not an effect of the bottleneck.

Across the full test set the two arms are separated by little more than label noise. On the 3-seed scaffold split the DirectGNN control reaches an $\ln x_2$ MAE of 1.70 ± 0.03, while the grounded physics arm (learned-σ COSMO-SAC closure) reaches 1.85 ± 0.05, for a global $\Delta\text{MAE} = \text{MAE(physics)} - \text{MAE(control)} = +0.14$ (positive meaning the closure adds cost).[2] At this granularity the difference between the two pipelines is small and near label noise. Once the error is resolved along chemical axes, that difference is an average over regimes that disagree in sign.

The solvent axis is the most pronounced. Binning test pairs by solvent class and recomputing ΔMAE per seed (Table 14) shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and $|\ln \gamma_2|$ is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class weakens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure's well-specified regime, are the seed-robust stratum in which the penalty reverses sign ($\Delta\text{MAE} = -0.22 \pm 0.11$ over seeds). This is exploratory rather than a confirmed improvement: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but is not seed-robust. The robust reading is the *direction* — the asymptotic physics penalty shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved counterpart of the synthetic closure-fidelity sweep (§K).

The solute axis shows a compatible but weaker pattern (Table 15). The closure costs most on ordinary neutral organics—oxygenated solutes, polyaromatics, and heterocycles—while the charged, high-$|\ln \gamma_2|$ stratum of salts is neutral. On this axis the penalty therefore does not track $|\ln \gamma_2|$: it falls on neutral organics, not on the ionic tail.

The novelty axis inverts the usual coverage expectation: the physics arm is neutral on the most novel chemistry (nearest-train Tanimoto ≤ 0.3) and hurts most on the most familiar

[2] The per-seed ΔMAE spans +0.05 to +0.25, so the point estimate is an imprecise 3-seed mean; being computed on unrounded per-seed means, the rounded 1.70/1.85 differ by 0.15 rather than 0.14.

Table 14: Per-solvent-class asymptotic physics penalty ΔMAE = MAE(physics) − MAE(control) on the scaffold split (mean ± sd over 3 seeds; positive means the closure adds cost).

Solvent class	ΔMAE (mean ± sd)
Water	$+0.52 \pm 0.35$
Carboxylic acid	$+0.39 \pm 0.16$
Sulfoxide	$+0.26 \pm 0.24$
Nitrile	$+0.21 \pm 0.19$
Alcohol	$+0.15 \pm 0.14$
Aromatic	$+0.03 \pm 0.63$
Hydrocarbon	$+0.02 \pm 0.50$
Halogenated	-0.29 ± 0.36
Amide	-0.22 ± 0.11

Table 15: Per-solute-class and novelty-stratum asymptotic physics penalty ΔMAE = MAE(physics) − MAE(control) on the scaffold split (positive means the closure adds cost). Entries carry the 3-seed spread where it was computed: the two Tanimoto strata and the charged/salt class. The other four solute classes are point estimates.

Solute stratum	ΔMAE
Oxygenated	$+0.32$
Polyaromatic	$+0.27$
Heterocycle	$+0.17$
Halogenated-aromatic	≈ 0
Charged / salt	-0.04 ± 0.19
Tanimoto ≤ 0.3 (novel)	-0.04 ± 0.18
Tanimoto > 0.8 (familiar)	$+0.51 \pm 0.09$

(> 0.8; Table 15). This is exploratory rather than a confirmed effect: the seed spread is not a within-bin sampling interval over which solutes fall in the stratum, and the Tanimoto binning is one of several strata families we examine (solvent class, solute class, similarity) without multiplicity control. Taken as a direction rather than a tested quantity, the inversion suggests that the ceiling on the physics path is not coverage or extrapolation but the closure's systematic misfit. That misfit is masked on novel solutes by the large errors both arms already incur there, and exposed on well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual, which is the same closure-misspecification reading as the rest of the paper. A multiplicity-controlled, within-bin-bootstrapped test is left to future work. Together, the maps indicate that the physics decomposition incurs cost where the closure is misspecified relative to the solvent chemistry, and that it can only become a net benefit in the narrow acceptor-dominated regime the closure was built to describe.

F.5 The data-efficiency sweep, and what survives its confounds

The strongest a-priori case for a physics prior is data scarcity: with few training molecules, an inductive bias toward the correct factorisation should outperform a data-hungry black box. We test it directly, subsampling the training set *by solute* at fractions 0.05–1.0 and training both the physics-grounded model and DirectGNN on the identical subsample, evaluated on the fixed scaffold test (Table 16, seed 42). The physics arm is *worse at every fraction*, and against the hypothesis the gap is *smallest* at the least data (ΔMAE = +0.08 at 5%); its R^2 is negative throughout. We do not read the growth of the gap with data (+1.0 at full) as a data-scale effect, because the sweep's endpoint does not reproduce the headline comparison. Its physics arm carries the NRTL closure, and at fraction 1.0 it gives 2.98 against 1.97 (control) on the same scaffold test where the three-seed runs of Table 4 give 1.7950 against 1.7022: ΔMAE an order of magnitude apart (+1.01 here, +0.09 there; the second is taken from the unrounded means, since the rounded column would give +0.10, §C). Both arms are worse here than their headline counterparts and the physics arm much the more so (+1.18 against +0.27 MAE), so the trend measures how the light per-fraction budget interacts with training-set size at least as much as how the physics prior does. The sweep is also single-seed, its physics-arm MAEs

Table 16: Data-efficiency sweep (train subsampled by solute, fixed scaffold test, seed 42). ΔMAE = MAE(physics) − MAE(direct).

train fraction	MAE		ΔMAE	R^2	
(by solute)	physics	direct	(phys−dir)	physics	direct
0.05	2.28	2.20	+0.08	−0.09	+0.07
0.10	2.44	2.07	+0.37	−0.25	+0.15
0.25	2.93	1.81	+1.12	−0.75	+0.38
0.50	3.12	1.98	+1.14	−0.98	+0.22
1.00	2.98	1.97	+1.01	−0.79	+0.22

are non-monotone, and its two arms differ from each other in capacity and in the per-arm optimisation settings of §2, and both are wider than the headline arms (hidden width 256 and 128 here, 64 there). What survives is the level and not the trend (Table 16, plotted in Fig. 8)—the physics arm is worse at every fraction under a matched light budget—which is *consistent with* the sign rule (Prop. 1) rather than a sharp confirmation of it. The one inverting metric is top-1 solvent ranking (0.42 vs 0.23 at 5%; the ranking-versus-accuracy split of §F.1), which does not translate into lower error. This is the last regime in which a physics prior should plausibly help; that it does not is consistent with the broad accuracy null and with the structural account—a non-identifiable split through a misspecified closure—as the operative explanation.

F.6 Diagnostics that do not work, and why

The drift is not the closure's signed first-order inverse—but the test is under-resolved. A candidate account of the compensating drift is the closure's linear-response inverse, $\hat{\sigma} - \sigma \approx J^{+}(m - g(\sigma))$ (with $J = \partial g/\partial\sigma$), predictable a priori from J before any solubility training. On the matched set the per-molecule drift shows no signed alignment with $J^{+}(m-g)$ (median cosine −0.00; Fig. 9).[3] We read this as a signed negative only, and do not build on it, for two reasons intrinsic

[3] The residual COSMO-SAC closure runs a segment-activity fixed point, so its autograd Jacobian is numerically unstable ($\|\partial g/\partial\sigma_{\text{solvent}}\| \sim 10^{15}$ at infinite dilution); J is computed by finite differences on the converged closure.

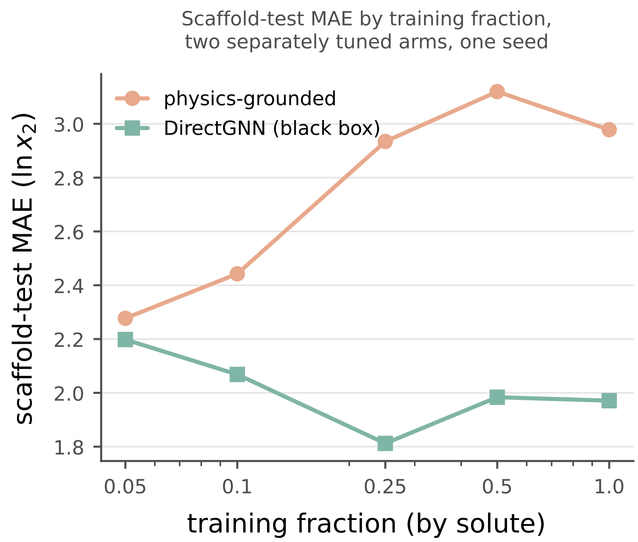

Figure 8: Data-efficiency curve: physics-grounded arm (salmon) and DirectGNN (teal) as a function of training-data fraction. *Scope:* one seed, five fractions, under a light per-fraction budget whose endpoint does not reproduce the headline comparison; the physics arm's MAEs are non-monotone. *Caveat:* the two arms differ in capacity and in per-arm optimisation, so nothing drawn here is attributable to the thermodynamic bottleneck, and this paper makes no accuracy claim in either direction (§4, §F.5).

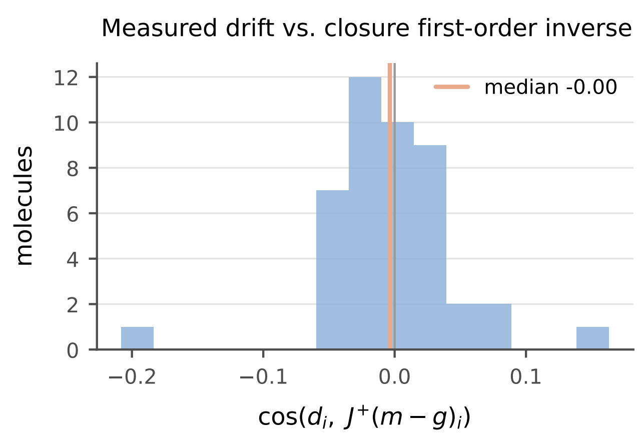

Figure 9: Per-molecule cosine between the measured drift $\hat{\sigma}-\sigma$ and the first-order closure compensation $J^{+}(m-g)$; the distribution is centred on zero (median −0.00).

to the test. First, the matched set is sparse (median one pair per molecule), so each per-molecule Jacobian subspace is one-dimensional and the dimension-matched row-space controls are uninformative—a resolved direction common across molecules makes a molecule's "own" and "other" subspaces coincide. Second, and more fundamentally, the single-pair $J^+(m-g)$ is a poor *estimator* of the first-order compensation: the activity Fisher is near rank-one (PR $\approx$ 1.4 of 51, §4.4), so the closure's first-order inverse is ill-posed along all but one direction. A null here is therefore uninformative about whether the drift is first-order, not evidence that it is not. The load-bearing claim is the under-determination of σ by activity (§4.4), not the geometry of the drift.

The instrument behind §4.4—the rank of the activity Fisher $J^\top J$ ($J = \partial \ln \gamma_2^\infty / \partial \sigma$), read as a participation ratio PR $= (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$—is computationally inexpensive and oracle-free, which is its appeal for probing any physics-grounded latent. It is also easy to misread in three ways that inflate a null or a rank in the favourable direction without any warning sign; we record them because the corrections are not self-evident.

(i) Finite differences inflate the participation ratio. PR is a *tail* statistic, governed by the small eigenvalues. A closure evaluated by an unrolled fixed point carries a small, step-*independent* gradient noise (from the convergence tolerance, not truncation), so central finite differences lift the exactly- and near-zero eigenvalues onto a floor and PR rises. On the differentiable COSMO-SAC-2002 layer, autograd and finite differences on *identical* profiles give PR = 1.00 and 1.25—a 25% inflation from noise alone, though the per-gradient cosine is 0.995 (the direction is right, the tail is not). It is worse on a typed (3×51) profile basis. A closure that ignores the typing (2002) has 102 exactly-redundant coordinates that finite differences decorrelate into a spurious floor, inflating its PR from 1.0 toward 2.0, while a closure that uses the typing (2010) has no such redundancy. The same step therefore inflates the two *asymmetrically* and can manufacture a rank ordering with no physical content. *Autograd is the appropriate instrument where the closure is differentiable; with only finite differences,* PR *is interpretable against a known-rank or permuted control, never absolutely.*

(ii) "Own vs. other" controls are powerless when the resolved direction is common. A dimension- and smoothness-matched control aligns a molecule's drift against its own predicted compensation and against other molecules'. It has power only if the predictions differ. When the aggregate Fisher is rank-one (§4.4), however, the resolved direction is *common*, so "own" and "other" point almost the same way and the diagonal must sit inside the null—$p \approx 0.5$ regardless of the truth. On the matched set sparsity compounds this: a median of one pair per molecule makes each per-molecule Jacobian subspace one-dimensional, so "own" and "other" subspaces coincide exactly. *A control's null is interpretable only once its power is established: if the compared predictions have median cosine near one, the test cannot reject.*

(iii) Energy-in-subspace baselines need the effective rank and a cluster null. A random zero-mean vector in D dimensions places $k_{\text{eff}}/(D-1)$ of its energy in a subspace of *effective* rank k_{eff} (the participation ratio of its singular values), which for near-collinear Jacobian rows is far below the nominal row count. On the matched set the per-molecule subspace is effectively one-dimensional, so the baseline is $\approx$2% and the observed 5.5% is a two-to-threefold *enrichment*, not the "no concentration" a raw own-versus-other equality suggests: own equals other because both one-dimensional subspaces are the same common direction, not because the drift avoids the resolved one. With that direction common, N enriched molecules are not N independent draws—the enrichment needs a cluster null (§4.2), without which its significance is over-stated by $\sqrt{N/N_{\text{eff}}}$. We therefore report the 5.5% as evidence in neither direction. *The baseline must be set by effective rank, and the null clustered by the shared direction.*

Each failure converts a spectral concentration statistic into a structural claim it does not support. The surviving statements of §4.4 avoid that conversion by carrying their null model and a power check inline.

G A second chemical closure: pKa through a Hammett LFER

The measured result of this appendix is stated in §4.6; what follows is the construction, the estimator and the sweep behind it. The instrument is not specific to COSMO-SAC. In a second real domain—aqueous pK_a of substituted aromatics—the fixed closure is a Hammett linear-free-energy relationship $g(\sigma_{\mathrm{H}}) = pK_{a,0} - \rho\,\sigma_{\mathrm{H}}$ over the Hammett substituent constant σ_{H}, written with its subscript throughout so that it cannot be confused with the σ-profile of §2, with Hansch–Leo tabulated constants as the external oracle. The two classic LFER failure modes fall cleanly onto the two channels of Lemma 1: *resonance saturation* (a curvature the linear closure cannot represent) lands in B_{closure}, while *ortho/steric* effects, absent from the tabulated constants, are information the intermediate lacks and land in B_{insuff}. On the OPERA experimental compilation[29]—the canonical full set under a uniform, closure-independent quality filter that drops pre-ionized microstates (App. H)—this yields a *measured* fidelity sweep. The clean meta/para series (n=158) is well specified (RMSE 0.55, B_{closure} below the insufficiency floor). The ortho/heteroaromatic regime (n=846) degrades to RMSE 2.26, of which a within-scaffold estimator places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure. That estimator is an out-of-fold residual, so both figures are one-sided, and the a-priori channel assignment—ortho information the tabulated constant lacks—is consistent with the measurement rather than certified by it (App. H); the sweep is the near-exact-to-degraded transition. The decisive test is the same oracle swap as on solubility—the learned $\hat{\sigma}_{\mathrm{H}}$ replaced by the tabulated $\sigma^{\star}_{\mathrm{H}}$ in the same closure, the sign of the change recorded per stratum—measured *in-distribution* across K=20 within-scaffold splits. It *flips*. On the clean meta/para pole feeding the reference constant *helps* (learned MAE 0.48 vs oracle 0.315; oracle better in 19/20 splits and in three of the four reaction series): the ceiling there is the noisy learned estimate, not the closure—a real-data grounding-helps corner and an internal positive control. On the ortho pole the same swap *hurts* (learned 0.83 vs oracle 1.62; oracle worse in **20/20** splits and in all four series, gap 90% CI $[+0.57, +0.95]$): the freely learned $\hat{\sigma}_{\mathrm{H}}$ absorbs the ortho/steric information the tabulated constant lacks. What reverses here is the sign of the grounding penalty; it reverses through the insufficiency channel—consistent with the one-sided bound above, though not isolated by it—rather than through the closure-error cancellation of §4.4: a second instance of the sign rule, not of that mechanism. The clustering unit is the Hammett reaction series, of which only G=4 exist, so the accompanying bootstrap resolves nothing finer than series-level sign agreement—read as a sign test on four units, p=1/16 on the ortho pole and $p\approx$0.31 on the clean one—and is quoted as such (App. H).

Both signs obey one *crossover law*—grounding helps a pole iff the learned arm's error there exceeds the fidelity-set oracle error—which a competence sweep confirms (App. H). The well-specified pole's threshold ($\approx$0.315) is never beaten by the learned arm, so grounding helps at every training budget. The degraded pole's threshold (1.62) is high, so the learned arm beats it, and grounding hurts, across the whole interpolation range (down to 12% of the data), reverting to helps only under scaffold *extrapolation*, where $\hat{\sigma}_{\mathrm{H}}$ breaks (2.6$\pm$2.0) back above the threshold. The flip is thus a within-distribution, fidelity-driven crossover whose one scope boundary is predicted by the same law. It is the sharp, two-sided form of the penalty sign rule (Prop. 1); its "a fitted intermediate can beat the true one" half is known—plug-in efficiency under correct specification,[34] pseudo-true fitting under misspecification[36]—sharpened here into a fidelity-set threshold. The trained physics-vs-DirectGNN comparison is weaker and is not load-bearing (App. H). On an independent chemical closure the instrument separates a helps-corner from a hurts-corner, each with a sign consistent across splits and series; the full construction, the estimator validation and the measured sweep are in App. H (Fig. 10).

H The pKa closure: full construction, measured sweep, and trained comparison

This section gives the full construction behind App. G: a fixed Hammett linear-free-energy relationship (LFER) $g(\sigma_{\rm H}) = {\rm p}K_{a,0} - \rho\,\sigma_{\rm H}$ over the Hammett substituent constant $\sigma_{\rm H}$ (with ${\rm p}K_{a,0}, \rho$ the per-scaffold Hansch–Leo–Taft reaction-series constants), for which Hansch–Leo tabulated constants provide an external oracle $z^\star$. The analogy is close: as with the σ-profile through COSMO-SAC, a learnable physical descriptor is pushed through a fixed, approximate, differentiable map. The two classic LFER failure modes fall cleanly onto the two error channels of Lemma 1: *resonance saturation*—a curvature in $\sigma_{\rm H}$ that the linear closure structurally cannot represent—is a mismap of a function of $z^\star$ and lands in $B_{\rm closure}$; *ortho/steric* field effects, absent from the tabulated constants, are information the intermediate lacks and land in $B_{\rm insuff}$. The chemical scope is a fidelity sweep: meta/para monosubstituted benzoic acids, phenols, and anilinium ions are the near-exact ($B_{\rm closure} \approx 0$) regime, ortho (insufficiency-dominated) and polyfunctional the low-fidelity one.

This decomposition is directly visible before any training. Pushed through the fixed Hammett closure, *p*-nitrophenol reads ${\rm p}K_a \approx 8.3$ against an experimental ≈ 7.1 (and *p*-nitroaniline overshoots similarly): the linear closure is off by ~ 1 unit exactly where through-conjugation ($\sigma_{\rm H}^-$) breaks it—$B_{\rm closure}$ observed in practice.

A controlled semi-synthetic construction on real tabulated $\sigma_{\rm H}$ (known ground-truth channels) validates the estimator and reproduces the paper's central distinction in this second closure (Fig. 10). Sweeping closure fidelity with a sufficient intermediate, the grounding gap $\mathcal{G}$ tracks $B_{\rm closure}$ (the plug-in estimate recovers the truth to within a few percent, e.g. 0.20 vs 0.21, 0.81 vs 0.81), the case where Theorem 1's proxy is valid. Sweeping the ortho/steric channel with a *well-specified* closure gives $B_{\rm closure} = 0$ at every knob while $\mathcal{G}$ rises with the insufficiency fraction, from -0.035 at zero insufficiency through zero near a fraction of 0.25 to $+0.34$ at 0.7: at the three fractions where it is positive it is positive with $B_{\rm closure}$ identically zero, which is the pKa realization of the conflation that makes $\mathcal{G}$ an indicator rather than a certificate. The two negative points are the same statement in the other direction—a well-specified closure with a nearly sufficient intermediate leaves the oracle marginally ahead. The oracle-$\sigma_{\rm H}$ pipeline (scaffold-resolved substituent$\rightarrow$ $\sigma_{\rm H}$ assignment) and this validation extend directly to real data. On the OPERA experimental ${\rm p}K_a$ compilation[29] (7,904 records under the same net-neutral filter), applying the same decomposition to the fixed Hammett closure yields a *measured* fidelity sweep. On the clean meta/para series (n=158) the closure predicts experimental ${\rm p}K_a$ to RMSE 0.55, with the closure at the input-insufficiency floor: the leakage-immune lower bound MSE $- B^{\rm up}_{\rm insuff}$ is -0.2 (95% CI $[-0.4, -0.0]$), i.e. below zero, as a lower bound on the non-negative $B_{\rm closure}$ must sit when the LFER predicts as well as a flexible within-scaffold regressor. There $B_{\rm closure}$ is not separately resolvable and is consistent with a well-specified closure (per-scaffold systematic offsets ≤ 0.18). The ortho/heteroaromatic regime (n=846) degrades to RMSE 2.26, dominated by input insufficiency. Here the pKa latent is a *scalar* Hammett constant, so—unlike the solubility regime—$B_{\rm insuff}$ is estimated *directly* rather than through a coarsening of the closure's argument (§2.2): a within-scaffold out-of-fold regression estimator gives $B_{\rm insuff} = 4.5$ (95% CI $[4.0, 5.1]$; $\approx$ 89% of the error—the tabulated $\sigma_{\rm H}$ carries no ortho information, exactly the a-priori channel assignment), leaving a closure misspecification $B_{\rm closure} = 0.6$ ($[0.3, 0.8]$, a stratified within-scaffold molecule bootstrap). Directly estimated is not point-identified: the estimator is an out-of-fold residual mean square, which is an upper estimate of $\mathbb{E}[\mathrm{Var}(\mathrm{p}K_a \mid \sigma_\mathrm{H})]$, so 4.5 is an upper figure and the 0.6 it leaves is a lower one—one-sided in the same direction as everywhere else in this paper, and narrower only because the scalar latent needs no binning (§2.2, App. G). This is the near-exact-to-degraded transition, measured rather than asserted. This places the pKa closure near the phase-diagram boundary in its clean regime.

Data quality control (uniform, closure-independent). The trained runs use the canonical full OPERA set (7904 records). A small number of entries are supplied as pre-ionized microstates—an [NH$^+$]-protonated aminopyridine at pK_a = -7.78, a pyrylium [O$^+$], among others—inconsistent with a neutral-scaffold Hammett assignment, and they degrade the clean pole (pyridinium RMSE 0.63 $\to$ 3.83). We drop them with one rule, *net formal charge* $\neq 0$, applied uniformly to both strata. It is closure-independent (it never references the oracle residual, so it cannot be circular) and keeps net-neutral substituents such as nitro [N$^+$][O$^-$]=O. This removes 5 of 163 meta/para rows (restoring the clean pole to RMSE $\approx$0.55) and 352 of 1198 ortho rows (leaving 846; the ortho RMSE is essentially unchanged, so the filter does not manufacture the sign), for 1004 Hammett-amenable rows.

Evidence for the flip. We train the physics arm (encoder $\to$ $\hat{\sigma}_\mathrm{H}$ $\to$ fixed LFER) on K=20 stratified 80/20-within-scaffold splits—a Bemis–Murcko split collapses the meta/para pole to two ring systems and is unstable—and record the paired per-molecule error against the deterministic oracle, per stratum. Meta/para: learned MAE 0.48 ± 0.09 vs oracle 0.315 ± 0.07, gap -0.165, oracle better in 19/20 splits, 90% CI $[-0.28, -0.04]$. Ortho: learned 0.83 ± 0.07 vs oracle 1.62 ± 0.08, gap $+0.79$, oracle worse in 20/20 splits, 90% CI $[+0.57, +0.95]$ (both intervals from the scaffold-cluster bootstrap).

What the two bootstraps do and do not resolve. We report the sign statements above rather than the bootstrap P values, because neither bootstrap here carries the resolution a two-decimal P implies. The cluster unit is the Hammett reaction series, and only G=4 exist (benzoic acid, phenol, anilinium, pyridinium): the resampling law then has $\binom{7}{4}$=35 distinct outcomes and a finest atom $(1/4)^4 \approx 0.004$, so the cluster-bootstrap P(oracle worse)=1.00 on ortho is exactly the statement that all four series means favour the learned arm, and P=0.004 on meta/para that exactly one of them does, the other three favouring the oracle. Read as a sign test on four units those are p=1/16 and $p\approx$0.31; the interval endpoints quoted above are percentiles of the same 35-outcome law and are no finer. This is the same small-G caution applied on the solubility side (§C), where the P values are likewise read as a coarse band rather than as two-decimal quantities. The split bootstrap adds nothing independent: the 20 splits resample train/test assignments over the same molecules (each molecule lands in $\approx$4 test sets), so it prices split-to-split noise rather than molecule sampling, and with all 20 ortho gaps positive it returns 1.000 by construction—a restatement of the 20/20, not a second confirmation. What the flip rests on is therefore the sign consistency across splits and series and the size of the ortho gap (+0.79 against an oracle MAE of 1.62, $\approx$49%), not on a P value. A leave-one-series-out refit, the estimate that would settle the series-level question at G=4, was not run.

The crossover law and its scope boundary. A competence sweep (train fraction $\in \{1.0, 0.5, 0.25, 0.12\}$ and, as the low-competence extreme, scaffold extrapolation; six seeds each) confirms the sign is set by the crossover of the learned error against the fidelity-fixed oracle threshold. On meta/para the learned arm rises from 0.46 to 0.98 as data shrinks but never reaches the 0.31 oracle threshold, so grounding helps throughout. On ortho the learned arm stays below the 1.62 threshold across the whole interpolation range (0.85 at full data, 1.31 at 12%), so grounding hurts throughout;

only scaffold extrapolation, where $\hat{\sigma}_{\mathrm{H}}$ breaks to 2.6±2.0, pushes it back above the threshold and reverts the sign. The flip is therefore a within-distribution, fidelity-driven crossover, and the extrapolation reversion is predicted by the same law. A trained physics-vs-DirectGNN comparison agrees in direction overall but has weak absolute errors and an untuned control; the load-bearing second-domain results are the decomposition above and the sign-consistent flip.

I Data provenance and reproducibility

The broad IDAC set (n=477), and the UD profile database. The UD σ-profile database is the 2261-compound COSMO set reported by Fingerhut et al.[26] and redistributed with the benchmark implementation of Ref.:[23] 1941 compounds from the University of Delaware collection assembled in Sandler's group, which extends the freely available VT set with revised molecular conformations for about a third of its compounds, plus 320 generated by the authors of Ref.[26] It ships both the Mullins-averaged single profile the 2002 kernel prescribes and the typed Hsieh three-profile set the 2010 kernel prescribes, with the cavity volume and, for most compounds, a tabulated dispersion parameter. The broad IDAC set is the intersection of that database with our expanded ThermoML infinite-dilution pull, at every temperature rather than only 298 K.

How the 14,900 *raw rows are extracted.* The pull walks the NIST ThermoML archive JSON records for four journals and keeps a row only where: the dataset has exactly two components; the property is an activity coefficient at infinite dilution, either declared so by a dataset constraint or carried at a solute mole fraction of zero; exactly one component is the non-solute; a temperature is resolvable for the value; and γ^{∞} is finite and positive. Components are mapped to SMILES through the record's own InChI. Duplicate (solute, solvent, temperature) records are then aggregated, with the record and DOI counts retained per row, which takes 14,910 extracted values to 14,900 rows. No solubility-side, method-side or quality-side filter is applied at this stage.

Construction ladder. 14,900 raw IDAC rows; 14,900 with a finite $\ln\gamma_2^{\infty}$ and temperature; 490 whose solute *and* solvent both resolve into the UD compound list by InChIKey (first-block fallback); 490 for which both the Mullins and the typed profile files exist and parse at the expected 51 and 153 bins; and 477 for which both components additionally carry a tabulated dispersion ϵ, which the *2010, dispersion on* arm needs and which we impose on every arm so that the three kernels are scored on identical rows. That leaves 78 solutes, 36 solvents and 35 distinct temperatures from 273.2 to 403.1 K, with $\mathrm{Var}(m) = 4.85$ and $\overline{m} = 3.42$—a harder and far less ideal regime than the corner's $\overline{\ln\gamma_2^{\infty}} = 0.75$. It is not disjoint from the corner: 57 of the 60 corner pairs appear in it at 298 K, evaluated there on UD rather than VT-2005 profiles. The two are therefore the same chemistry measured against two profile databases, not independent samples, and we never combine them. The 3 corner pairs it excludes are exactly the 3 that carry no tabulated dispersion ϵ: bromobenzene and iodobenzene in 1, 2-ethanediol, and 2-bromopropane in pyridine. The "dispersion-defined 57" of §4.2(iii) and the "57 shared with the broad IDAC set" are therefore one set and not two.

Why neither set can observe B_{insuff} directly. $B_{\mathrm{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^{\star}, T)]$ would be point-identified only by two or more independent determinations at a common closure argument, and neither design supplies one. VT-2005 assigns exactly one profile per compound; the 44 matched compounds have distinct profiles (closest pair L_2=3.4 against a median profile norm of 44), and no pair carries a replicate, so on the corner no two rows sit at a common $z^{\star}$. The broad IDAC set comes closer without reaching it: 359 of its 477 rows share a molecule pair with another row, but at a different temperature and hence at a different $(z^{\star}, T)$. On these two sets B_{insuff} is accordingly reached only as an upper bound—by coarsening in the law-of-total-variance cells, and out of fold in the fifteen fitted rows of Table 9—and B_{closure} correspondingly only as a lower bound; where the latent is

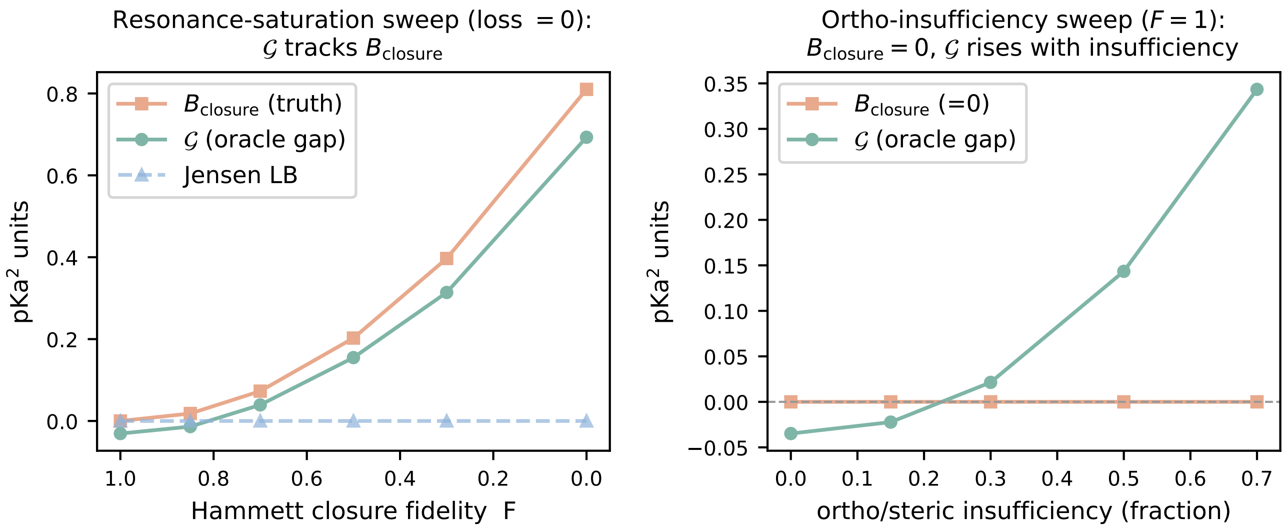

Figure 10: pKa through a Hammett LFER on a construction with known ground-truth channels. *Left:* closure fidelity varied with a sufficient intermediate; grounding gap $\mathcal{G}$, closure bias B_{closure}, and the assumption-light Jensen bound. *Right:* an ortho/steric insufficiency channel varied with a *well-specified* closure; $B_{\text{closure}} = 0$ at each knob while $\mathcal{G}$ rises with the insufficiency fraction, crossing zero near 0.25. Scope: §H.

a scalar the headline estimate is the out-of-fold residual, one-sided in the same direction (§2.2, App. H).

The matching rule, and its asymmetry with the corner. The corner is matched by canonical SMILES and this set by InChIKey with a 14-character first-block fallback, which is looser and, since a $z^\star$ imported across a tautomer, protonation or stereochemical variant inflates MSE and hence B_{closure}, looser in the anti-conservative direction. We therefore count it. Of the 90 distinct molecules, 86 resolve by full InChIKey; four resolve only through the first block (`C=CC1CC=CCC1`, `CCC(C)N`, `CC(Cl)CCl`, `CCC(C)C(C)=O`), touching 12 of the 477 rows. In all four the discrepancy is the stereochemical layer and not a tautomer or a protonation state: each query key ends `-UHFFFAOYSA-N`, an unspecified stereocentre, against a stereo-labelled UD entry with the same connectivity. On the 465 exactly matched rows (178 distinct pairs) the decomposition is unchanged, read at the headline cell of the grid (eight equal-count bins, unbiased within-bin variance, edge values to the upper bin), which is what recomputes these two lines from the deposit: deployed convention MSE 1.930, $B_{\text{insuff}} \lesssim 0.699$, margin $+0.53$ at $P \approx 0.9$; full convention 1.926, 0.495, $+0.94$ at $P > 0.95$.

Composition, and where the squared error sits. The set's chemistry is small in both channels: n-alkanes, aromatic hydrocarbons and ketones supply 243 of the 477 solute rows, and water and the four ethylene-glycol oligomers 293 of the solvent rows. Its overlap with this paper's solubility corpus is correspondingly asymmetric—only 0.46% of corpus rows carry a solute from this set, while 34 of its 36 solvents cover 59% of corpus rows—which is what the main text means by the set covering the solvent side of deployment and not the solute side. The error concentrates in the same places: under the deployed convention 45% of the squared error sits in the four ethylene-glycol solvents and 32% in the 37 rows carrying water as solute, and under the full convention 49% sits in those same four solvents. The fidelity null of §4.2(iii) is therefore a small-molecule and glycol-heavy null.

Design and stratification. The 477 rows are 185 distinct solute–solvent pairs: 118 pairs carry one row, 38 carry five or more, and the largest carries 15. Counted per row the design ratio is $\dim(z^\star, T)/n = 103/477 \approx 0.22$; counted per distinct pair, 0.56, against 1.7 on the corner. Molecule-pair identity coarsens $(z^\star, T)$, so $\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$ is a valid B_{insuff} upper bound over the 359 rows that support it; it is 0.046, under 1% of $\mathrm{Var}(m)$ and thirteen times tighter than the binning bound on those same rows (0.61 deployed, 0.39 full). We report it and do not promote it, because all 67 multi-row pairs come from a *single* publication each: those rows are one laboratory's temperature series, so the bound omits the between-determination component of the superpopulation (§2.2) while the binning bound, whose eight bins mix all 15 sources, does not. It bounds a sub-component, so the reading that holds is that the binning bound is slack rather than that B_{insuff} is near zero. The same fact —that the temperature axis carries under 1% of $\mathrm{Var}(m)$ at a fixed profile pair—makes the choice between conditioning on $z^\star$ and on $(z^\star, T)$ numerically immaterial on this set, though only the latter is licensed by Lemma 1 and it is the one we use. Collapsing to one row per pair (each the row nearest 298 K) gives MSE $= 1.679$, $B_{\text{insuff}} \lesssim 0.495$ and a margin of $+0.69$ at $P > 0.95$ under the full convention (1.540, 0.635, $+0.27$, $P \approx 0.8$ deployed). By method, 334 rows are chromatographic, 37 chemical or heterogeneous equilibration and 106 carry no method field; by source, 15 publications in four journals (*J. Chem. Eng. Data* 203, *J. Chem. Thermodyn.* 114, *Fluid Phase Equilib.* 108, *Thermochim. Acta* 52), the largest single publication contributing 105 rows. On the 334 chromatographic rows alone, re-binned within the stratum, the activity error is MSE $= 1.783$ against $B_{\text{insuff}} \lesssim 0.36$ under the full convention—a margin of $+1.06$ ($P > 0.95$), wider than the full set's $+0.95$—and 1.474 against $B_{\text{insuff}} \lesssim 0.58$ under the deployed one, a margin of $+0.31$ ($P \approx 0.9$), narrower than the full set's $+0.51$. The cut therefore does not move the sign under either convention, and it does not uniformly widen the margin either. A DOI-clustered bootstrap of the full-set margin gives $P > 0.95$ with 90% $[+0.50, +1.57]$ (full) and $P \approx 0.9$ with $[-0.07, +1.74]$ (deployed), and a pair-clustered one $P > 0.95$ with $[+0.71, +1.27]$ and $[+0.23, +1.16]$ respectively. This is the stratification the 60-pair corner cannot support, and it is why the broad IDAC set carries the verdict: the quality cut that *inverts* the enlarged 298 K corner (§5, limitation (i)) leaves the sign alone here under either convention.

Sources. Solubility: BigSolDB 2.0,[21] partitioned by Bemis–Murcko scaffold—molecules grouped by ring system plus linker, side chains stripped. Crystal $T_m/\Delta H_{\text{fus}}$: an open melting-point pool plus a group-contribution prior. True σ-profiles: the VT-2005 database, the Virginia Tech 2005 compilation of quantum-chemically computed screening-charge profiles for ∼1,400 compounds[6] (ingested to `sigma_profiles.csv`, 1432 compounds, with the COSMO cavity volume V_{cosmo} for the combinatorial term). These are quantum-chemically computed for a single reference conformer and protonation state; mismatch to the experimental measurement conditions is noise in $z^\star$ that inflates B_{insuff}, but under a Berkson-type displacement it also loads onto B_{closure} through the curvature of g, so we treat it as an unsigned limitation and *not* as conservative for the closure-dominance verdict (§4.2). Concretely, if the true latent is $z^\star$ displaced by a mean-zero (Berkson-type) ε, then even a perfectly specified g has $B_{\text{closure}} = \mathbb{E}\big[(\mathbb{E}_\varepsilon[g(z^\star{+}\varepsilon)] - g(z^\star))^2\big]$, a pure curvature effect that vanishes only for affine g—and under classical rather than Berkson error, not even then. We do not bound that curvature share. Infinite-dilution activity coefficients: ThermoML, the NIST/IUPAC archive of experimental thermophysical data.[22] Enlarging the reference σ-profile set would not materially grow the matched activity corner: of the 298 K ThermoML IDAC pairs, only two solutes lack a VT-2005 profile while their solvent has one—the binding constraint is joint IDAC×profile coverage, not profile availability. What does enlarge the corner is a broader extraction on the IDAC side: the expanded ThermoML pull released with the code matches 115 pairs at 298 K. We do not adopt it as a replacement corner because its verdict is quality-cut sensitive: it widens the margin from 0.24 to 0.35 but *inverts* it to −0.13 on the 100 pairs labelled gas chromatography, and that sensitivity is unresolved at 298 K. It is not a verdict on the extraction, which matters because the broad IDAC set is built from the same pull: there the same cut leaves the sign alone under either convention (+0.51 → +0.31 deployed, +0.95 → +1.06 full). What differs is resolution, not provenance—at 298 K a 15-row cut inverts a margin already inside the design's resolution, whereas on 477 rows a 143-row cut moves one without inverting it—so we use the enlarged pull where it is stratifiable and keep the strict n=60 corner where the paradox's own oracle is defined. The deployed corner is also narrowly sourced: the curated IDAC compilation yields 74 VT-2005-matchable pairs under canonical SMILES (60 at 298 K plus 14 off-temperature), all by gas chromatography and all from three publications, 33 pairs from one, so neither a method-stratified nor a source-clustered re-estimate is available on it. The corner is matched to VT-2005 by RDKit canonical SMILES on solute and solvent, the rule the deployed σ-oracle itself uses; matching by InChIKey instead admits 85 pairs over 46 solutes and 19 solvents at 298 K in place of 60/41/17 (both counts are recorded in `decomposition.json`). We keep the stricter rule so that the $z^\star$ charged against each label is the profile the deployed pipeline would supply, rather than one imported across a tautomer or protonation variant—the same mismatch that makes the reference imperfect above. The decomposition was not re-estimated on the InChIKey-matched set; whether the verdict moves on those 25 extra pairs is untested. The broad IDAC set uses the looser InChIKey rule, so the two sets are matched under different standards; that asymmetry is quantified above (four molecules, 12 rows, verdict unchanged on the exactly matched subset) rather than left implicit. Second-domain pKa: the OPERA experimental compilation[29] (`pKa_QR.sdf`, 7904 QSAR-ready records, public-domain US EPA; cited, not redistributed); the uniform net-formal-charge filter reduces this to the 1004 Hammett-amenable rows (158 meta/para, 846 ortho/heteroaromatic) analysed in §H. The Bemis–Murcko scaffold split follows the main text. Because the seeded split is not stable across pipeline versions (§2), the numbers in this paper are computed on a *single, pinned* split, whose per-file SHA-256 each run's manifest records and whose files are deposited with the checkpoints rather than versioned—with the one run whose split provenance was not logged, recorded in the footnote.[4] Melting-point tables are auto-detected as °C or K to avoid a +273 K double-conversion.

Grounding-stream provenance, and one build that violates the guard. The σ-stream builder drops any pool molecule whose Bemis–Murcko scaffold appears in the validation or test split and aborts rather than proceed when a split file is missing; a correct build retains 1319 of the 1427 VT-2005 molecules that parse into the pool. The stream file was subsequently rebuilt with the exclusion pointed at the splits of a small debug dataset: it carries 1425 rows, of which 106 (99 train, 7 validation) have a scaffold in the real test ∪ validation union of 2702 scaffolds. The overlap is not confined to scaffold relatives: 91 of those 106 rows are themselves held-out solutes by canonical SMILES — 65 test solutes and 26 validation solutes — and the correctly guarded build leaves none of them. The crystal stream's build on disk records the correct 2702-scaffold exclusion, so among the streams feeding the runs reported here the defect is confined to the σ one. Per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm. The rebuilt file postdates every prediction file behind Fig. 2 and Table 4 and predates the two single-seed oracle-application controls (§C) and the output-residual arms (§4.5). That ordering is consistent with the headline arms having used the guarded build and the later controls the

[4]The three-seed surrogate isolation (§4.4) was produced on a separate run whose per-run split provenance was not logged; its matched set is the same n=44 molecules, derivable from the committed split independently of the model, so it is consistent with the reported split. We record the split hash in the runner so future runs are auditable directly.

leaking one, but these are timestamps on local copies of cloud-run outputs and settle nothing on their own. Three facts bound what the leak can do. It is common-mode across the paradox contrast, the learned-σ and σ-oracle rows being one checkpoint evaluated two ways, so it cannot generate the gap between them; Direct-GNN uses no grounding stream, so a leak can only favour the physics arm that loses to it; and on the n=44 matched set of §4.4 a correct guard would have removed one molecule (piperidine). What it does compromise is auditability, and with it the reach of the split guarantee: an arm trained on that build would have had reference σ-profiles — though no solubility labels — for 65 test solutes, and the guarantee stated in §3 is therefore a property of the released builder rather than of every stream file we trained on. Closing this requires hashing each stream build beside the split hash, which was not done for these runs.

The two decomposition sets, deposited row by row. Because the closure/insufficiency verdict rests on a newly constructed set, and because the repository record is a placeholder (limitation (x)), both sets are deposited with the Supporting Information as row-level tables that need no code to inspect. The results the placeholder leaves uncheckable are, specifically, the broad IDAC set with its design and provenance audit, the 2010 kernel's own-latent floors, the fitted kernel residual and the estimator grid. **Table S1** (`broad_idac_set_477.csv`) carries one row per measurement: pair identifier, solute and solvent SMILES, the UD compound key matched to each and which rule matched it (full InChIKey or first-block fallback), temperature, the experimental $\ln \gamma_2^\infty$, the *four* closure outputs—2002 and 2010-dispersion-off, each in the full and the residual-only combinatorial convention, so that the deployed-convention kernel contrast of §4.2(iii) is recomputable from the file—the ThermoML method field, the source DOI and journal, and a flag for the 57 rows shared with the corner. Some analysis artifacts label that same table `representative_set_477.csv`. **Table S2** (`corner_set_60.csv`, printed in full as Table 17) carries the same fields for the 60 corner pairs, with three differences that follow from how the corner is defined. Its closure outputs are the VT-2005 ones the deployed σ-oracle uses (columns `g_full`, `g_res`) alongside the four UD-profile outputs that make it commensurable with Table S1. Its matching rule is canonical SMILES throughout, so it carries neither the per-row match-rule columns nor the UD compound keys. It also names each compound in words as well as in SMILES. It additionally carries a per-row record count, 1 throughout. All 60 rows sit at 298 K, all are chromatographic, and their three source DOIs divide the set 33/20/7. The counts, margins, stratified re-estimates and clustered bootstraps reported for either set are recomputable from these two files with the estimator definitions of §4.2; what they do *not* restore is the provenance of the closure evaluations themselves, which still requires the code.

Compute ledger. The headline σ-grounding comparison (Fig. 2, Table 4) is a full-corpus GPU run (three seeds); the closure/insufficiency decomposition (§4.2), the Fisher audit (§F.2), the noise floor, the encoder probe, and the temperature and mechanism diagnostics of §E are CPU-reproducible from deposited artifacts. The prediction CSVs, the model checkpoints and the larger analysis artifacts will be deposited separately, in a Zenodo archive whose DOI is registered at submission, rather than in the repository. The broad IDAC set's design and provenance audit—distinct-pair counts, within-pair variance across temperature, pair-blocked out-of-fold estimators, method and source-DOI composition, the method-stratified and cluster-resampled margins, the kernel contrast split into the shared 57, the complementary 420 and the 80 complementary rows that are also at 298 K, the corner floor recomputed on UD profiles, and both sets' floors under the 2010 kernel's own typed latent—regenerates in one CPU run of `scripts/analysis/run_b_insuff_representative_audit.py` into `results/b_insuff/representative_audit.json`. A sec-

Table 17: **Table S2.** The corner (n=60), row by row, deposited so the set can be inspected without running code. Read in two blocks of 30, left then right.

#	solute[a]	solvent	m	g_{full}	g_{res}	broad	#	solute	solvent	m	g_{full}	g_{res}	broad
1	`Brc1ccccc1`	`OCCO`	3.81	2.33	2.40	n	31	`CCc1ccccc1`	`c1ccncc1`	0.36	0.35	0.39	y
2	`C#CCCCCC`	`c1ccncc1`	0.44	-0.42	-0.34	y	32	`CO`	`C1CCNCC1`	-0.84	-2.38	-2.21	y
3	`C1=CCCCC1`	`c1ccncc1`	0.91	0.75	0.75	y	33	`CO`	`CC(C)(C)N`	-1.24	-3.15	-2.99	y
4	`C1CCOC1`	`c1ccncc1`	-0.07	0.03	0.03	y	34	`CO`	`CCC(C)N`	-0.97	-3.20	-3.02	y
5	`C=CCCCCC`	`c1ccncc1`	1.27	0.93	1.03	y	35	`CO`	`CCCCCCN`	-0.97	-3.23	-2.87	y
6	`CC#N`	`c1ccncc1`	0.38	0.25	0.30	y	36	`CO`	`CCCCN`	-1.05	-3.00	-2.79	y
7	`CC(C)(C)N`	`CO`	-1.39	-2.07	-1.85	y	37	`CO`	`CCCNCCC`	-0.37	-2.33	-1.97	y
8	`CC(C)(C)N`	`c1ccccc1`	0.28	0.39	0.38	y	38	`CO`	`CCNCC`	-0.67	-2.35	-2.15	y
9	`CC(C)=O`	`c1ccncc1`	0.17	-0.13	-0.12	y	39	`CO`	`Cc1cccc(C)n1`	-0.49	-0.78	-0.49	y
10	`CC(C)Br`	`c1ccncc1`	0.30	0.22	0.21	n	40	`CO`	`Cc1ccccn1`	-0.24	-1.07	-0.86	y
11	`CC(Cl)CCl`	`c1ccncc1`	-0.07	-0.05	-0.05	y	41	`CO`	`Cc1cccnc1`	-0.06	-1.06	-0.85	y
12	`CCC(C)N`	`CO`	-0.29	-1.99	-1.71	y	42	`CO`	`Cc1ccncc1`	-0.39	-1.11	-0.90	y
13	`CCC(C)N`	`c1ccccc1`	0.14	0.42	0.42	y	43	`CO`	`NC1CCCCC1`	-1.11	-3.21	-2.98	y
14	`CCCCCCC`	`c1ccncc1`	1.91	1.60	1.71	y	44	`Cc1ccc(C)cc1`	`c1ccncc1`	0.44	0.39	0.45	y
15	`CCCCCCCC`	`c1ccncc1`	2.07	1.74	1.91	y	45	`Cc1cccc(C)c1`	`c1ccncc1`	0.43	0.37	0.43	y
16	`CCCCCCCCC`	`c1ccncc1`	2.17	1.86	2.12	y	46	`Cc1cccc(C)n1`	`CO`	-0.06	0.14	0.64	y
17	`CCCCCCCCCC`	`c1ccncc1`	2.26	1.97	2.32	y	47	`Cc1ccccc1`	`OCCO`	4.53	2.61	2.67	y
18	`CCCCCCCCCCC`	`c1ccncc1`	2.32	2.08	2.52	y	48	`Cc1ccccc1C`	`c1ccncc1`	0.34	0.33	0.37	y
19	`CCCCCCN`	`CO`	2.62	-2.35	-1.66	y	49	`Cc1ccccn1`	`CO`	-0.29	-0.21	0.12	y
20	`CCCCCCN`	`c1ccccc1`	2.79	0.51	0.59	y	50	`Cc1cccnc1`	`CO`	-0.20	-0.36	-0.03	y
21	`CCCCCOC(C)=O`	`c1ccncc1`	0.11	0.09	0.26	y	51	`Cc1cccnc1`	`c1ccccc1`	-0.04	-0.01	-0.01	y
22	`CCCCCl`	`c1ccncc1`	0.43	0.37	0.38	y	52	`Cc1ccncc1`	`CO`	0.00	-0.41	-0.08	y
23	`CCCCN`	`CO`	-0.78	-2.18	-1.85	y	53	`Clc1ccccc1`	`OCCO`	3.69	2.29	2.35	y
24	`CCCCOC(C)=O`	`c1ccncc1`	0.14	0.04	0.14	y	54	`Fc1ccccc1`	`OCCO`	3.47	2.03	2.06	y
25	`CCCN(CCC)CCC`	`CCCCCCCCCCCCCCCC`	0.01	-0.03	0.07	y	55	`Ic1ccccc1`	`OCCO`	3.74	2.45	2.54	n
26	`CCCN(CCC)CCC`	`CO`	2.17	1.96	3.15	y	56	`NC1CCCCC1`	`CO`	1.57	-2.17	-1.81	y
27	`CCCN(CCC)CCC`	`c1ccccc1`	1.03	0.54	0.79	y	57	`NC1CCCCC1`	`c1ccccc1`	1.86	0.45	0.46	y
28	`CCCNCCC`	`CO`	-0.09	-0.31	0.40	y	58	`c1ccc2ccccc2c1`	`OCCO`	5.47	2.71	2.85	y
29	`CCCNCCC`	`c1ccccc1`	0.39	0.28	0.35	y	59	`c1ccccc1`	`OCCO`	3.47	2.11	2.12	y
30	`CCNCC`	`CO`	-1.11	-0.64	-0.30	y	60	`c1ccncc1`	`CO`	0.18	-0.28	-0.09	y

[a] Columns: m, the experimental $\ln \gamma_2^{\infty}$ at 298 K; g_{full} and g_{res}, the VT-2005-input COSMO-SAC-2002 output under the full and residual-only conventions; *broad* (the deposit's `in_broad_idac_set`), the 57 pairs that also appear in the broad IDAC set, the three that do not being the bromine- and iodine-containing pairs of §4.2(iii). The 477-row broad IDAC set is the companion machine-readable Table S1.

ond CPU run, `scripts/analysis/run_b_insuff_convention_audit.py` into `results/b_insuff/convention_audit.json`, produces the convention-rule material: both sets' margins and clustered bootstraps under each combinatorial convention, the per-cell broad-set grid of Table 6, the InChIKey match-rule audit and its exactly matched re-estimate, the pair-conditional bound and its single-source diagnosis, the typed latent's dimensions, and the two deposited row-level tables. Those files are among the artifacts limitation (x) records as uncheckable until the repository is released; the two deposited tables are the part of that material which can be checked without it.

Seeds and determinism. The controlled comparisons of §4.1 use seeds $\{42, 43, 44\}$ and are reported on a cross-arm intersection lock (rows supervised and finite in every arm); the surrogate isolation (§4.4) uses seeds $\{0, 1, 2\}$; the training-time and channel-swap oracle controls, the output-residual arms (§4.5), the data-efficiency curve (App. F.5) and the loss-simplification ablation (App. A) are seed 42 only. The decomposition is seed-independent (its closure has no learnable parameters). One artifact caveat: a save-time bug truncated the released per-row prediction file for the seed-44 oracle solubility arm to 295 rows, a small fraction of the test set. The three-seed headline summaries (Fig. 2, Table 4) were aggregated at run time, before that file was written, so they are unaffected; only the one on-disk artifact is partial, and any per-row re-analysis of the solubility arms therefore uses the two complete seeds $(42, 43)$. The learned-σ and DirectGNN solubility files and the activity-level (n=60, n=44) artifacts are complete for every seed.

J Black-box probe programme: full tables

All three studies use the same current-split DirectGNN checkpoints (seeds 42/43/44, test MAE 1.749/1.674/1.684) whose manifests pin the train/val/test SHA-256 to the split on disk. The probed object throughout is the control's 788-dimensional pair representation. Every number here is CPU-computed; none required a GPU.

Crystal-term decodability (Table 18). Probe fitted on 152 solutes / 14,348 rows, scored on 13 held-out solutes / 527 rows, solute overlap zero. Φ^* is 92.2% between-solute variance, so the effective n is 13 clusters.
Controls: positive—model's own prediction $R^2 = 0.980$, temperature $R^2 = 0.9997$; negative—random target -0.0006, molecule-blind constant profile -0.015, raw RDKit descriptors -0.514 (set A) and -0.928 (set B). Null $p_{95} = +0.105/+0.117/+0.096$ and max $= +0.316/+0.380/+0.346$ over 1000 draws.

Two features of Table 18 bear on how such probes should be read. First, the between-seed spread (0.616 range, sd 0.334) exceeds every individual value: a single-seed run would have been unreadable, and seed 43 alone would have read as a near-miss. Second, a decision rule of the common form $R^2(\text{model}) - R^2(\text{raw}) \geq 0.10$ would have declared success here on all three seeds under descriptor set B—including seed 42, whose probe is worse than a molecule-blind constant—and on two of three under set A. The subtrahend is an unregistered analyst choice worth more than the effect; we report $R^2(\text{raw})$ as a level beside $R^2(\text{model})$ and never as a difference.

Temperature structure (Table 19). Slopes are fitted at fixed (solute, solvent); a slope pooled across solvents would confound activity differences with temperature. Ceiling, measured on the data before any model: $r = +0.239$ on train (1353 pairs / 151 solutes) and -0.125 held out (66 pairs / 13 solutes, permutation $p = 0.581$).

Organisation of the representation (Table 20). Excess over a permutation null matched to the factor's level, on the full test set.
As a share of the solute-identity ceiling: scaffold 100.3/99.9/99.8%, $\log P$ 13.6/8.2/4.7%, TPSA 9.0/10.3/7.1%, MolWt 5.3/2.3/3.1%. That 99.8–100.3% scaffold range is the organisation measure §5 points here for. The scaffold share is pinned near one by the split rather than by the representation: of the 124 scaffolds spanned by the 147 test solutes only 11 hold more than one solute, so only 34 solutes share a Bemis–Murcko scaffold with another, carrying 492 of the 5608 rows (8.8%), and the two partitions coincide on 91% of the data. The raw variance shares run the other way (seed 42: scaffold 0.636 against solute 0.639), so the *raw* ratio falls short of one by 0.4/0.7/0.9% across the three seeds, and it reaches one only through the differing size-matched permutation nulls (0.022 against 0.026); no interval is attached to that 0.4–0.9% gap. The row therefore bounds what the representation adds beyond scaffold and does not measure within-scaffold discriminability (§5). A direct measure on the 34 solutes that do share a scaffold sizes what it bounds: two same-scaffold solutes in a common solvent sit at 0.21–0.40 of the between-scaffold representation distance across the three seeds (112 pairs over 11 scaffolds; 90% scaffold-cluster intervals inside $[0.14, 0.57]$). Molecules sharing a scaffold are compressed rather than unresolved.

Temperature is decodable from this vector at $R^2 = 0.9997$ yet occupies only 3–4% of its variance (rows above): decodability of a quantity is not evidence that a representation is organised around it, which is why the crystal-term probe is run as a held-out transfer test rather than an in-sample decode.

On the 331 rows carrying a measured Φ^*, Φ^* accounts for 18.3/42.5/18.0% of that subset's own solute-identity ceiling. This does not contradict the held-out null above. Φ^* is a solute-level quantity and the representation is organised by solute identity, so any solute-level variable shares variance with it in sample; the decodability probe above asks the different and sharper question of whether the mapping trans-

Table 18: Crystal-term decodability per seed, with the permutation null, the drop-one-solute jackknife and the leverage carried by the worst solutes.

seed	R^2(model)	perm. p	jackknife range	sign flips	top-3 leverage
42	−0.506	0.894	[−1.151, −0.228]	0/13	56.2%
43	+0.111	0.055	[−0.286, +0.333]	3/13	59.8%
44	+0.027	0.177	[−0.253, +0.237]	6/13	57.2%

Table 19: The model's temperature response against the empirical slope and against $-\Delta H_{\text{fus}}/R$, with the curvature of both in $1/T$.

	seed 42	seed 43	seed 44
corr(model, empirical slope)	+0.294	+0.406	+0.195
corr(model, $-\Delta H/R$)	+0.066	−0.283	−0.028
median |model − empirical| slope	1329	1088	1007
curvature in $1/T$ (data: 6.8×10^5)	2.0×10^6	2.1×10^6	2.3×10^6

Table 20: Share of representation variance per organising factor, as an excess over a permutation null matched to the level at which that factor varies.

factor	seed 42	seed 43	seed 44	null level
solute identity (147 levels)	+0.613	+0.587	+0.597	row
solute scaffold (124 levels)	+0.614	+0.586	+0.595	row
scaffold beyond arbitrary solute grouping	+0.088	+0.082	+0.082	solute
solvent identity (70 levels)	+0.278	+0.301	+0.271	row
temperature	+0.040	+0.029	+0.037	row
$\log P$ (solute)	+0.083	+0.048	+0.028	solute
TPSA (solute)	+0.055	+0.061	+0.042	solute
MolWt (solute)	+0.032	+0.013	+0.018	solute

fers to solutes the probe never saw. In sample the organisation is partly Φ^*-aligned; out of sample that alignment does not carry.

K Closure fidelity sets the reference-input penalty: a controlled synthetic family

To isolate the mechanism we construct a controlled synthetic experiment in which only the closure's fidelity varies (input quality, sample size, and estimator held fixed). A teacher draws a latent $z^\star$ and a target through a known map T; a fixed candidate closure g_F reconstructs it under a tunable misspecification of scalar fidelity F ($F = 1$ recovers T exactly, smaller F pushes g_F away). The reference-input penalty $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$ is a measure of B_{closure} in this construction—the free head recovers $\mathbb{E}[m \mid z^\star]$ while g_F is the misspecified map. Because ΔR^2 *equals* B_{closure} by design here, the sweep verifies that the instrument responds to fidelity as constructed; it is a check on the construction, not an independent test of the hypothesis. As F falls the penalty grows monotonically: averaged over six misspecification forms (linear, quadratic, cubic, absolute-value, sinusoidal, cross-term) F=1.00 gives $\Delta R^2 \approx 0$, F=0.38 gives ≈ -0.33, and an over-rotation stress test (F=−0.50) gives ≈ -2.0 (Fig. 11); all 24 family × form sweeps are individually monotone, and the ordering is identical across four unrelated teacher families (linear, monotone-nonlinear, kinetics-exponential, PDE-field), whose curves are visually coincident. That coincidence says the teacher factor carries no information *in this*

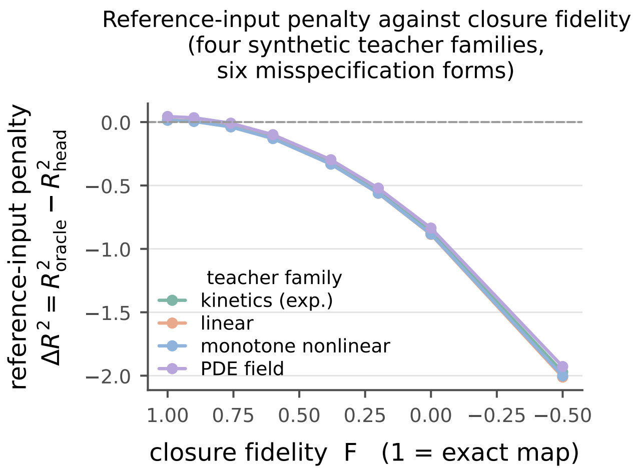

Figure 11: Closure-fidelity sweep on synthetic, non-chemical teacher functions. As the closure is made more misspecified (fidelity F falling from 1; axis reversed), the reference-input penalty $\Delta R^2 = R^2_{\mathrm{oracle}} - R^2_{\mathrm{head}}$ falls monotonically below zero. Each curve is one of four teacher families, averaged over six misspecification forms, which the panel does not resolve. No point is marked for the real closure. *Caveat:* here ΔR^2 *equals* B_{closure} by design, so the sweep is a check on the construction and not an independent test of the hypothesis; and the four curves are visually coincident because the teacher factor carries no information in this construction, not because the effect is domain-general. Scope: §K.

construction; it is not evidence that the effect is domain-general, and the panel does not resolve the six misspecification forms, which do spread (Fig. 11). Placing the real system on the sweep only illustratively (the real-closure-to-F map is a synthetic construction, not a calibrated measurement), the COSMO-SAC closure over predicted σ-profiles falls at $F \lesssim 0.4$: its solubility $\Delta R^2 \approx -0.36$ is what a closure of this fidelity predicts, consistent with B_{closure} dominating B_{insuff}. The two ΔR^2 values are measured on different targets—synthetic m on the sweep, $\ln x_2$ through the SLE solver and the crystal head in the real system—which is the other reason the placement carries no calibration.

The same construction shows why the grounding gap is an indicator, not a certificate (§2.3). Closure fidelity and input sufficiency are varied separately: when $z^\star$ is sufficient, $\mathcal{G}$ tracks B_{closure} as fidelity varies (Theorem 1's proxy is exact); when $z^\star$ is lossy but the closure is *well specified*, $B_{\mathrm{closure}} = 0$ throughout yet $\mathcal{G} > 0$ grows with the discarded variance. A positive grounding gap can thus be pure input insufficiency, not closure error—so the closure-dominance verdict rests on the law-of-total-variance bound of the decomposition, computed under the deployed convention (the bound is convention-specific even though the estimand it bounds is not; §2.2), which grades it likely rather than certified (§4.2), and $\mathcal{G}$ only corroborates it.

L Positioning against adjacent literatures

§1 places the instrument against the model-discrepancy problem of Kennedy and O'Hagan[19] and its calibration critique,[20] which is the frame the paper works in. Four further literatures bear on it without being that frame. Separating whether predictions are right on average from whether they are sharp is Murphy and DeGroot's reliability–resolution decomposition:[37,38] the closure/insufficiency split is an object of the same kind, an identity that partitions a squared error, but on a *fixed* operator with an external reference for the intermediate rather than on a forecast–outcome pair. A misspecified model still converges to a pseudo-true limit under White's quasi-maximum-likelihood account,[36] which is what Δ_∞ of Proposition 1 names for a composed predictor. That a decomposition may be only weakly constrained by the data is the subject of parameter sloppiness[39,40] and of structural versus practical identifiability[41,42]—the setting of Lemma 2. That apparent skill may reflect pipeline choices rather than physics is the concern of underspecification,[43] leakage[44] and compositional risk,[45] which is why every arm difference in this paper is reported as a difference between two separately tuned pipelines (§4). Finally, that a constrained prior lowers variance at the cost of a bias confined to the directions the data leave undetermined is the null-space reading of regularisation[46,47] and of null-space priors in inverse problems,[48] which is the frame in which Proposition 1 is stated.

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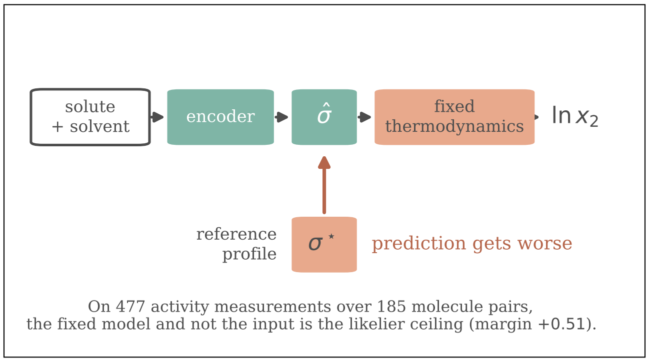
TOC Graphic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On 477 activity measurements over 185 molecule pairs,
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